

W2000 风力发电机组

操作运行手册

Operating Manual

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上海电气
SHANGHAI ELECTRIC

上海电气风电设备有限公司
SHANGHAI ELECTRIC WINDPOWER EQUIPMENT CO.,LTD.

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序言 Foreword

W2000 型风力发电机组为三叶片、上风向、水平轴、变速变桨距风力发电机组。风力发电机组的额定功率为2.0MW。

该风力发电机组风能利用效率高、噪声小、电网兼容性好、经久耐用、外型美观；能在变速条件下运行，在低风速段可提高有效发电，高风速段不会使风力发电机组和部件超载；采用双馈异步发电机和先进的IGBT 变流器的组合将电网电压、频率与发电机的转速隔离，从而使风力发电机组能与任何电网连接。

风力发电机组能在无操作人员值班条件下安全运行；所有部件均能在设计风况下安全可靠运行。

为了确保风力发电机组长期安全可靠的运行，事先要仔细的阅读操作运行手册并留意其中给出的建议。

请将这份操作运行手册看作风力发电机组安全装备的一部分，因此在风力发电机组随机资料中必须要保留一份该手册副本。

请注意手册中出现的警告标记。



警告！

忽略此警告将导致人员的受伤或致死，或其它导致财产损失的事故。

当以下某个事件发生，而控制设备未显示任何错误信息或检查到中央控制室未收到任何报警信号，请将相关信息告知我们：

安全装备尤其是攀爬系统故障或损坏、运行程序发生故障、不正常的噪声或意外事故等。

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W2000 is a three-bladed, up-wind, variable-pitch variable-speed, horizontal axis wind turbine, and its rated power is 2.0MW.

The wind turbine has the following advantage: wind energy utilization high, noise small, grid compatibility good, adequate durability, appearance good, and it can operate with variable speed so that the generating efficiency of the wind turbine is improved at the low wind speed and the wind turbine won't overload at the high wind speed. This wind turbine can isolate the generator speed with the grid voltage and frequency using double fed induction generator and advanced converter with IGBT, thus, the wind turbine can connect with any power grid.

This wind turbine can operate securely without any operating personnel on duty, and every part of the turbine can work reliably under the designed wind conditions.

The operating manual and the suggestions offered in it shall actually be read to insure the wind turbine can operate reliably for a long time.

Please regard the operating manual as a part of wind turbine safety equipment, so the copy of this manual shall be reserved in the wind turbine materials.

Attention! Ignoring this warning will cause injuries to persons resulting in death and/or accidents causing property damage.

When one of the following accident occurred without any error information displayed on the control equipment or the central control room receiving any warning information, please tell us the related information: safety equipment especially the climb system fault or damage, operation program fault, noise abnormal or other accident, etc.

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操作运行手册 Operating Manual

共 4 页

第 1 页

目 录

序言 Foreword	1
1 概述 General	1
1.1 对操作人员的总体要求 General requirements to the operating personnel	1
1.2 对操作及相关人员的特殊要求 Additional requirements to the operating personnel	1
2 安全指导 Accident prevention	3
2.1 通用安全指南 General safety instructions.....	3
2.1.1 总体要求 General.....	3
2.1.2 例行检查 Wind turbine inspection	4
2.1.3 个人防护装备 Personal protective equipment (PPE)	5
2.1.4 防坠落装备 Fall protection	6
2.1.5 个人安全保护 Personal securing	7
2.1.6 防护装备的存放和检查 Storage and inspection of the protective equipment	7
2.1.7 机舱顶上的安全绳系着点和行为方式 Attachment points and behaviour on the rooftop.....	8
2.1.8 其它装备 Inspection equipment	8
2.2 紧急情况下的行动——逃生通道 Behaviour in case of emergency - escape routes	9
2.2.1 生命链 Chain of survival.....	9
2.2.2 紧急情况下的行动 Behaviour in case of emergency	9
2.2.3 风机失火时的行动 Behaviour in case of fire at the wind turbine	11
2.2.4 雷暴中的行动 Behaviour in case of thunderstorm	13
2.2.5 叶片结冰时的应对措施 Behaviour in case of ice formation on the rotor blades	13
2.3 急救 First aid	14
2.3.1 概要 General.....	14
2.3.2 滑落后的受伤处理 Hanging trauma after the fall into the harness	15
2.3.3 安全报告 Safety report.....	16
3 风机的基本技术参数 Description of the wind turbine	16
4 风机的基本操作概念 Description of the operation concept	17
5 进入、攀爬和离开风机 The entering, the ascending and the leaving of the wind turbine	18
6 相关部件及其操作 Functioning principle and operation of the components.....	21
6.1 变流器 Converter.....	21
6.2 滑环单元 Slip ring unit.....	22
6.3 安全传感器系统 Safety sensor system	22
6.4 照明和电源 Lighting and power supply	23
6.5 维护小吊车 Maintenance crane	23
6.6 低温加热系统 Heating system.....	24

6.6.1 系统组成 Composition.....	24
6.6.2 运行方式 Operation method.....	24
6.6.3 运行工况 Working condition.....	25
6.6.4 停机再启动 Restart.....	25
6.6.5 安全说明 Safety instruction.....	26
6.6.5.1 控制柜 Control cabinet.....	26
6.6.5.2 暖风机 Fan heater.....	27
7 控制系统 Control System.....	28
7.1 概述 General.....	28
7.2 控制操作 Control operation.....	30
7.2.1 控制面板 Control panel.....	30
7.2.1.1 塔底控制柜柜门控制面板 Control panel of the cabinet located at tower base.....	30
7.2.1.2 机舱控制柜柜门控制面板 Control panel of the cabinet located in nacelle.....	31
7.2.2 塔底控制柜显示屏操作 Operations of display screen in control cabinet at tower base	33
7.2.2.1 登陆及权限 Login and permission.....	33
7.2.2.2 实时数据监控 Monitoring of realtime data.....	34
7.2.2.3 服务模式 Service mode.....	35
7.2.2.3.1 手动变桨（权限：90）Pitch the blade manually(permission:90).....	36
7.2.2.3.2 手动维护（权限：90）Maintain manually(permission:90).....	37
7.2.2.4 状态码记录 Status code record.....	37
7.2.2.4.1 状态码历史列表 History list of status code.....	38
7.2.2.4.2 状态码激活次数列表 List of status code activation number.....	38
7.2.2.5 输入输出列表（IO） Input/Output list.....	39
7.2.2.6 安全链 Safety chain.....	39
7.2.2.7 EntherCAT 及 Profibus 通讯 EntherCAT and Profibus communication.....	40
7.2.2.8 历史数据统计 Historical data statistics.....	41
7.3 运行模式 Operational states.....	43
7.3.1 初始化模式 Initialization mode.....	44
7.3.2 停机模式 Stop mode.....	45
7.3.3 服务模式 Service mode.....	45
7.3.4 待机模式 Standby mode.....	46
7.3.5 自检测模式 Self-test mode.....	46
7.3.5.1 自检程序 1 Self-test program 1.....	47
7.3.5.2 自检程序 2 Self-test mode program2.....	47
7.3.5.3 自检程序 3 Self-test mode program3.....	48



7.3.6 空转模式 Idling mode	48
7.3.7 启动模式 Start mode	48
7.3.8 切入模式 Cut-in mode.....	49
7.3.9 发电模式 Generation mode	49
7.3.10 共振模式 Resonant mode.....	50
7.3.11 变桨润滑模式 Pitch-lubrication mode	50
7.4 系统控制 System control.....	50
7.4.1 制动系统 Braking system.....	51
7.4.1.1 正常停机 Normal stop.....	51
7.4.1.2 快速停机 Fast stop	53
7.4.1.3 紧急停机 Emergency stop.....	54
7.4.2 变桨系统 Pitch system	55
7.4.3 偏航系统 Pitch system	58
7.4.3.1 自动偏航 Automatic yaw	58
7.4.3.1.1 自动偏航启动 Start of automatic yaw	58
7.4.3.1.2 自动偏航的方向 Direction of automatic yaw.....	59
7.4.3.2 手动偏航系统 Manual yaw.....	61
7.4.3.3 偏航解缆 Cable unwinding	62
7.4.3.3.1 偏航解缆启动 Start of cable unwinding	62
7.4.3.3.2 偏航解缆结束 End of cable unwinding	62
7.4.4 液压系统 Hydraulic system.....	63
7.4.4.1 液压风轮制动 Hydraulic rotor brake	63
7.4.4.2 液压偏航制动 Hydraulic yaw brakes.....	63
7.4.4.3 液压泵马达控制 Control of hydraulic motor	64
7.4.5 安全系统 Safety system	64
7.5 电网检测 Grid monitoring.....	66
7.5.1 电网电压检测 Grid voltage monitoring.....	66
7.5.2 电网电流检测 Grid current monitoring	67
7.5.3 电网频率检测 Grid frequency monitoring.....	67
7.5.4 电网有功功率检测 Grid active power monitoring.....	67
7.5.5 电网无功功率检测 Grid reactive power monitoring.....	68
7.6 温度控制 Temperature control	68
7.6.1 齿轮箱温度控制 Control of gearbox temperature	69
7.6.2 发电机温度控制 Control of generator temperature.....	70
7.6.3 变压器温度控制 Control of the voltage transformer temperature.....	71

7.6.4 控制柜温度控制 Control of the control cabinets temperature	71
7.6.5 机舱温度控制 Control of the nacelle temperature	72
7.6.6 塔底温度控制 Control of the tower base temperature	73
7.7 润滑控制 Control of lubrication	73
7.7.1 主轴润滑控制 Control of rotor bearing lubrication	73
7.7.2 发电机润滑控制 Control of generator lubrication	73
7.7.3 齿轮箱润滑控制 Control of gearbox lubrication	74
7.7.4 偏航润滑控制 Control of yaw bearing lubrication.....	74
7.7.5 变桨润滑控制 Control of pitch bearing lubrication	75
7.8 手动控制 Manual control.....	75
7.8.1 手动启动 Manual start.....	75
7.8.1.1 功能描述 Function description	75
7.8.1.2 操作方法 Operating method	75
7.8.2 手动停机 Manual stop	76
7.8.2.1 功能描述 Function description	76
7.8.2.2 操作方法 Operating method	76
7.8.3 手动复位 Manual reset	76
7.8.3.1 功能描述 Function description	76
7.8.3.2 操作方法 Operating method	77
7.8.4 服务模式下的手动控制 Manual control during service mode.....	78
7.8.4.1 功能描述 Function description	78
7.8.4.2 操作方法 Operating method	80
7.9 变频器冷却系统 Convertor cooling system.....	81
7.9.1 功能描述 Function description	81
7.9.2 安装、维护操作方法 Operations of installing and maintenance	81
附录 A 状态代码列表 Appendix I State code list.....	83
附录 B 数据记录 Appendix B Data records.....	111
附录 C I/O 列表 AppendixC I/O List.....	117



1 概述 General

本操作运行手册解释了风力发电机组（以下简称风机）的操作人员如何在风机内部或者外部操作所有的设备以及需参照的文件。

This operating manual explains how you must operate all devices outside and inside the wind turbine and refers to relevant documentation.

1.1 对操作人员的总体要求 General requirements to the operating personnel

风机的所有工作必须由具有资格的人员来执行。（参照标准IEC 364 或CENELEC HD 384或DIN VDE 0100，以及IEC 664 或者DIN VDE 0110，并按照相应的国内事故预防管理条例）。在这些基本的安全条例文献中所指的具有资格的操作人员要求如下：

- 熟悉产品的装配、调试和维护过程；
- 具有相应的操作资格证；
- 能够根据技术培训、知识、经验和自身对于相关标准(BGI 657)的学习来识别可能的危险。

All activities shall be carried out by qualified personnel (observe the standards IEC 364 or CENELEC HD 384 or DIN VDE 0100 and IEC 664 or DIN VDE 0110 as well as national accident prevention regulations). Qualified technical personnel in the sense of these fundamental safety documents means persons:

- who are familiar with the installation, the assembly, the start-up and the handling of the product.
- who have corresponding qualifications for those activities.
- who recognise possible dangers due to their technical training, knowledge and experiences as well as due to their knowledge of relevant norms (BGI 657).

1.2 对操作及相关人员的特殊要求 Additional requirements to the operating personnel

对风机操作人员及相关工作人员来说除了掌握总体性的技术要求（即参加风机机电培训）之外，还需要掌握一些特殊的技能并拥有一定的安全意识：

• 除了考虑维护工作（检查、维护、维修）的技术层面，维护承包商必须进行风险评估，特别是操作人员在一起工作的时候的风险评估。

- 所有操作人员应按照风机上的警告和安全指令（参照相关的随机文件）进行操作。
- 在风场工作，至少应有两个人熟悉风机的名称、特性和确切位置。
- 操作人员应该熟悉救生和安全装备（在高空作业安全装备），来保证自身和他人的安全。

根据地方法规，必须练习救生和安全装备的使用。

- 工作人员进入风机前，必须确保他们在身体健康状况上没有勉强的因素。
- 电气设备、在其附近工作及相关的操作人员中，必须有足够数量的人员经过培训并进行指导，以便在出现电击、烧伤事故时进行急救。对这些操作人员应进行急救培训和电气方面的指导

应至少每两年重复进行一次。

- 每个维护小队中至少有一个人完成火灾应对强化培训，而另一个人完成医疗急救强化培训。
- 应阅读和理解完整的操作运行手册。
- 应知晓关闭风机的操作步骤，以及防止接通或意外启动风机时相关部件的操作和保护措施（作为运行手册的部分内容）。
- 应知道风机及其部件的操作程序和规定，以保证在风机受损的情况下，操作人员能做出正确的决定，以避免危险情况扩大。
- 根据所做工作的种类和范畴，工作人员应受到相应的培训和具备必要的技能。

Apart from the general technical requirements for the operating personnel of the wind turbine (the electromechanical training); the works in the area of a power station of the described dimension require additional special knowledge and care:

- In addition to taking care of the technical side of the servicing (inspection, maintenance, repair), the company entrusted with the maintenance has to draw up a risk assessment, that shall be taken into consideration and used when the operating team is put together.
- All operating personnel shall know and observe the warnings and safety instructions regarding the wind turbine (attached to the wind turbine).
- At least two of the persons present at the wind turbine shall know the name, the individual characteristics and the exact position of the wind turbine.
- The operating personnel shall be familiar with handling life-saving and securing appliances (rescue from heights) for securing others and themselves. Handling shall be practiced according to the local law.
- There should be no insist doubts according to the health of every coworker at the wind turbine.
- Of the persons working at, with or near electrical equipment, there shall be a sufficient number trained and instructed so that they can render first aid in case of an electric shock and / or burns. These persons shall be trained in first aid and at least be instructed in electrical engineering. The first aid training shall be repeated every two years.
- Furthermore, one person of each operating team shall complete an extended training as a fire watch, and another person an extended training for the medical first aid.
- The complete operating manual shall actually be read and understood.
- The operation steps for switching off the wind turbine and its components for workings and securing against switch-on or inadvertent start-up shall be known (part of the operating manual).
- Operating procedure and regulations of the wind turbine and its components shall be known so that in the event of damage the correct decisions for avoiding other dangers can be made.
- The personnel shall be trained depending on the type and extent of entrusted activities for

carrying out these works. They shall have the required competence.

2 安全指导 Accident prevention

下列安全和事故预防说明包含了在风机中或与风机相关工作的有关要求。

进行与风机相关的工作时，运行操作人员必须遵守相应的提示信息、规章、操作指导以及相关国家法规，特别是：

- 关于手动操作时的健康和安全管理法规。
- 关于使用个人防护装备（PPE-BV）的健康和安全管理法规。
- 事故预防规章和安全管理规范。

The following safety and accident prevention instructions contain requirements for handling regarding works and presence in and at the wind turbine.

When working at the wind turbine, the operating personnel are obliged to observe information, regulations and instructions, as well as national laws, especially:

- Local ordinance on health and safety when handling manual loads
- Local ordinance on health and safety when using personal protective equipment (PPE – BV)
- Accident prevention regulations and safety rules

2.1 通用安全指南 General safety instructions

除了上述的法规之外，所有对风机的操作、检查（即便是短暂的）以及在风机运行影响范围内的相关人员，都必须遵守下列的安全指南。

In addition to the mentioned regulations, the following safety instructions apply to the works and checks at the wind turbine as well as for any, even short-term, presence in the range of action of the wind turbine.

2.1.1 总体要求 General

- 所有 2MW 机组在 10 分钟平均风速大于 17m/s 时，禁止登塔。
- 风轮直径为 87m、93m 和 99m 的 2MW 机组，在 10 分钟平均风速大于 14m/s 时，禁止进入轮毂。风轮直径为 105m 和 111m 的 2MW 机组，在 10 分钟平均风速大于 12m/s 时，禁止进入轮毂。

- 所有 2MW 机组在 10 分钟平均风速大于 12m/s 时，禁止出舱。
- 在风机内的所有区域，禁酒。在这里所指的禁酒是指血液中的酒精含量上限为 0.0‰，即禁止血液中残留任何酒精、麻醉品和违禁药物的残留成分。
- 在风机内的所有区域，禁烟。



- 严禁使用明火（如，打火机或者火柴等）

• If the 10min mean wind speed exceeds 17 m/s, you shall leave the wind turbine. For 87m/93m/99m rotor diameter 2MW wind turbines, if the 10min mean wind speed exceeds 14 m/s, going into the hub is forbidden. For 105m/111m rotor diameter 2MW wind turbines, if the 10min mean wind speed exceeds 12 m/s, going into the hub is forbidden. If the 10min mean wind speed exceeds 12 m/s, going out the nacelle is forbidden.



• No alcohol is allowed in the entire area of the wind turbine. Here the legal 0.0 ‰ blood alcohol limit applies. Residual blood alcohol, drugs consumption and consumption of certain medicine are forbidden as well.

- No smoking is allowed in the entire area of the wind turbine.
- Never use naked flame (a match or a lighter, for example)!



2.1.2 例行检查 Wind turbine inspection

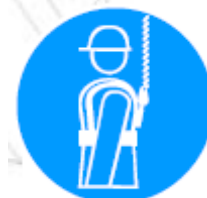
未经许可任何的人禁止进入风机。风机的铁门必须加锁，且该铁门任何情况下都可以从内部打开而不需要钥匙。应该在风机外备有一把钥匙以供消防人员进入风机。



许可进入风机的人员包括：

• 委托从事风机相关工作，穿戴个人防护装备（PPE）及防坠落装备，并达到要求的人员。

• 获得操作人员许可，穿戴个人防护装备（PPE）及防坠落装备，受过安全培训，并由操作人员陪同的人员。



作为基本规则，同时进入 1 台风机的人数必须不少于两个。

沟通的方式必须提前澄清：

- 对讲机的频率相同（电池充好电）。
- 如果通讯中断，不会混淆手势表达意思。
- 相互交换好手机号码。

检查风机时，切记要排除在风机上工作时人员跌落和物品坠落的危险！



The presence of unauthorised persons in the wind turbine is not allowed. A locked steel door impedes access. This door can always be opened from inside, even without a key. But a key should be deposited outside the door so that the fire fighters can enter the wind turbine.

Authorised are:

- Personnel, who are entrusted with works at the wind turbine, who wear the personal protective equipment (PPE) and fulfill the requirements.
- Third persons, who have the permission of the operator, wear the required personal protective

equipment, have been instructed with respect to safety and are accompanied by operating personnel.

As a basic rule, the wind turbine may only be accessed by at least two authorised persons at a time. The second person may stay outside the wind turbine, if the two contact each other at adequate intervals.

The ways of communication must be clarified in advance.

- Same frequency of the walkie-talkie (are accumulators charged?)
- Unmistakable hand signals, if communication is interrupted.
- Mobile phone numbers have been exchanged.

When inspecting the wind turbine and when working on and inside the wind turbine, always take into account the risk of falling and the risk of falling objects.

2. 1. 3 个人防护装备 Personal protective equipment (PPE)

进入风机塔架的每一个人都应该配备下列的个人防护装备（PPE）：

• 紧身合体的工作服。布料的可燃性低，合适的棉含量为 $320 \sim 380 \text{g/m}^2$ ，或者其它的特制布料，例如在升降时的特制衣服。



• 结实的工作鞋，安全等级 S3，并有脚趾保护层和防穿孔鞋底。



• 防滑手套。



• 带有防脱绳的硬质安全帽，合适的安全帽是由热固性塑料制成的（例如，Petzl）。



• 带有耳罩和头顶灯的在现场使用的攀爬头盔。



- 与安全帽相配的，带有 P3 锁紧系统的减噪等级为 SNR 35dB 的耳罩。



- 进行可能对眼睛造成损伤的工作时要配戴护目镜。



- 在光源失效情况下工作、维修照明灯、在光线差的区域内工作时佩戴头灯。



Every person which enters the tower of the wind turbine, shall carry along the following personal protective equipment (PPE):

- Tight-fitting clothing and appropriate working clothes. Of low flammability, preferably cotton with a cloth weight of about 320 to 380 g/m² or special clothing, for example, during the roping down.
- Sturdy working shoes, safety class S3 with a protective cap and a puncture resistant sole.
- Anti-slip glove.
- A hard hat with a chin strap, preferably a climbing helmet made of thermosetting plastic (for example, Petzl)
- A climbing helmet with ear protection and a lamp for the use on the building site
- Ear protection compatible with the hard hat with the fastening system P3 with a noise reduction index of SNR 35 dB.
- Safety goggles for workings that can cause damages of the eyes.
- A forehead or a hard hat lamp for:
 - working during the failure of light
 - repairing lamps
 - working in badly lighted areas

2.1.4 防坠落装备 Fall protection

- 达到 DIN EN 361 要求的带胸部吊环的攀爬保护安全带





- 安全绳，包括长 1m 的绳索（具有缓冲功能）及至少 50mm 开口的弹簧钩。



- 在爬梯上时佩戴的防跌落滑块。



- 工作固定绳，使用工作固定绳可以让工作人员腾出双手来工作。



- Safety harness in accordance with DIN EN 361 with catching eyelets for climbing protection on the chest area.

- Fall arrest lanyards, 1 m of rope with an energy absorber and a spring hook with an opening of a least 50 mm.

- Fall arrest slider with a fall brake for the arrestor bar at the ladder.

- Work positioning lanyard. This lanyard holds a worker in place while leaving the hands free for working.

2.1.5 个人安全保护 Personal securing

- 穿戴好个人防护和防坠落装备（以下简称防护装备）。
- 调节带缓冲功能的安全绳钩紧度，保证跌落高度不能超过 0.4m。
- 套上滑块，调整滑块与人体距离，保证滑块功能。
- Fasten safety harness tightly around your body (see PPE).
- Adjust the holding rope and the double energy absorber to the fall arrester so that the drop height cannot exceed 0.4m.

- Fasten the fall arrester (the fall prevention device) and check the functioning.

2.1.6 防护装备的存放和检查 Storage and inspection of the protective equipment

防护装备应该吊挂在干燥和凉爽的屋子里，不得与腐蚀性液体接触和与尖锐锋利的物品放置

在一起，以免损坏。

如果防护装备上出现可见的损坏或者检验日期过期了，就不能再用，应立即更换。



注意！

在开始工作前谨记：

检查材料和装备可能的损坏；检查装备生产日期或检验日期。

The protective equipment shall be stored hanging in a dry and cool room. It shall not come into contact with corrosive fluids and sharp objects that might cause material damage.

In case damage is visible or the date of inspection is overdue, the corresponding equipment shall not be used anymore. It shall be replaced immediately.

Attention!

Keep in mind before commencing work:

- Check the material/equipment for possible damages.
- Check the date of production or inspection.

2.1.7 机舱顶上的安全绳系着点和行为方式 Attachment points and behaviour on the rooft

在工作过程中，若工作人员不需要变动位置，则两个带缓冲功能的安全绳钩都要钩住系着点。

当工作人员需要从一端走到另一端时，一个带缓冲功能的安全绳钩必须总是钩住系着点。

工作人员最多可以下坠 4.0m



When the person does not change its position during working, both ends of the double energy absorber shall be hooked in.

When the person walks from one side to the other, one end of the double energy absorber must always be fastened.

You can fall 4.0 m to the maximum.

2.1.8 其它装备 Inspection equipment

在风机上工作时，除了个人防护装备之外，须配备以下装备：

绳降装备

- （绳子的长度=2×轮毂中心高×(1+10%)，真空包装），永久性的或至少在检查或者维护风机的时候位于紧急出口。

对讲机或其他通讯设备

以便在风机内与第二个人、运输车辆或服务中心取得联系。此类



无线通讯工具或者其他通讯工具必须遵守关于频带、发射功率和调制的国家法规。如果必要，需获取相关的许可。

In case of wind turbine inspection, the following equipment must be carried along in addition to the PPE:

- Rope-down device (length of the rope = hub height multiplied by two plus 10%, vacuum-packed), permanently but at least located at the emergency exit during inspection or maintenance.
- Radio or other communication device in order to be able to speak from inside the wind turbine with the second person, with the transportation vehicle or with the service station.

Follow the national legislation regarding the frequency band, the transmitting power and the modulation. If necessary, obtain the relevant permits.

2.2 紧急情况下的行动——逃生通道 Behaviour in case of emergency - escape routes

本节包含了在形成烟雾、火灾以及发生严重的机械或电气故障等危及生命时的个人安全行为。

为了应对紧急情况，营救人员可能使用的通道必须保持畅通。严禁阻塞任何紧急出口或者其他出口，即使是短暂的。

This section contains information about personal safety in case of smoke formation, fire, the occurrence of serious mechanical or electrical defects, or life-threatening situations.

In order to cope with emergencies, the possible access ways for rescue personnel must always be kept free from obstruction. Even short-term obstructing of emergency exits or other exits is prohibited.

2.2.1 生命链 Chain of survival

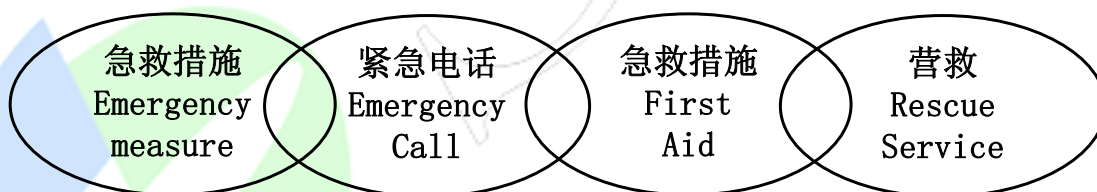


图 1 生命链
Fig1 Chain of survival

2.2.2 紧急情况下的行动 Behaviour in case of emergency

第一逃生通道——爬梯

- 保持冷静，将情况通知受影响的同事（现场人员）。
 - 顺着塔筒内的爬梯尽快下到塔底，并离开塔筒。
- （塔底的入口门应该总是可以从内部打开，即使没有钥匙）



第一逃生通道
The primary escape route

第二逃生通道——机舱起吊门（紧急出口）

- 将绳降装备固定在机架的吊耳上。
- 将绳降装备的弹簧钩与安全带扣牢。
- 打开位于机舱内发电机旁的起吊门及机舱门盖。
- 将被营救人员或自己从紧急出口机舱盖绳降到地面。

若可能的话，被绳降人员应该用一根引导绳将其固定牢靠，然后再下降。

- 如果正常照明损坏，而且应急灯也不能正常启动时，不可使用明火（例如，火柴或者打火机）。否则将增加烧伤、失火、爆燃等意外灾难的可能性。

- 首先保证自身的安全，然后再帮助受伤的或者失去自救能力的同事（现场人员）。

- 呼叫和通知救援车辆或者公司救援协调中心，并提供以下信息：

- 紧急情况的位置（在哪一台风机上）
- 发生什么事件
- 有多少人受影响
- 发生什么类型的紧急情况
- 等待回复（勿挂机）



第二逃生通道
 The secondary escape route

The primary escape route is the ladder inside the tower.

- Stay calm - inform affected colleagues (present persons) about the situation.
- Down to bottom of the tower through the ladder as soon as possible, and leave the tower drum.
 (The entrance door at the tower base can always be opened from inside, even without a key).

The secondary escape route is the rope-down device in the nacelle.

- Fasten the rope-down device to the transportation brackets of the main frame.
- Fasten the safety harness to the spring clip of the rope-down device
- Open lifting hatch and the door of nacelle cover beside generator in the nacelle
- Down to bottom of the tower through the ladder as soon as possible, and leave the tower drum.
 The person who is being roped down should be secured with a guide rope from below, if possible.
- If the normal lighting breaks down and the emergency lighting does not switch on do not use naked flame (for example, a match or a lighter). The risk to add additional hazards from burns, fire or deflagration is too high.
- First secure yourself then help injured or disoriented colleagues (present persons).
- Call and inform the supply vehicle or the company rescue coordination centre providing the

following information:

- Where is the emergency (on which wind turbine)?
- What happened?
- How many persons are affected?
- What kind of an emergency is it?
- Wait for questions (stay on call).

2.2.3 风机失火时的行动 Behaviour in case of fire at the wind turbine

最重要的原则:

在机舱内，受过训练的安全人员必须首先照顾同事。

不可让烟雾扩散!

- 关闭失火舱室的门。
- 离开风机之后，关闭所有通道，特别是通往塔筒的通道。
- 警告同事。
- 戴上呼吸保护面罩。



如果可能，使用便携式灭火器（机舱、塔底各有一个），快速彻底的灭火。

如果能够将火扑灭，确保灭火剂的气体 and 烟灰能够尽快的排出和通风。

如果需要离开风机：通常情况下总是向下逃生！

始终保持冷静，因为恐惧中的人不会照顾他人！

不可太着急打开门或者舱盖，应该小心打开，如果发现门后有浓密的烟雾层应迅速将其关闭！

只有在有足够的时间来保证逃生前提下，才可在风机内呼叫协调中心

如果公司的协调中心没有回音，则：

- 打火警电话：119
- 是什么失火？
- 失火的位置？
- 谁在打火警电话？

如果不能在风机内发出警报，在离开风机之后立刻报警。

离开风机时，听从安全员的指令：

- 与在现场的其他人一起迅速离开，但不可慌张。
- 如果可以通过塔筒离开风机，请使用爬梯，且从容不迫并正确固定好安全装备。
- 如果有电梯，不可使用。万一断电，电梯的马达会停止运转，灯光也会熄灭。那么到时候只能在黑暗中更换到爬梯上。
- 如果从机舱的紧急出口离开风机，安全员应首先安排绳降现场的所有其他人，然后再绳降自己。

- 贴近地面。如果必要，应当匍匐的离开，因为有毒的高温气体首先会在舱室顶部聚集，然后才扩散到舱室的剩余空间。
- 必须疏散和封锁距离风机半径为 200m 的区域。
- 查看是否有人失踪。
- 为赶到现场的救援人员指明方向。

如何保护自己？

- 在舱门外就近处留一把紧急钥匙。
- 上风机前熟悉安全规则、逃生方案，时刻注意风机内的安全警示标记。

The most important rules are:

The trained safety guard immediately takes care of the coordination at the nacelle.

Smoke must not spread:

- Close the door of the room where the fire broke out.
- After leaving the wind turbine, close all passageways, especially the passage to the tower.
- Alert your colleagues.
- Put on your respiratory protection mask.

If it is possible to use portable fire extinguishers (both of the nacelle and the tower bottom have one), fight the fire as quickly and as extensively as possible.

If the fire could be extinguished, make sure that the gas of the extinguishing agents and floating soot are ventilated out as quickly as possible.

When you need to leave the wind turbine: Generally flee downwards!

Always hold on tight, because people in panic do not look out for others!

Never fling open doors and lids too quickly, but open them cautiously and close them immediately, if there is a dense smoke layer behind them!

Only if it is still possible without time delay, call the coordination centre from the wind turbine.

If the coordination centre of the company does not respond, make:

- Emergency call reporting fire: 119
- What is burning?
- Where is it burning?
- Who is calling?

If alerting is not possible from the wind turbine, do it as soon as you have left the wind turbine.

Leave the wind turbine according to the instructions of the safety guard:

- Quickly, but not hastily, together with the other present persons.
- If it is possible to leave the wind turbine through the tower, use the ladder. Take your time and fasten your fall prevention device properly.
- If there is a lift, do not use it. In case of power failure, the lift motor will stop, and the light will



probably go out. Then you would have to change over to the ladder in the darkness.

- If the wind turbine is left via the emergency exit at the nacelle, the safety guard will rope down all present persons and colleagues first, before roping down himself.
- Keep near the floor. If necessary you should leave the area by crawling on the floor, because the toxic and hot fire gases first accumulate under the ceiling of the room and subsequently spread to the rest of the room.
- An area within a radius of at least 200m from the wind turbine has to be evacuated and closed off.
- Check if someone is missing.
- Give directions to the arriving rescue workers.

How can you protect yourself?

- Leave an emergency key outside near the door.
- Practice for an emergency at least once a year (see local law).

2.2.4 雷暴中的行动 Behaviour in case of thunderstorm

万一遇到雷暴，所有人员都必须离开风机。

立刻寻找掩体

- 可以利用法拉第笼（例如，金属硬顶汽车）作为掩体。
- 在塔底应该能够安全，但是不可触摸任何金属物体（例如，塔筒壁）。

In case of thunderstorm everybody must leave the wind turbine.

Look immediately for a shelter.

- You can use a Faraday cage (for example, a car with a metal roof) as a shelter!
- You should also be safe in the tower bottom, but do not touch any metal things (for example, the tower wall).

2.2.5 叶片结冰时的应对措施 Behaviour in case of ice formation on the rotor blades

在寒冷的冬季，大雾、雨、雪都可能使风力机叶片、塔架、机舱顶在很短的时间内结冰。叶片结冰会对叶片的气动性能有严重影响，叶片覆冰运行的功率曲线会大大偏离正常功率曲线，叶片结冰还会对机组和人员的安全造成威胁，当机组结冰严重时，应停机对机组进行保护。在冬季结冰期，风电场运行须采取适当的防措施，避免风力机甩冰、落冰造成人员伤亡或财物损失。

在可能发生叶片结冰风电场，建议风电场在安全距离设立防护栏杆，禁止进入；在进入风场的入口或人员可能经过的道路口设置警示牌，提醒该区域冬季可能会有冰块甩落的风险，应注意安全。

一般情况，在风电场结冰期，应与机组保持安全距离。结冰期安全距离即为不会被风力机甩冰，滑冰所威胁的安全距离。机组运行时，安全距离为叶片甩冰最大距离；停机时，安全距离为风轮半

径+20m。叶片甩冰最大距离可以按照下面公式估算：

$$1.5 \times (\text{风轮直径} + \text{轮毂中心高})$$

如果必须进入结冰的风力机，应该远程停机后，沿着垂直于风轮面的方向小心接近风力机，在快到达塔架门时，应注意避开上方叶片、塔架或机舱上可能滑落的冰块。

登风力机的护具和防护帽应在安全距离外就进行穿戴，避免在覆冰机组下方停留。

出于安全考虑，覆冰停机后再启动，应风机组的安全情况进行排查后，才可以远程手动复位。机组覆冰再启动前的安全排查除了风电场一般的安全检查，还应检查确认机组安全距离内没有人、畜经过，机组叶片覆冰情况已经有改善。

In the wintertime freezing fog, rain or snow can create considerable ice layers on the rotor blades or on the tower of the wind turbine within a very short period of time due to temperatures above 0 °C Ice formation on the rotor blades can cause death and considerable worsens the aerodynamic profile characteristics. The risk of ice formation is greater when rotors are rotating in comparison with when they are standing still. Safety measures should be introduced to avoid the potential risks that may triggered by the falling ices.

In the freezing period, Placing fences and warning signs as appropriate for the protection of site personnel and the public.

In general, a safe distance should be keep to avoid the falling ice. For calculating the safe distance when turbine is running ,the following formula for calculating the maximum ice throw distance should be used, rotor radius plus 20m could be take as safe distance when the turbine is shut down.

Maximum ice throw distance: $1.5 \times (\text{hub height} + \text{rotor diameter})$

Restricting access to turbines by site personnel while ice remains on the turbine structure. If siee personnel absolutely must access the turbine while iced, satety precautions may include remotely shutting down the turbine, cautiously approach the wind turbine from the leeward side protected by the tower shadow. watch out the possible falling ice.

Standard protective gear and helmet should be worn at the safety distance. Do not stay under the coverage of rotor .

Restarting the turbine remotely when routine safety check is complete. Satety check should at least including the safety distance check and blade ice check beside the routine checks.

2.3 急救 First aid

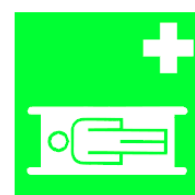
2.3.1 概要 General

在运输车辆、补给车辆、或者变电站处存放一副担架，万一有人受伤时，能直接送往医院。

在机舱内和塔底入口平台处必须存放急救箱，以备有人需要急救时用和对小伤口进行处理。



急救箱
first aid kit



担架
stretcher

在使用完急救药品后，应迅速补充并检查有效期。

受伤人员可以从塔筒内的爬梯或者机舱内的起吊孔（第二逃生通道）处绳降到地面。

急救时必须遵守以下事项：

- 保护事故现场
- 照顾伤者
- 通知协调中心
- 听从指令

Keep a stretcher available on the transportation vehicle, on another supply vehicle or at the transformer station, in case an injured person needs to be transported with a helicopter.

In case a person needs first aid and for the treatment of smaller wounds, the nacelle and the area of the entry platform must be supplied with a first aid kit.

Replace used materials immediately after consumption. Check the expiry date.

Injured persons can be roped down completely through the transportation well inside the tower or via the second escape route outside the tower.

The following always applies to giving first aid:

- Secure the site of the accident.
- Attend to the injured person.
- Inform the coordination centre.
- Follow instructions.

2.3.2 滑落后的受伤处理 Hanging trauma after the fall into the harness

工作人员在摔倒并悬挂在安全带上时，会导致呼吸衰竭，并在短时间（2~30 分钟）内进入昏迷。要迅速将伤员的受伤相关信息通知救援协调中心和将要抵达的救援人员。

确保执行下面的救援方法：

- 协助伤者停靠在最近的平台上。
- 让一个救援人员保持垂直前倾姿势，让伤者斜靠在救援人员身上。
- 如果伤者仍有意识，应反复拍打受伤部位的肌肉（收紧和放松肌肉，尤其是脚上的肌肉）。



A fall into the harness with subsequent hanging in the harness results in a circulatory collapse (orthostatic hock) and in unconsciousness after only a short time (2 to 30 minutes). Give information about hanging trauma to the rescue coordination center and the ambulance officer that will arrive!

The rescue is always carried out downwards!

- Put the hanging trauma on the nearest platform
- After the rescue, put a conscious person in an upright squat position, before removing the harness
- If the accident victim is responsive, he/she should be urged to pump his muscles (contract and loosen up the muscles, especially those of the feet)

2.3.3 安全报告 Safety report

在进入和离开风机时，操作人员要将相关信息记录在塔底的日志记录本上，并向协调中心报告进入和离开风机情况。

When entering or leaving the wind turbine, the operating personnel shall report their coming and leaving to the coordination centre.

3 风机的基本技术参数 Description of the wind turbine

表 1 风机的基本参数
Table1 Description of the wind turbine

型号 Type	SEWIND W2000
适用风区类别 Classification	T2A+/TC3A
额定功率 Rated Power	2,000kW
轮毂中心高 Hub height	70m/80m/90m
叶片 Rotor blade	SE87/SE93/SE99/SE105/SE111
风轮直径 Rotor diameter	87m/93m/99m/105m/111m
叶片数目 Number of rotor blades	3
风轮相对于塔架位置 Rotor position related to the tower	上风向 Upwind
风轮转速范围 Rotor speed range	8.33~16.8 rpm/7.87~15.95rpm(111m)
发电机转速范围 Generator speed range	983~1983 rpm
发电机原理 Generator principle	双馈异步发电机，4 极 Double-fed Induction Generator, 4 poles
额定转速 Nominal rotation speed	15.35 rpm/15rpm(105m)/14.5rpm(111m)
叶片锥角 Rotor blade cone angle	3.5° /5° (111m)
主轴倾角 Rotor axis inclination	5.0°
切入风速 Cut-in wind speed	3.0 m/s
切出风速 Cut-out wind speed	25.0 m/s / 20.0m/s(105m) / 22.0m/s(111m)
工作温度范围 Operating temperature range	常温环境 Normal climate -20° C ~ +4 0°C 低温环境 cold climate -30° C ~ +40°C
偏航方式 Type of wind adjusting	主动对风偏航系统 active yaw system by wind direction

更多的风机技术参数可以参阅相关技术文件。

More technical data of the wind turbine can be found in the data sheet.



4 风机的基本操作概念 Description of the operation concept

W2000 型风力发电机组是三叶片、上风向、水平轴、变速变桨距风力发电机组。

变桨系统是风机的主制动系统-即气动刹车，每个叶片配备了独立的电动变桨驱动器和备用电源来保证冗余和容差能力，在失电或故障情况下，变桨驱动器将叶片转动到顺桨位置，使风轮安全停车。系统的第二制动系统为机械制动，由安装在齿轮箱的高速端的刹车卡钳实现，在变桨系统将风轮转速降至设定转速后，使风轮达到完全停止的待机状态。在急停状态下，此机械制动器的功能是快速降低风轮转速。

特定情况下（例如，对轮毂或在轮毂内进行维护工作），必须将驱动链固定以防止风轮转动。可以手动操作位于风轮和主轴之间的风轮锁定盘。将 2 个风轮锁定销插入风轮锁定盘并锁紧。可在 12 个不同的位置锁定，即可以在间隔 30° 的任意位置锁定风轮锁定盘。



注意！

只有维护人员才可以使用风轮锁定！

变桨系统和机械制动系统由风机控制系统和安全系统共同控制。安全系统总是比控制系统有更高的优先级。安全系统是依据失效安全模式设计的，也就是在失电及系统故障时能能保证风机安全可靠的停下来。

在机舱顶上装有测风设备（风向传感器和风速传感器），当机舱轴线偏离风向超过设定值时，控制器将启动偏航电机，实现风机主动对风。风机正常工作时，机舱由偏航制动器固定。偏航过程中，制动器制动压力降低，使偏航电机能正常转动。

解缆时，偏航制动器制动压力也降低，也保证偏航电机正常转动。

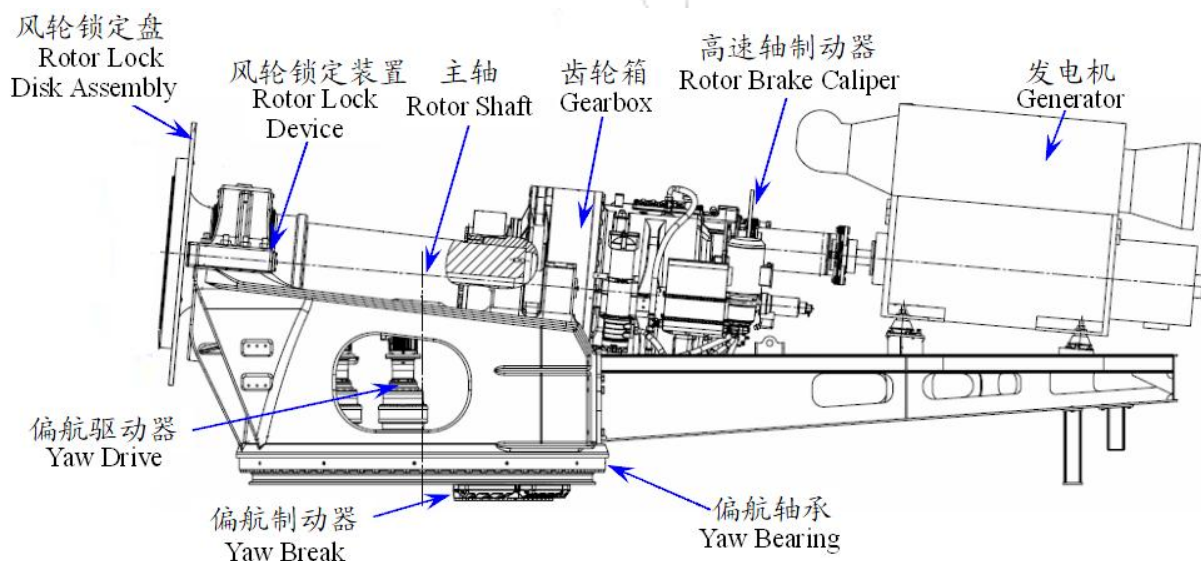


图 2 机舱结构图

Fig2 Structure chart of the nacelle

W2000 is a three-bladed, up-wind, variable-pitch variable-speed, horizontal axis wind turbine.

The pitch system is the main braking system of the wind turbine. It is designed to have independent electrical drives for each rotor blade and a battery back-up to ensure redundancy and fault tolerance. The second braking system of the wind turbine is an additional mechanical brake integrated in the high speed side of the drive train. The primary function of the mechanical braking system is to bring the rotor to a complete stand-still after it has been decelerated to idling speed by the pitch system. The mechanical brake is also used during emergency stops to decelerate the wind turbine as quickly as possible.

In certain cases (for example, the maintenance works at or in the hub), it is also possible to secure the drive train mechanically against rotation. Between the rotor hub and the main shaft, there is a lock disc for securing. This rotor lock is actuated and activated manually. Two rotor lock spindles have to be plugged into this disc and fastened. The drive train can be locked in 12 different positions, so that the rotor can be locked in steps of 30 °.

Attention!

Only the maintenance personnel are allowed to use the rotor lock!

The pitch system as well as the mechanical brake are controlled by the control system and the safety system of the wind turbine. By design the safety system will always have a higher priority than the control system. The safety system is fail-safe, meaning that it can assure the wind turbine stop reliably even if the electricity is cut off or system fault occurred.

Measuring instruments (wind direction sensors and wind speed sensors) are situated on top of the nacelle. If the wind direction changes, the sensors give the information to the control system. Then the control system yaws the nacelle into the ideal position. Nacelle is fixed by the yaw brake when the wind turbine works normally.

The brake pressure will be decreased during yawing and cable unwinding to secure the motor rotate normally.

5 进入、攀爬和离开风机 The entering, the ascending and the leaving of the wind turbine

在进入、攀爬和离开风机时，请按照以下步骤进行：

进入：

- 打开塔架门，进入塔架底部平台，打开塔底照明。
- 在塔底将塔架门用插销锁住，因为未经许可的人员可能会进入风机。
- 关闭远程控制。

攀爬：

- 在攀爬之前，应该停机。
- 进入服务模式，打开塔架照明（参阅第 7.2.1.1 及 7.2.1.2 中的面板上的状态灯及开

关定义)。

注意！



塔底控制柜和机舱控制柜各有一个维护开关，任何一个均可进入服务模式。

系统维护结束后，应保证两个维护开关均处于维护切除位置。

- 遵守所有的安全规则（参阅第 2.1 条“通用安全指南”），穿戴好个人防护装备。一定要将滑块套入滑轨，防止坠落。
- 在攀爬风机过程中，进行目测检查（查看维护手册中的目测循环检测列表）。
- 每段segment塔筒通过塔筒法兰flange由螺栓bolt联接connection固定fix，由于法兰的存在，会导致攀爬通道缩小，在攀爬时要注意避开。
- 每段塔筒设有一个休息平台，在步入平台之前，必须先将安全绳弹簧钩钩住连接环或爬梯，然后才能脱离滑块。
- 进入平台后，必须在关上平台门盖后，才能松开安全绳弹簧钩。这样就可以在平台上自由活动休息了。

进入机舱

- 登上塔顶平台后，在关上平台门盖后，可将滑块及携带物品卸下，轻装进入机舱。
- 通过通往机舱的爬梯进入机舱，注意此时要开启机舱照明开关。
- 进入机舱后，首先要关闭机舱门。

工作

注意！



只允许具有相应资格的人员打开机舱控制柜，操作有关电气部件。

只允许具有相应资格的人员操作风轮锁定装置。

离开

- 必须确保风轮锁定装置已解除。
- 必须确保机舱控制柜上的紧停开关、维护模式开关处于切除位置。
- 必须确保行车已固定好。
- 确认控制柜照明电源、吊车电源对应的空气开关已断开。
- 清洁现场，带走所有的维护工具及其它物品。如果太重，可通过舱内起吊装置通过吊物孔将其放下至塔底。
- 关闭并锁好所有的门盖（控制柜门、吊物门、机舱罩门等）。
- 离开机舱，关闭机舱入口门，关闭机舱照明灯。
- 进入塔顶平台，遵守所有的安全规则（参阅第 2.1 条“通用安全指南”），穿戴好个人防护装备。
- 登上爬梯，扣上安全绳扣，打开平台门盖。

- 将滑块套入滑轨，打开安全绳扣，顺着爬梯爬下。



注意！

一定要关闭各平台门盖。

When entering, ascending and leaving the wind turbine, please consider the following:

Entering:

- Open the tower door, go to the towerbottom platform, and turn on the lighting in the towerbottom
- Do not leave the tower door ajar because unauthorized persons could enter the wind turbine.
- Deactivate the remote control.

Ascending:

- Before ascending the wind turbine it shall be be stopped by the control.
- Switch it off into the service mode (see the defination of state lighting and swith in chapter 7.2.1.1 and 7.2.1.2).

Attention!

Both contol cabinet located at the tower base and in the nacelle have a service swith, each of them can set wind turbine in the service mode. After system service, make sure both of the service keys are not activated.

- All safety instructions should be observed (see chapter 2.1 “General safety instructions”). Wear the personal protective equipment, and fasten the fall arrester to avoid falling.
- During the ascending of the wind turbine make a visual inspection (see the list of S-frequency maintenance in the maintenance manual).
- Every segment of the tower is connected each other with flanges fixed by the bolts, and the ascending route is reduced by it, so you shall avoid it when you ascend.
- There is a platform for rest between every segment of the tower. You shall hook the spring clip with the connection ring or the ladder before you divorce the spring clip from the fall arrester to enter into the platform.
- After you entering the platform, you shall close the hatch of platform before you loosen the safety spring clip. And then you can rest on the platform.

Enter nacelle:

- After entering the platform at the top of tower and closing the hatch of platform, you can unlade the fall arrester and other things to entering the nacelle.
- Entering nacelle through the ladder going to it, pay attention to open the lamp in nacelle.
- Close the door of nacelle after entering the nacelle.

Working:



- Attention: only qualified personnel are allowed to carry out works at the electrical components and to open the cabinets.

- Attention: only qualified personnel are allowed to arrest the rotor lock.

Leaving:

- Deactivate the rotor lock if it is activated!
- Make sure the emergency stop and service switch key in the control cabinet in nacelle is not activated.
- Make sure the travelling crane is fixed.
- Make sure the lighting power of control cabinet and the electrical resource of the crane is cut off.
- Take away tools and other things, and the rope-down device can be used to put down them if they are too heavy.
- Lock all hatches and openings.
- Leave the nacelle and close the door entering into the nacelle and turn off the lamp in nacelle.
- Enter top platform of the tower. All safety instructions should be observed (see chapter 2.1 “General safety instructions”). Wear the personal protective equipment.
- Climb the ladder and close the door. Open hatch of the platform.
- Put fall arrester into slip course, open the safety rope and clamb down through the ladder.

Attention! Close all hatches of every platform.

6 相关部件及其操作 Functioning principle and operation of the components

本章所描述的部件在运行一段时间后（或由于其它原因）需要重新调整。

The following components described in this chapter need to be adjusted after a period of operating or for other reasons.

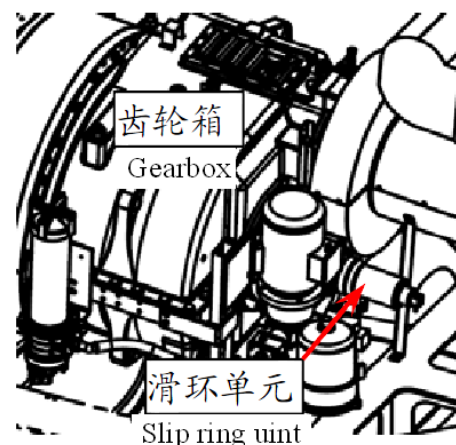
6.1 变流器 Converter

变流器位于塔底。变流器控制发电机发出的电能以 50Hz 的电网频率和 690VAC 电压传输到电网。

变流器的运行和监控由风机控制系统直接控制。

The converter is located at the tower base. It adjusts frequency and voltage of the generator to 50HZ and 690VAC.

Operation and monitoring of the converter are carried out by the wind turbine control system.



6.2 滑环单元 Slip ring unit

滑环单元是机电部件，包括固定部分、滑环及转动部分。固定部分与齿箱箱体相联，电源和数据通讯线从固定部分接入。通过滑环将电源和数据等传输到转动部分，最后通过导线传到轮毂。变桨系统的电源供给和控制信号由滑环单元传递。

The slip ring unit is an electromechanical component comprised of fixed part, slip ring and rotating element. The fixed part is combined with the gearbox. Power supply and control signals accessed from fixed part, transferred to rotating element through clip ring, and finally transferred to the hub using wires.

Power supply and control signals for the pitch are transferred through the slip ring unit.

6.3 安全传感器系统 Safety sensor system

在塔底和机舱内有很多传感器。这些传感器将不同的信息传递到控制系统，并为风机的控制和安全系统所采用。

传感器系统包含以下传感器：

- 测量控制柜和机舱罩内的空气温度，以及机舱外环境空气温度的温度传感器
- 测量介质（油、水）温度的温度传感器
- 各种管道系统中的压力传感器
- 在润滑剂不足或润滑油液位太低时，发出警报的液位检测传感器
- 高速轴制动器摩擦片厚度检测传感器
- 偏航控制的位置编码器
- 机舱振动传感器
- 风轮转速编码器

以上传感器为控制系统提供数据，由控制系统确定当前的工作模式及风机相关的各设备的工作情况，在正常情况下时风机正常工作发电，在危险情况下将风机安全停机。

There are various sensors located in the tower base and in the nacelle. They transfer the most different information to the control system and thus serve for monitoring and safety of the wind turbine.

Sensor system consists of following sensors:

- Temperature sensors measuring the air temperature in the control cabinets and in the nacelle as well as the outside air temperature
- Temperature sensors measuring the medium temperature (oil, water, seal)
- Sensors monitoring the pressure in different line systems
- Lubricant and oil level monitoring sensors that give an alarm when lubricant level is too low
- Sensors controlling the thickness of the brake pad
- Rotational speed sensors giving values for yaw control
- Sensors controlling tower vibration

- Vibration sensors located in the nacelle
- Speed sensors

The sensors give values to the control system. The control system checks values of the sensors and determines present working status of system and the components. Wind turbine will be stopped by the control system in case of danger.

6.4 照明和电源 Lighting and power supply

在塔筒第 1 段和第 2 段各有 2 盏灯，在塔筒第 3 段和第 4 段各有 3 盏灯，在机舱内有 6 盏灯，在每个控制柜内都装有灯。

在每一个塔筒段和机舱内都各有 2 个电源插座，在每个控制柜内都有 1 个电源插座。

There are two lamps on both tower levels 1 and 2. There are three lamps on the tower level 3 and 4 and six lamps in the nacelle. Lighting is also installed in each control cabinet.

There are two sockets on every tower level and in the nacelle and one socket in every control cabinet.

6.5 维护小吊车 Maintenance crane

机舱内布置了一个小吊车。禁止以下的吊车操作：

- 超过最大起吊重量
- 以一定角度起吊重物
- 起吊松散的、未捆绑好的重物
- 拽、拖曳重物
- 没有人监控就吊起并悬置负载
- 在吊起和放下负载的过程中，触碰吊链
- 通过拖曳垂链移动重物
- 抓住正在下落的重物
- 将人员与重物或者附加提升装置一起吊起
- 人员站在吊起并悬置的重物下
- 当吊链松弛时，允许重物突然下放
- 当提升时，控制电缆与吊链缠绕在一起
- 在清除可能的危险之前启动吊车

当工作完成后，将吊钩或操作盘移开行走区，通过接触器或断路器切断电源，以防伤人。在危险情况下，可以使用吊车急停装置，将所有的动作都停止下来。急停按钮位于控制盘上，为了操作此按钮功能，必须将按钮按到底，然后按钮自动锁住。为了解锁急停按钮，按照箭头所示方向旋转并释放按钮。

A board crane is situated in the nacelle, avoid the following crane operations:

- Exceeding the maximum carrying force
- Pulling loads at an angle
- Pulling loosening fixed or obstructed loads
- Tearing loose, towing or pulling loads
- Leaving suspended loads unsupervised
- Touching the chain while lifting or lowering the load
- Moving loads by dragging the control pendant
- Catching falling loads
- Carrying persons with load or lifting attachment
- Persons standing under suspended loads
- Allowing loads to drop when the chain is in a slack condition
- Allowing the control cable to be wound around the chain when lifting motions are performed
- Activating the crane before the hazard has been eliminated

When the work has been completed, position the hook assembly or the bottom block outside the travel area. Switch off the power supply at the mains connection or at the isolating switch. The crane has an emergency stop device with which all motions can be stopped in the event of a hazard. The emergency stop button is situated at the control pendant. To actuate the button, press it until it reaches the end stop. Then the button locks automatically. To unlock the emergency stop button, turn the push-button in the direction of the arrows and release.

6.6 低温加热系统 Heating system

6.6.1 系统组成 Composition

对低温型机组，机舱内布置有 8 个暖风机对机舱进行整体加热，以保持机舱温度始终处于适当的范围内。机舱内还配置有低温控制柜，以实现暖风机的启停控制。另外，塔底还配置有低温型塔底控制柜，相对于常温型塔底控制柜而言，柜内增设了部分加热元件及相应的控制回路，可实现低温条件下塔底控制柜柜体的加热和液晶显示屏的局部加热。

Eight fan heaters are situated in the nacelle of cold climate type wind turbine to heat the whole nacelle to keep the nacelle at a suitable temperature. These fan heaters are controlled by an extra heating control cabinet located in the nacelle. In addition, a control cabinet of cold climate type is located at tower base instead of one of normal type, which adds some heater elements and theirs control circuits, to heat the control cabinet and the LCD at the tower base at a cold temperature.

6.6.2 运行方式 Operation method

机舱内设置的 8 个暖风机，低温加热控制系统将其分为四组，由机舱低温控制柜实施调节控制，使其能根据设定的温度实现梯度启动和关闭：



1) 当外部温度的降低使机舱内温度低于 $X1^{\circ}\text{C}$ (可调整)时, 机舱低温控制柜将促使第一组 (2 个) 暖风机启动工作; 如机舱内温度继续降低至 $X2^{\circ}\text{C}$ (可调整) 时, 第二组暖风机也投入运行, 此时共有两组 (4 个) 暖风机工作; 照此类推, 当温度低于 $X3^{\circ}\text{C}$ (可调整)时, 三组 (6 个) 暖风机投入运行; 当温度低于 $X4^{\circ}\text{C}$ (可调整)时, 四组 (8 个) 暖风机全部投入运行。

2) 反之, 温度升高时, 暖风机逐次关闭。

Eight fan heaters situated in the nacelle is divided into four groups. The heating control cabinet makes these groups on and off in echelon according to the given temperatures:

1) When the nacelle temperature becomes lower than $X1^{\circ}\text{C}$ (can be adjusted), the first group which includes two fan heaters are turned on. When the nacelle temperature continue to fall to $X2^{\circ}\text{C}$ (can be adjusted), the second group of fan heaters are turned on. At this time, there are two groups which includes four fan heaters in total working. Likewise, when the nacelle temperature is lower than $X3^{\circ}\text{C}$ (can be adjusted), there are three groups of fan heaters which includes six fan heaters working; when the nacelle temperature is lower than $X4^{\circ}\text{C}$ (can be adjusted), all the fan heaters are turned on.

2) Conversely, when temperatur rise, the fan heaters are turned off in echelon.

6.6.3 运行工况 Working condition

在低温条件下 ($\leq -20^{\circ}\text{C}$), 风电机组的运行状态和低温加热控制系统运行工况会有些不同, 具体如下:

1) 环境温度低于 -40°C 时, 机组达到极限生存温度, 各部件的安全无法保证。

2) 环境温度介于 $-40 \sim -30^{\circ}\text{C}$ 时, 机组停止运行, 但低温加热控制系统仍需保持工作状态, 以保证加热设备一直工作, 为机组重新投入运行做好准备。

3) 环境温度高于 -30°C 时, 机组正常运行, 各加热设备根据设定温度由低温加热控制系统控制启停, 以确保各运行部件处于合适的运行温度范围内。

In the condition of low temperature($\leq -20^{\circ}\text{C}$), the wind turbine and the heating control system may run in a different way. Specific as follows:

1) When the environment temperature is lower than -40°C , the wind turbine is below the limit survival temperature, all parts of the wind turbine can not keep safety.

2) When the environment temperature is between -40°C and -30°C , the wind turbine stops, but the heating control system shall still keep working, to keep heaters heating, and to prepare for the restart of the wind turbine.

3) When the environment temperature is higher than -30°C , the wind turbine works normally, each heater is controlled on and off by heating control system to make sure all parts of the wind turbine is running at a suitable temperature.

6.6.4 停机再启动 Restart

在低温条件下, 机组停机再启动, 视低温加热控制系统的运行情况不同, 处理方式也不尽相

 上海电气 上海电气风电设备有限公司 SHANGHAI ELECTRIC SHANGHAI ELECTRIC WINDPOWER EQUIPMENT CO.,LTD.		W2000 风电机组	MD-030.00-A	
操作运行手册 Operating Manual			共 130 页	第 26 页
同。具体如下： 1) 机组停机，但停机期间低温加热控制系统始终保持工作，即机组的加热设备一直处于工作状态，则一旦环境温度高于-30℃且其他运行条件均能满足，机组可立即投入运行。 2) 机组停机，但停机期间因某种原因（如电网故障）导致低温加热控制系统无法运行，即所有的加热设备均无法工作，则当故障排除（如电网恢复）时，低温加热控制系统需立即开始工作，控制加热设备投入使用。 如此时温度低于-20℃，则机组不可立即投入运行，需视情况加热一段时间，机组再投入运行。 At a cold climate, the restart procedure of the wind turbine can be different dependent on the different running conditions of the heating control system. Specific as follows: 1) When the wind turbine stops, the heating control system still keeps working, namely, the heaters of the wind turbine still keep working. In this condition, as soon as the environment temperature is higher than -30℃ and other conditions are all met, the wind turbine can be restarted at once. 2) When the wind turbine stops, the heating control system can not work due to some reason (such as a grid fault), namely, all the heaters can not work. In this condition, as soon as the fault is removed, the heating control system shall starts working at once, to control the heaters. At that time, if the temperature is below -20℃, the wind turbine can not restart until the heaters have heated the wind turbine for a certain time. 6.6.5 安全说明 Safety instruction 6.6.5.1 控制柜 Control cabinet 机舱低温控制柜柜内设置有电气器件，出于保护目的，柜门平时都是锁着的。在每年可能出现极端低温（低于-20℃）的月份里，接通电源启用低温加热控制系统。机舱低温控制柜与低温型塔底控制柜会根据实际温度自动调节加热设备的工作状态，从而保持机舱温度和塔底控制柜内部温度始终在合适的范围内。而在不可能出现极端低温（低于-20℃）的月份里，可以切断低温加热控制系统的电源。  警告 控制柜平时必须锁着，只有专业工作人员才允许打开柜门进行操作。 未经 SEWIND 公司同意，不允许修改低温加热控制柜的任何参数设置。  注意 柜体内的技术文件，如电路图、接线表、使用维护说明书等，这些文件请小心保存。它们是风电机组的一部分，当风电机组被转卖时，这些文件要一并交给新的运行单位。 There are electric elements installed in the heating control cabinet located in the nacelle. To guarantee security, the door of the cabinet is usually locked. During the months when the extremely low				



temperature (lower than -20°C) may be expected, the heating control system shall be powered on. Then the heating control cabinet located in the nacelle and the control cabinet of cold climate type located at tower base will adjust the heaters' working state based on the actual temperature to keep the temperature in the nacelle and in the control cabinet located at tower base at a suitable temperature. Otherwise, during the months when the extremely low temperature (lower than -20°C) can not be expected, the heating control system can be powered off.

Attention:

The doors of the control cabinets must be locked at usual time. Only professional staff is permitted to open the doors.

Without permission of SEWIND, any parameters of the heating control cabinet must not be modified.

Attention:

The technical documents in the control cabinets, such as circuit diagram, wiring table, maintenance manual etc., shall be preserved carefully. They are part of the wind turbine, and if the wind turbine is sold, they shall be handed over as well.

6.6.5.2 暖风机 Fan heater

每一个暖风机装备了热保护，在过热情况下，加热器会自动关闭。在每个加热器上面都设有一个复位按钮。



警告

暖风机所有开关、旋钮出厂前已设定好，只有专业工作人员才允许对电暖风机进行操作。

Every fan heater is equipped with thermal protection that automatically switches off in case of overheating. A reset button is located at the top of every fan heater.

Attention:

All the switches of the fan heaters are set before leaving the factory. Only professional staff is permitted to operate on the fan heaters.

7 控制系统 Control System

7.1 概述 General

W2000 型风力发电机组控制系统由机舱控制柜和塔底控制柜两部分组成。

塔底控制柜作为控制系统的主站，控制器采用德国倍福(Beckhoff)公司的嵌入式 PC，信号采集通过总线端子模块实现。塔底控制柜的通讯接口有两个，一个 Profibus 通讯接口，实现与变流柜之间的通讯；另一个多模光纤通讯接口，实现与机舱控制柜之间的通讯。机舱控制柜内还有一个站点，用于收发塔底部分的数字量及模拟量输入输出信号、完成温度测试、电能监控等功能。塔底控制柜配有触摸屏一台，预装系统，用于系统管理和监控。

机舱控制柜有两个通讯接口，一个是与塔底控制柜实现光纤通讯的模块；另一个是 Profibus 通讯接口，实现与变桨系统之间的通讯。机舱控制柜按照数字量与模拟量的划分方式，将总线端子模块分为了两个站点，实现了对机舱内相关物理量的监测与处理。机舱控制柜配有嵌入式面板型 PC 一台，预装 WinXP 系统，作为 HMI，对系统进行管理和监控。

2MW 控制系统的硬件配置结构如图 4 所示：

The control system of W2000 is comprised of a control cabinet located in nacelle and a control cabinet located at tower base.

The main control cabinet located at tower base uses embedded PC produced by Beckhoff company in Germany as its controller, and the signals are collected through bus terminal. There are two communication interfaces in the control cabinet located at tower base: one is profibus communication interface for communication with converter, and another one is multimode fibre communication interface for communication with control cabinet located in the nacelle. There is a station in the control cabinet located in nacelle, which send-receive digital signals, analog signals from tower base to accomplish temperature test, electric energy monitoring, etc. There is a touch panel with WinCE system in the control cabinet located at tower base to finish system management and monitoring.

There are two communication interfaces in the control cabinet in nacelle: one is multimode fibre communication interface for communication with control cabinet located at tower base, another is profibus communication interface for communication with pitch system. Bus terminal in the control cabinet located at nacelle is divided into two stations according to digital and analog signals to accomplish monitor and process the physical quantities. There is an embedded PC installed in the control cabinet located in nacelle with WinXP system. And system is managed and monitored by it as HMI.

The structure chart of control system hardware configuration for 2MW is shown as Fig4.

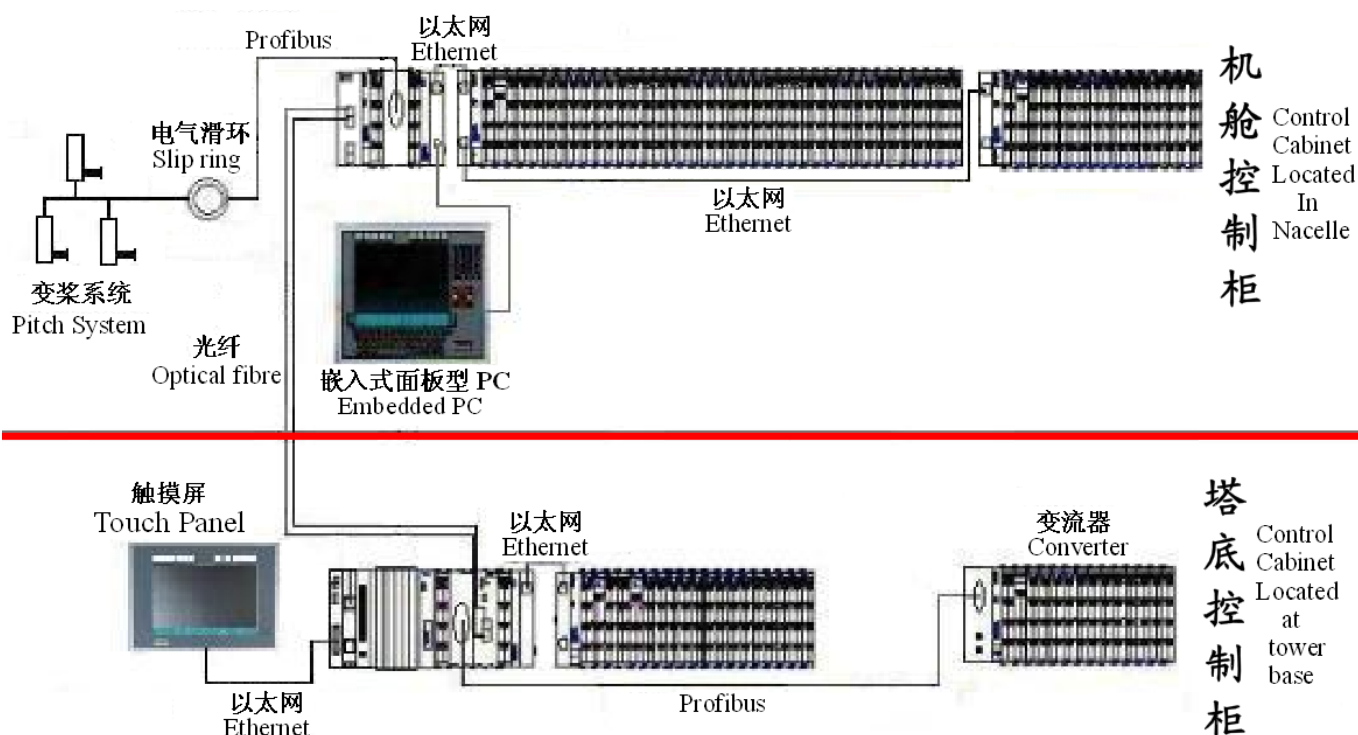


图 3 系统配置结构图

Fig 3 Structure chart of system configuration

塔底控制柜主要控制塔底柜冷却风扇、塔底通风风扇、变流器风扇及塔底主电源等执行机构；机舱控制柜主要控制液压站、齿轮箱、发电机、风轮、偏航及风扇、照明等执行机构。

硬件系统由倍福(Beckhoff)系列总线端子构成，主要采用的总线端子有数字量输入端子、数字量输出端子、模拟量输入端子、特殊功能端子、供电端子、安全模块端子。这些总线端子与总线耦合器一起组成站点，各个站点之间通过通讯线缆相连接，与变流柜从站、变桨从站以及 CPU 模块构成整个风机的控制系统。

The control cabinet at tower base main controls the cooling fan, draught fan and converter fan at tower base, it also controls the main power source at tower base, etc. The control cabinet at nacelle main controls the actuating mechanism such as hydraulic station, gearbox, generator, wind rotor, yaw, fan, and lighting, etc.

The hardware system of the wind turbine is consisted of bus terminal including digital input /output terminal, analog input terminal, special terminal, power supply terminal and safety terminal. These bus terminals constitute bus station together with bus coupler, and these stations are connected with each other through communication cable. The whole wind turbine control system is consisted of bus station, converter slave station, pitch slave station and CPU module.

7.2 控制操作 Control operation

7.2.1 控制面板 Control panel

7.2.1.1 塔底控制柜柜门控制面板 Control panel of the cabinet located at tower base

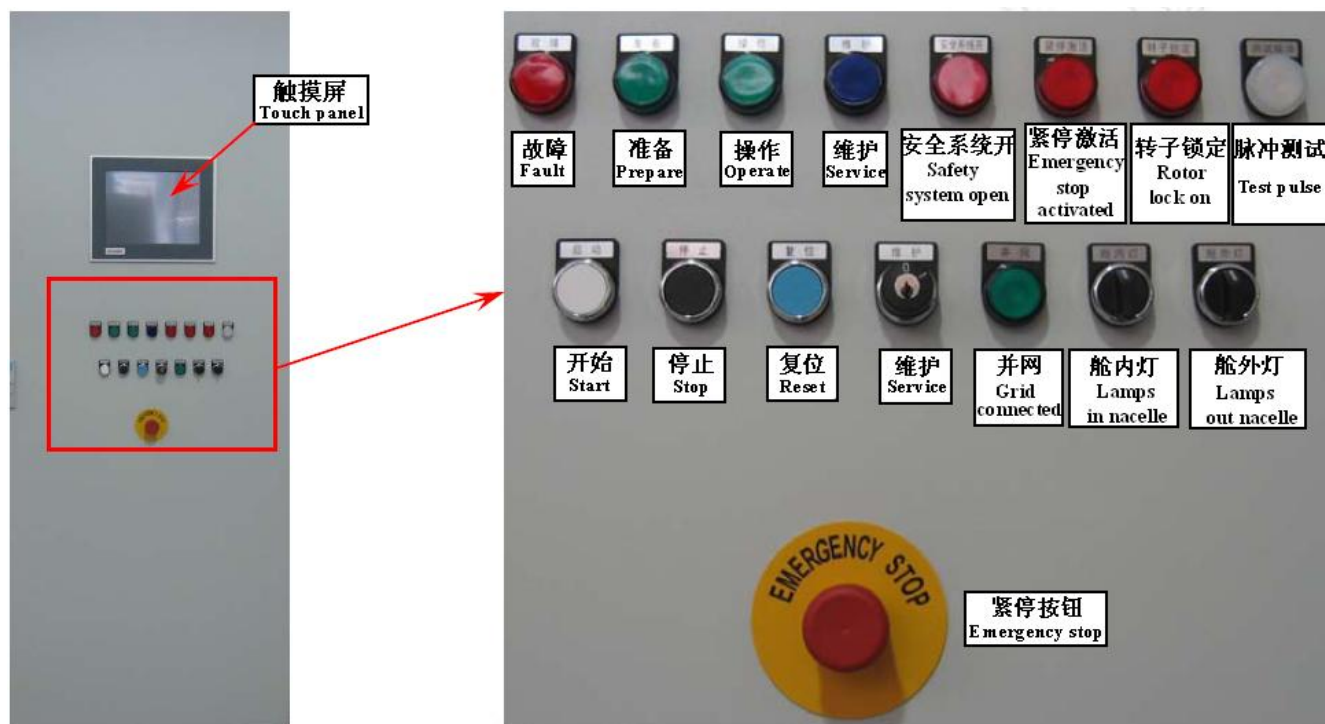


图 4 塔底控制柜柜门面板

Fig4 Control panel of the control panel at tower base

塔底控制柜柜门面板由触摸屏、指示灯、按钮和开关组成，如图 5 及表 2 所示：

The control panel of control cabinet located at tower base is comprised of touch panel, lamp, button and switch, shown as Fig5 and table2.



表 2 塔底控制柜柜门面板指示灯、按钮及开关

Table2 Lamp, button and switch on the panel of control cabinet at tower base

类别 Class	编号 Number	名称 Name	颜色 Colour	含义/功能描述 Meaning/Function Description
指示灯 Lamp	1	故障指示灯 Fault lamp	红灯 Red	系统报错, 运行停止 System fault occurs, operation stop
	2	准备指示灯 Preparation lamp	绿灯 Green	系统处于待机模式, 准备运行 Stand-by mode , ready to run
	3	操作指示灯 Operation lamp	绿灯 Green	系统处于正常运行状态 Operation normal
	4	维护指示灯 Service lamp	蓝灯 Blue	系统处于服务模式 Service mode
	5	安全系统开指示灯 Safety system	红灯 Red	安全系统紧停触发 Emergency stop has opened the safety chain
	6	紧停激活指示灯 Emergency stop	红灯 Red	紧停开关触发 Emergency stop switches is actuated
	7	转子锁定指示灯 Rotor lock on	白灯 White	风轮、转子锁定 Rotor locking has been enabled
	8	测试脉冲指示灯 Test pulse	白灯 White	PLC 处于运行状态 PLC works correctly
	9	并网指示灯 Grid-connected lamp	绿灯 Green	系统处于并网运行状态 Grid-connected operation state
按钮 Button	1	启动按钮 Start	白色 White	按下, 系统启动 Press down, system start
	2	停止按钮 Stop	黑色 Black	按下, 停机 Press down, stop
	3	复位按钮 Reset	蓝色 Blue	按下, 系统复位 Press down, system reset
	4	紧停按钮 Emergency stop	红色 Red	按下, 紧停触发 Press down, emergency stop activated
开关 Switch	1	维护开关 Service switch		钥匙开关, 使系统进入服务模式 Key switch, switch wind turbine into service mode
	2	舱内灯开关 Switch for lamps in nacelle		塔底内部空间照明 Lighting inside the tower base
	3	舱外灯开关 Switch for lamps out nacelle		塔底外部楼梯照明 Lighting for the ladder outside the tower base

7.2.1.2 机舱控制柜柜门控制面板 Control panel of the cabinet located in nacelle

机舱控制柜柜门面板由嵌入式面板型 PC、指示灯、按钮和开关组成, 如图 6 及表 3 所示:

The control panel of control cabinet located in nacelle is comprised of embedded PC, lamp, button and switch, shown as Fig6 and table3.

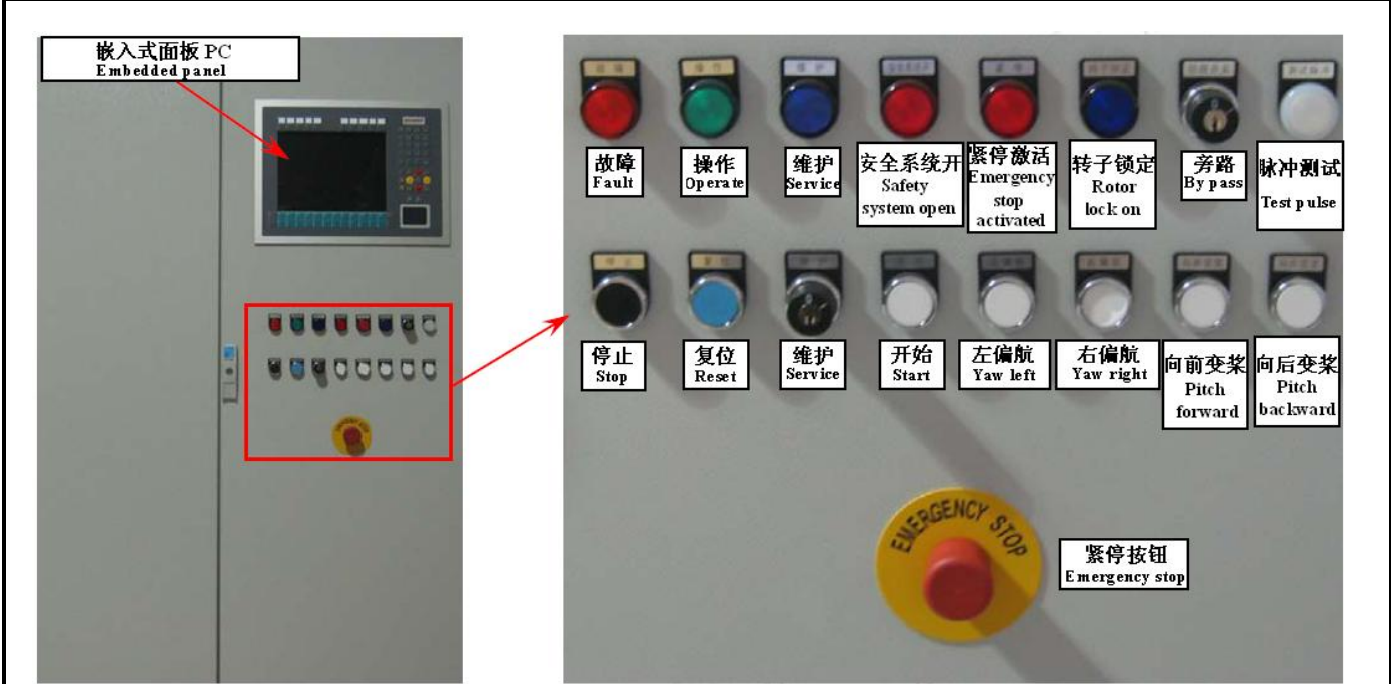


图 5 机舱控制柜柜门面板
Fig5 Control panel of the control panel in nacelle

表 3 机舱控制柜柜门面板指示灯、按钮及开关

Table3 Lamp, button and switch of control cabinet located in nacelle

类别 Class	编号 Number	名称 Name	颜色 Colour	含义/功能描述 Meaning/Function Description
指示灯 Lamp	1	故障指示灯 Fault lamp	红灯 Red	系统报错，运行停止 System fault occurs, operation stop
	2	操作指示灯 Operation lamp	绿灯 Green	系统处于正常运行状态 Operation normal
	3	维护指示灯 Service lamp	蓝灯 Blue	系统处于服务模式 Service mode
	4	安全系统开指示灯 Safety system	红灯 Red	安全系统紧急停止触发 Emergency stop has opened the safety chain
	5	紧急激活指示灯 Emergency stop	红灯 Red	紧急开关触发 Emergency stop switches is actuated
	6	转子锁定指示灯 Rotor lock on	白灯 White	风轮、转子锁定 Rotor locking has been enabled
	7	测试脉冲指示灯 Test pulse	白灯 White	PLC 处于运行状态 PLC works correctly
	2	停机按钮 Stop	黑色 Black	按下，停机 Press down, stop
	3	复位按钮 Reset	蓝色 Blue	按下，系统复位 Press down, system reset

表 3 (续)
Continued Table3

类别 Class	编号 Number	名称 Name	颜色 Colour	含义/功能描述 Meaning/Function Description
按钮 Button	4	紧停按钮 Emergency stop	红色 Red	按下, 紧停触发 Press down, emergency stop activated
	5	左偏航按钮 Yaw left	白色 White	按下, 风机左偏航 Press down, wind turbine yaw left
	6	右偏航按钮 Yaw right	白色 White	按下, 风机右偏航 Press down, wind turbine yaw right
	7	向前变桨按钮 Pitch forward	白色 White	按下, 叶片向前变桨 Press down, pitch forward
	8	向后变桨按钮 Pitch backward	白色 White	按下, 叶片向后变桨 Press down, pitch backward
开关 Switch	1	维护开关 Service switch		钥匙开关, 使系统进入服务模式 Key switch, switch wind turbine into service mode
	2	旁路开关 By pass key switch		钥匙开关, 变桨 91° 旁路开关锁定 Key switch, pitch 91° by pass key switch locked

7.2.2 塔底控制柜显示屏操作 Operations of display screen in control cabinet at tower base

7.2.2.1 登陆及权限 Login and permission

显示屏的工具栏如下图所示:

Tool bar of the display screen is shown as the following figure:



图 6 塔底控制柜显示屏的工具栏

Fig6 Tool bar of display screen in control cabinet located at tower base

在进入主画面后, 点击登录按钮, 弹出用户登录界面, 如下图所示:

After enter the main screen, login interface will be popped up when the login button is clicked, shown as the following figure:



图 7 登录界面
Fig7 Login interface

点击红色方框，可以切换用户：

LEVEL1 的权限为 50：可操作手动启停风机，查看状态码和历史数据等。

LEVEL3 的权限为 90：可操作手动变桨，手动启停泵，风扇和加热器等。

输入密码后，点击确定按钮，权限从 0 变为 90。

Click the red box, user can be switched:

The permissions the LEVEL1 is 50: can start or stop wind turbine manually; can view state code and history data, etc.

The permissions the LEVEL3 is 90: can pitch the blade manually; can manually start or stop the pump, fan and heater, etc

After inputting code, the permissions will change from 0 to 90 when the OK button is clicked.

7.2.2.2 实时数据监控 Monitoring of realtime data

实时数据监控界面，如图 9 所示：

The monitoring interface of realtime data is shown as fig9:

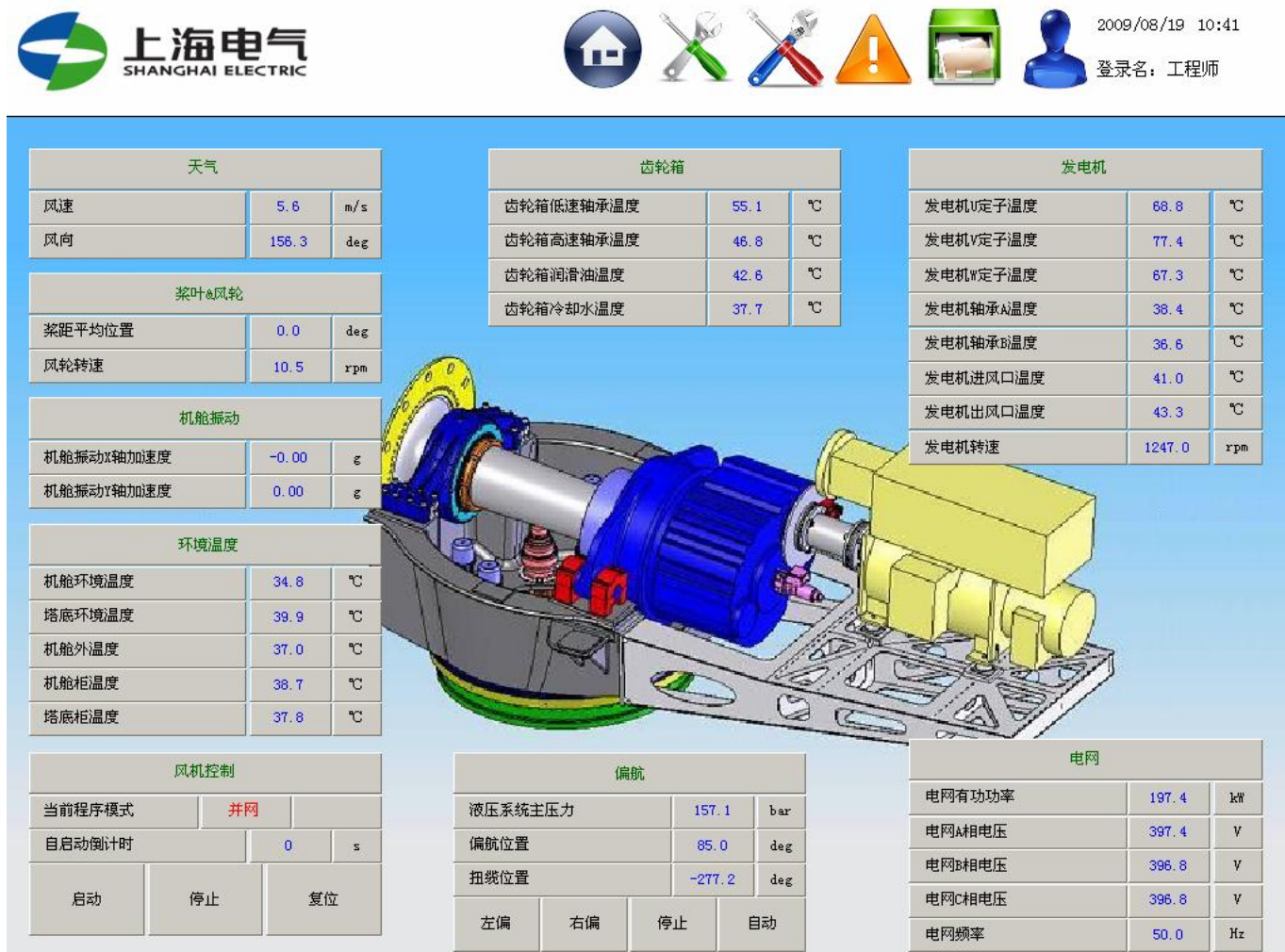


图 8 实时数据监控界面

Fig8 Monitoring of realtime data

实时数据包括天气、桨叶&风轮、机舱振动、环境温度、风机控制、齿轮箱、发电机、偏航及电网的相关数据及操作按钮。

Realtime data includes the data of weather, blade&wind rotor, nacelle vibration, ambient temperature, wind control system, gearbox, generator, yawing, power grid and some operational buttons.

7.2.2.3 服务模式 Service mode

通过操作维护开关进入服务模式，服务模式操作界面如图 10 所示。

The operational interface of service mode entered with service switch key is shown as fig10.



上海电气

SHANGHAI ELECTRIC

上海电气风电设备有限公司

SHANGHAI ELECTRIC WINDPOWER EQUIPMENT CO.,LTD.

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MD-030.00-A

操作运行手册

Operating Manual

共 130 页

第 38 页



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安全链

IO列表

通讯状态

变桨系统状态

状态码

	编号	激活时间	复位时间			编号	激活次数(天)	激活次数(全)	
1	40010	2009-08-17-10:38:44.920	2009-08-17-10:40:55.236	 	1	10001	0	14	 
2	300709	2009-08-17-10:38:44.360	2009-08-17-10:41:00.356		2	10002	0	11	
3	300343	2009-08-17-10:38:44.360	2009-08-17-10:41:00.436		3	30001	0	15	
4	300342	2009-08-17-10:38:44.360	2009-08-17-10:41:00.436		4	30002	0	15	
5	300341	2009-08-17-10:38:44.360	2009-08-17-10:41:00.436		5	30003	0	15	
6	40010	2009-08-17-10:22:27.240			6	30004	0	15	
7	300709	2009-08-17-10:22:26.440			7	30005	0	17	
8	300343	2009-08-17-10:22:26.440			8	30006	0	18	
9	300342	2009-08-17-10:22:26.440			9	30011	0	0	
10	300341	2009-08-17-10:22:26.440			10	30061	0	0	
11	40010	2009-08-17-10:16:21.921			11	30081	0	2	
12	300709	2009-08-17-10:16:21.360			12	40010	3	29	
13	300343	2009-08-17-10:16:21.360			13	50001	0	1	
14	300342	2009-08-17-10:16:21.360			14	50002	0	14	
15	300341	2009-08-17-10:16:21.360			15	50010	0	1	
16	40010	2009-08-16-21:09:22.880	2009-08-17-09:13:39.760		16	50020	0	11	
17	300909	2009-08-16-21:08:59.200	2009-08-17-09:13:39.360		17	50025	0	11	
18	300908	2009-08-16-21:08:59.120	2009-08-17-09:13:39.360		18	50030	0	3	
19	300907	2009-08-16-21:08:59.120	2009-08-17-09:13:39.360		19	50035	0	0	
20	300342	2009-08-16-21:08:37.680	2009-08-17-09:13:45.040		20	50036	0	6	

状态码说明

风轮制动垫故障

图 11 状态码记录界面

Fig11 Interface of status code record

7.2.2.4.1 状态码历史列表 History list of status code

显示状态码编号，激活时间和复位时间。最多显示 100 条历史记录。点击任意一条状态码都会在“状态码说明”栏中显示中文注释。

The list displays number of status code, activation time and resetting time. It can show 100 historical record at most. Click any status code, the Chinese annotation will be shown in the ‘status code instruction’.

7.2.2.4.2 状态码激活次数列表 List of status code activation number

显示所有状态码的 24 小时激活次数和累计激活次数。

This list display 24 hours activation number and accumulative total activation number of all the status code.



7.2.2.5 输入输出列表 (IO) Input/Output list

点击 IO 列表按钮，进入图 13 所示的界面，显示 DI 和 DO 的状态。

The interface shown as fig13 will be entered and the status of DI and DO will be shown if the IO list button is clicked.



The screenshot shows the '数字量输入输出列表' (Digital Input/Output List) interface. It features a top navigation bar with the Shanghai Electric logo, a home icon, and various tool icons. The main content area contains two tables: 'DI列表' (DI List) and 'DO列表' (DO List). The DI table lists 20 digital input channels with their names, addresses, and status (all FALSE). The DO table lists 20 digital output channels with their names, addresses, and status (various TRUE and FALSE values).

DI列表				DO列表			
	名称	通道	值		名称	通道	值
1	塔底UPS电池OK	213DI1, CH1	FALSE	1	塔底柜内冷却风扇	214D01, CH1	FALSE
2	塔底UPS供电模式	213DI1, CH2	FALSE	2	塔底故障灯	214D01, CH2	TRUE
3	塔底启动按钮	213DI1, CH3	FALSE	3	塔底准备灯	214D01, CH3	FALSE
4	塔底停止按钮	213DI1, CH4	FALSE	4	塔底操作灯	214D01, CH4	FALSE
5	塔底复位按钮	213DI2, CH1	FALSE	5	塔底维护模式灯	214D02, CH1	FALSE
6	塔底远程停止	213DI2, CH2	FALSE	6	塔底安全链断开灯	214D02, CH2	FALSE
7	塔底紧停	213DI2, CH3	FALSE	7	塔底紧停激活灯	214D02, CH3	FALSE
8	备用	213DI2, CH4	FALSE	8	塔底风轮锁定灯	214D02, CH4	FALSE
9	塔底维护模式开关	213DI3, CH1	FALSE	9	塔底并网灯	214D03, CH1	FALSE
10	箱变避雷保护OK	213DI3, CH2	FALSE	10	安全系统PLC紧停	214D03, CH2	TRUE
11	备用	213DI3, CH3	FALSE	11	塔底冷却风扇	214D03, CH3	FALSE
12	塔底柜内加热器OK	213DI3, CH4	FALSE	12	塔底脉冲测试灯	214D03, CH4	TRUE
13	塔底熔断器反馈	213DI4, CH1	FALSE	13	塔底主电源	214D04, CH1	TRUE
14	塔底避雷保护OK	213DI4, CH2	FALSE	14	变频柜吸风风扇	214D04, CH2	FALSE
15	塔底风扇保护OK	213DI4, CH3	FALSE	15	备用	214D04, CH3	FALSE
16	变频柜吸风风扇反馈	213DI4, CH4	FALSE	16	备用	214D04, CH4	FALSE
17	变压器超温	213DI5, CH1	FALSE	17	机舱故障灯	455D01, CH1	TRUE
18	轻瓦斯报警	213DI5, CH2	FALSE	18	机舱操作灯	455D01, CH2	FALSE
19	塔底主电源OK	213DI5, CH3	FALSE	19	机舱维护模式灯	455D01, CH3	FALSE
20	重瓦斯跳闸	213DI5, CH4	FALSE	20	机舱安全链断开灯	455D01, CH4	FALSE

图 12 I/O 列表界面
Fig12 Interface of I/O list

7.2.2.6 安全链 Safety chain

点击安全链按钮，进入图 14 所示的界面，显示安全链的状态。当安全链正常时，安全链显示颜色为绿色；当安全链断开时，安全链显示颜色为红色。



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安全链

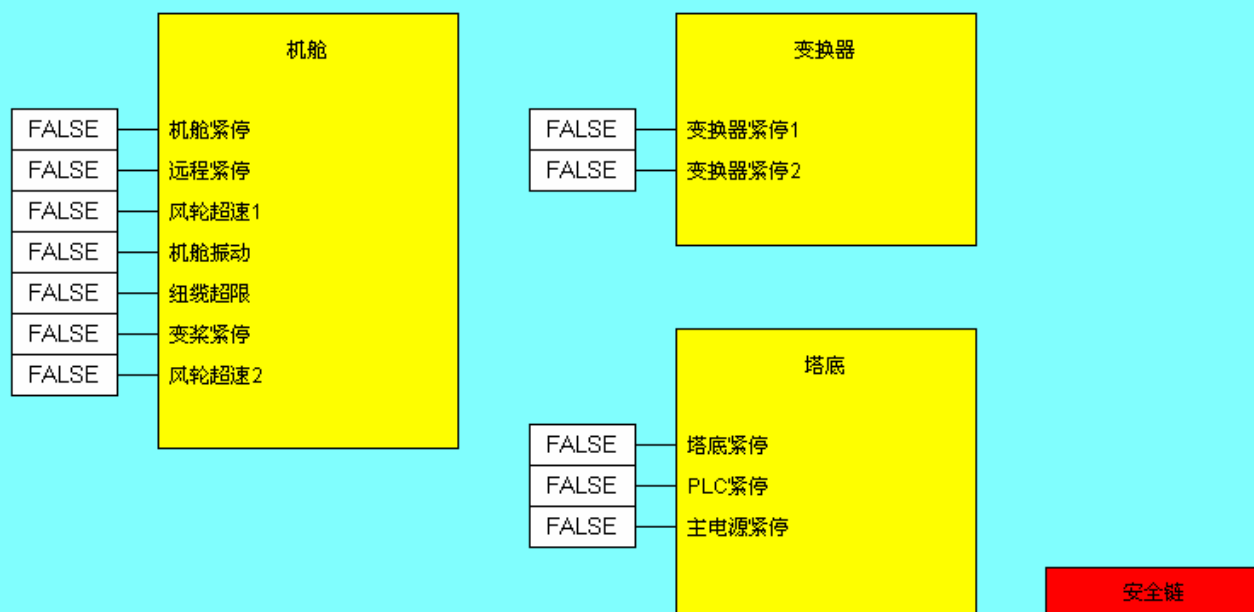


图 13 安全链界面

Fig13 Interface of the safety chain

Click button of the safety chain, you will enter the interface shown as fig7.11 and see the status of the safety chain. The colour of the safety chain is green when it is normal, and the colour will turn into red when the safety chain opens.

7.2.2.7 EntherCAT 及 Profibus 通讯 EntherCAT and Profibus communication

点击通讯状态按钮，进入如图 15 所示的界面，显示机舱和塔底的 Ethercat 和 Profibus 的通讯状态。

Click the communication button, you will enter the interface shown as fig15, and the communication status of the Enthercat and Profibus located in nacelle and tower base will be shown.



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上海电气风电设备有限公司
SHANGHAI ELECTRIC WINDPOWER EQUIPMENT CO.,LTD.

W2000 风电机组

MD-030.00-A

操作运行手册

Operating Manual

共 130 页

第 41 页



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通讯状态

塔底

EtherCAT

BK1120_TowerBase_CoupleState	0
Bk1120_TowerBase_State	0
CX1100_TowerBase_State	0
EK1521_TowerBase_State	0

Profibus

EL6731_Converter_Master_State	0
EL6731_Converter_Master_Error	0
EL6731_Converter_Master_ActMd	0
BK3150_Converter_Slave_Dpstate	0

机舱

EtherCAT

EK1501_Nacelle_State	0
BK1120_Nacelle_Digital_CoupleState	0
Bk1120_Nacelle_Digital_State	0
BK1120_Nacelle_Analog_CoupleState	0
Bk1120_Nacelle_Analog_State	0

Profibus

EL6731_Pitch_Master_State	0
EL6731_Pitch_Master_Error	0
EL6731_Pitch_Master_ActualModule	0
Bk1120_Nacelle_Analog_State	0

图 14 通讯状态界面

Fig14 Interface of the communication status

7.2.2.8 历史数据统计 Historical data statistics

历史数据统计显示操作数据的历史记录，点击总报表，年报表和月报表按钮，显示不同时间跨度的历史数据，如图 16 所示。共记录 16 列数据，包括：

1) 日期 Date

2) 电量 Energy(kWh)

3) 可利用率 Availability

4) 工作时间 WorkTime(min)

5) 故障时间 FaultTime(min)

6) 停机时间 StopTime(min)

7) 发电时间 PD.Time(min)

8) 偏航时间 YawTime(min)

9) 顺时针偏航时间 CWYawTime(min)

10) 逆时针偏航时间 CCWYawTime(min)

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- 11) 维护时间 ServiceTime(min)
- 12) 停机次数 StopNr
- 13) 并网次数 CutinNr
- 14) 偏航次数 YawNr
- 15) 主断路器闭合次数 CircuitOnNr
- 16) 维护次数 ServiceNr

Historical data statistics shows the history record of the operation data, click the button of total report form, yearly report form and mothly report form, the history data of different time range will be displayed, shown as fig7.13. There are 16 columns in all, including:

- 1) date : Date
- 2) electric quantity: Energy(kWh)
- 3) availability rate: Availability
- 4) working time: WorkTime(min)
- 5) fault time: FaultTime(min)
- 6) stop time: StopTime(min)
- 7) generate electricity time: PD.Time(min)
- 8) yawing time: YawTime(min)
- 9) clockwise yawing time: CWYawTime(min)
- 10) anticlockwise yawing time : CCWYawTime(min)
- 11) servicetime: ServiceTime(min)
- 12) stop number: StopNr
- 13) grid-connect number: CutinNr
- 14) yawing number: YawNr
- 15) number of the main circuit breaker closing: CircuitOnNr
- 16) service number: ServiceNr





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数据统计

总报表

年报表

月报表

	DATE	Energy	WorkTime	FaultTime	StopTime	ProdTime	YawTime	
13	8-13	975	488	274	375	429	42	▲
14	8-14	1138	1100	303	305	1082	47	
15	8-15	5918	1315	25	119	1308	94	
16	8-16	5892	1184	244	239	1181	103	
17	8-17	311	77	565	562	72	9	
18	8-18	0	0	0	0	0	0	
19	8-19	0	0	0	0	0	0	
20	8-20	0	0	0	0	0	0	
21	8-21	0	0	0	0	0	0	
22	8-22	0	0	0	0	0	0	
23	8-23	0	0	0	0	0	0	
24	8-24	0	0	0	0	0	0	
25	8-25	0	0	0	0	0	0	
26	8-26	0	0	0	0	0	0	
27	8-27	0	0	0	0	0	0	
28	8-28	0	0	0	0	0	0	
29	8-29	0	0	0	0	0	0	
30	8-30	0	0	0	0	0	0	▼

◀ ▶

图 15 历史数据统计界面

Fig15 Interface of historical data statistics

7.3 运行模式 Operational states

W2000 风机控制系统的主流程中包括十一种运行模式：初始化模式、停机模式、服务模式、待机模式、自检测模式、空转模式、启动模式、切入模式、发电模式、共振模式、变桨润滑模式。流程图如图 17 示。

W2000 wind turbine has eleven operational states: initialization mode, stop mode, service mode, standby mode, self-test mode, idling mode, start mode, cut-in mode, generation mode, resonant mode, pitch-lubrication mode. The floe chart is shown as fig17.

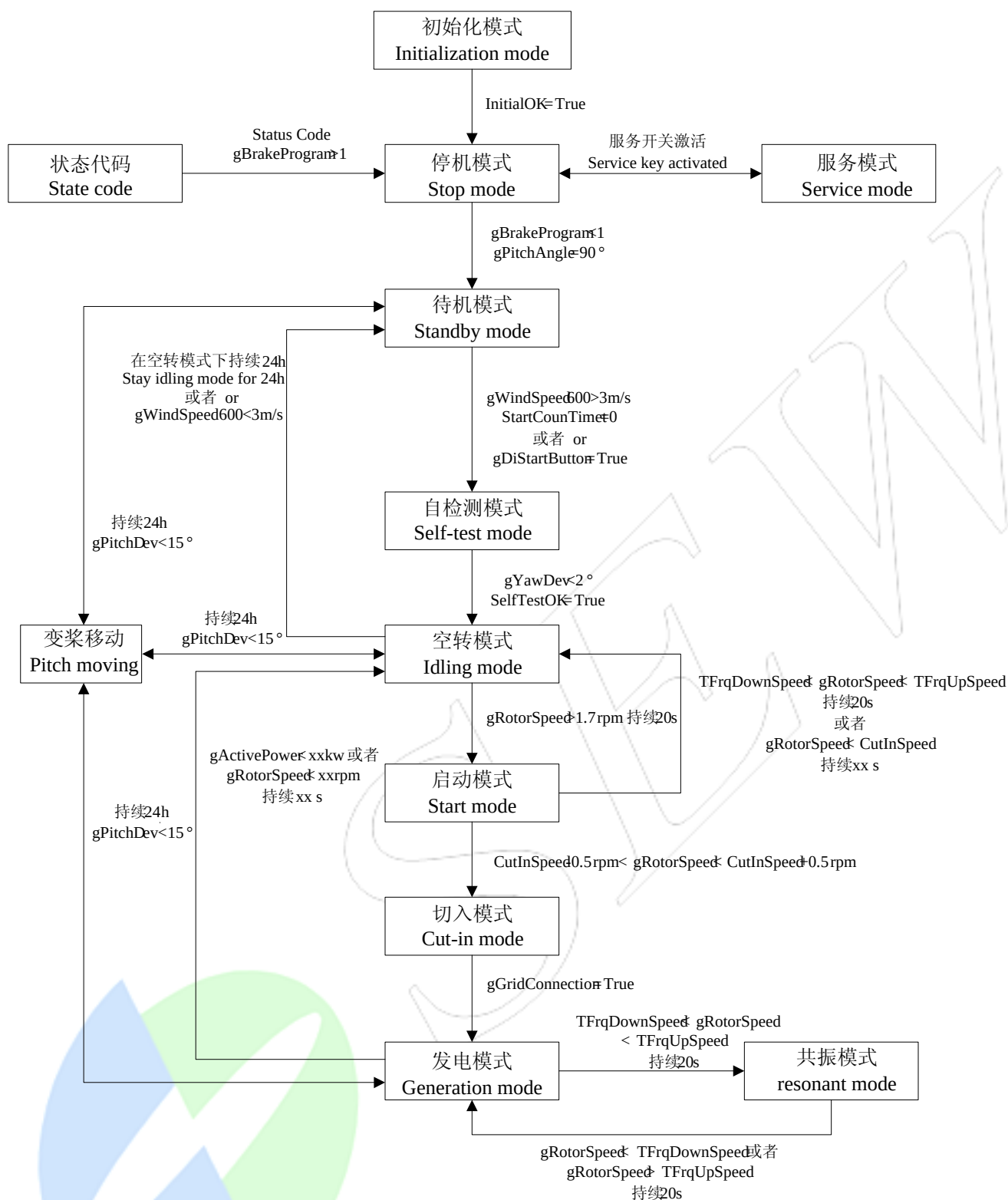


图 16 运行流程图
Fig 16 Flow chart of operation

7.3.1 初始化模式 Initialization mode

当风力发电机组控制器在重新启动、参数修改、程序更改等情况发生之后，需要进入初始化模式。



- 1) 执行初始化程序读取参数文件，如果文件读取错误，将置出状态代码“初始化文件错误”；
- 2) 将读取到的参数赋给相对应的运行变量；
- 3) 初始化完成，进入停机模式。

After restarting controller of wind turbine, modifying parameter value or changing the program, etc, the wind turbine needs to go to initialization status.

- 1) Execute initialization program and read variable file. If the file readed fault, a state code 'initialization file fault' will be shown;
- 2) Assign the reading parameters to the corresponding variables;
- 3) Initialization finished, wind turbie goes to stop mode.

7.3.2 停机模式 Stop mode

由状态代码产生停机程序，机组将进入停机模式，在停机模式下：

- 1) 对于能够自动复位的状态代码，通过自动复位可进行复位；
- 2) 可通过手动复位的方式对状态代码进行复位，不同的状态代码对应了不同的复位权限；
- 3) 可通过 SCADA 系统对风机进行手动复位；
- 4) 操作“服务开关”，风机可实现服务模式和停机模式之间的切换；
- 5) 执行复位后，如果停机程序 <1 ，则风机进入待机模式。

The wind turbine goes to stop mode when the state code triggers the stop program.

- 1) Some state code can reset automatically;
- 2) The state code can be reset manually, different state code has different reset permissions;
- 3) Wind turbine can be reset manually through SCADA system;
- 4) Wind turbine can be switched between service mode and stop mode through operating 'service switch key';
- 5) After reset executed, wind turbine will go to standby mode if $gBrakeProgram < 1$.

7.3.3 服务模式 Service mode

服务模式用于对风机进行维护和试运行，当机舱控制柜门或塔底控制柜门上的维护钥匙开关打开时，系统进入服务模式。服务模式下的操作请见 7.8.4 节。当维护钥匙开关锁定时，系统退出服务模式进入停机模式。

Wind turbine will be maintained and trial ran at the service mode. System will go to service mode when the service switch key on the panel of control cabinet located in the nacelle or the tower base is activated. The operation at service mode is discussed detailedly in chapter 7.8.4. The system will out of service mode when the service switch key is locked.

7.3.4 待机模式 Standby mode

- 1) 执行自动偏航;
 - 2) 如果 60s 平均风速大于 3m/s, 启动倒计时程序。倒计时时间由自启动倒计时和高速轴刹车倒计时剩余时间之间较大的数值作为倒计时时间。启动倒计时后, 当风速大于 3m/s 时倒计时计数器减 1, 当风速小于 3m/s 时, 倒计时计数器加 1;
 - 3) 当机组执行了紧急停机程序, 则高速轴刹车倒计时激活, 设置倒计时时间为 120s。当风轮转速小于 1rpm 时, 倒计时开始。倒计时满 120s 后, 高速轴刹车倒计时关闭, 高速轴刹车倒计时剩余时间置 0;
 - 4) 按下柜门面板或 HMI 上的“启动按钮”, 则可跳过自启动倒计时。但是如果高速轴刹车倒数计时剩余时间大于 0, 则必须要等到高速轴刹车倒计时完成后才能进入自检测模式。
- 1) Execute yawing automatically;
 - 2) Countdown program will start when the average of 60 seconds wind speed exceeds 3m/s. Countdown time uses the larger one between self-starting countdown time and high shaft brake countdown time. The countdown time will add one when the wind speed exceeds 3m/s, and it will subtract one when the wind speed less than one;
 - 3) If wind turbine executes emergency stop program, the high speed shaft braking program will be activated, and the countdown time will be set as 120 seconds. Countdown time will start when the rotor speed less than 1rpm. The high speed shaft brake countdown time will stop and rest of the countdown time will be set as 0 when the countdown time runs 120s.
 - 4) Self-starting countdown time can be jumped if the “START” button on the panel or HMI is pressed. But if the rest time of high speed shaft countdown time exceeds 0, system goes to self-test mode only if the high speed brake program finished.

7.3.5 自检测模式 Self-test mode

在自检测程序模式下, 控制器将对变桨系统进行测试, 主要包括变桨速度、变桨位置以及高速轴刹车检测。自检测流程图如图 18 所示。

用户可以选择在自检测程序开始之前, 通过 HMI 上的按钮选择跳过测试, 如果进入了自检测程序, 检测过程将不能中途退出。

在所有的部件完成测试之后, 风机转入空转模式。

Contoller will test the pitch system at the self-test mode, including pitch rate, pitch position and high speed shaft brake. The flow chart of self-test is shown as fig18.

Users can control program jump this test through the button on HMI before the self-test program start, and it can't drop out during the test process.

After all of the components finishing test, the wind turbine will go to idling mode.

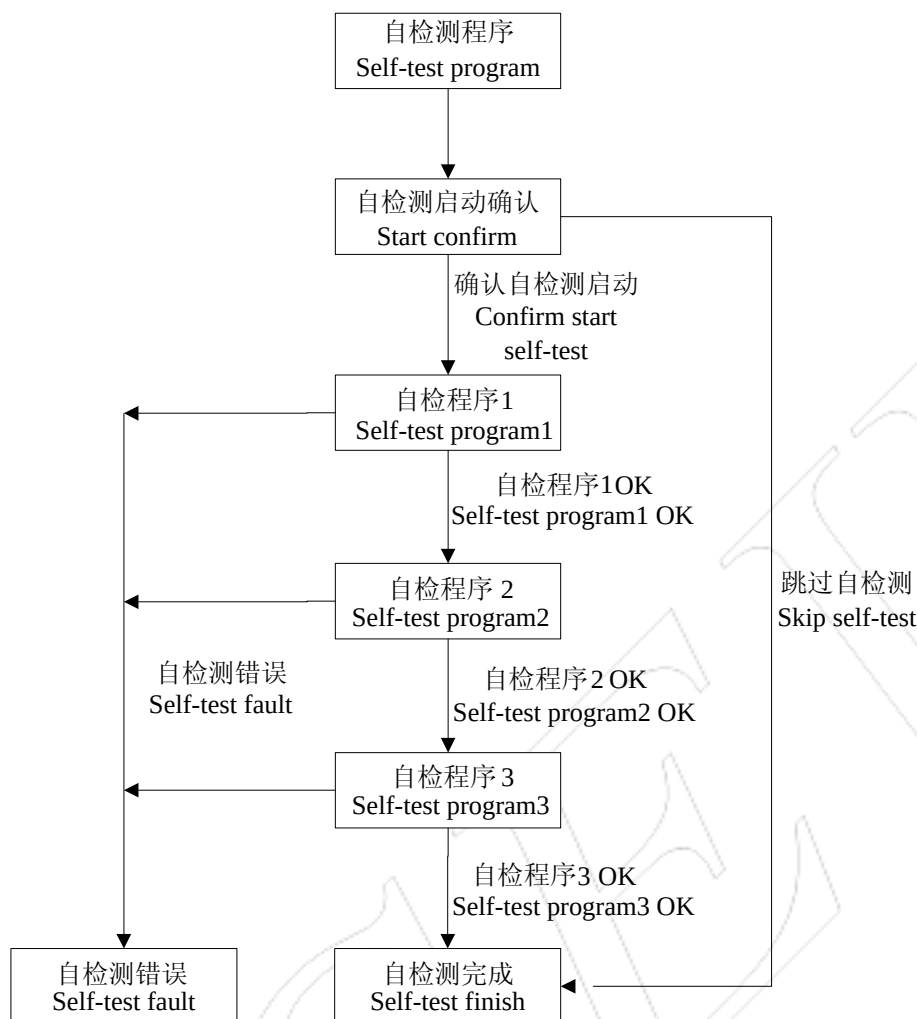


图 17 自检测流程图

Fig17 Flow chart of self-test mode

7.3.5.1 自检程序 1 Self-test program 1

- 1) 液压风轮制动阀关闭;
- 2) 如果在 20s 内, 平均桨距位置大于 72° , 状态代码“自检 1 故障”激活;
- 3) 如果在 20s 内, 平均桨距位置在 $70 \pm 2^{\circ}$ 范围内, 则进入自检程序 2。

1) Hydraulic rotor brake valves close;

2) State code 'self-test program 1 fault' will be activated if the average pitch angle exceed 72° in 20 seconds;3) Self-test program 2 will be entered when the average pitch angle within $70 \pm 2^{\circ}$ in 20 seconds.

7.3.5.2 自检程序 2 Self-test mode program2

- 1) 如果液压系统主压力小于 135bar 或大于 170bar, 或风轮转速大于 1rpm, 状态代码“自检 2 故障”激活;
- 2) 进入自检程序 3。

- 1) State code 'self-test program 2 fault' will be activated if the main pressure less than 135bar or exceeds 170bar, or the rotor speed exceed 1rpm;
- 2) Go to self-test program 3.

7.3.5.3 自检程序 3 Self-test mode program3

- 1) 液压风轮制动阀打开;
- 2) 检查每个转速信号的输入值, 要求:
 - 风轮转速 1 大于 0.1rpm
 - 风轮转速 2 大于 0.1rpm
- 3) 如果不满足上述条件, 则“自检 3 故障”激活;
- 4) 自检测完成。
 - 1) Hydraulic rotor brake valves open;
 - 2) Check the input of every speed signal:
 - Rotor speed 1 exceeds 0.1rpm
 - Rotor speed 2 exceeds 0.1rpm
 - 3) 'Self-test 3 fault' activated of the above conditions are not satisfied;
 - 4) Self-test finished.

7.3.6 空转模式 Idling mode

- 1) 设定最大变桨速度为 $10^{\circ}/s$, 设定桨距角最小值为 35° , 设定转速最大值为 2rpm;
- 2) 在 120s 内, 如果风轮转速大于 1.7rpm 持续 8s, 则进入启动模式;
- 3) 如果控制器在空转模式下超过 24 小时, 或者 10min 平均风速小于 3m/s, 则退出空转模式, 回到待机模式。
 - 1) Set the maximum pitch speed as $10^{\circ}/s$, set the minimum pitch angle as 35° , set the maximum rotor speed as 2rpm;
 - 2) If the rotor speed exceeds 1.7rpm continues 8s in 120 seconds, system goes to start mode;
 - 3) If the controller stay at idling mode exceeds 24 hours, or the average speed of 10 minute is less than 3m/s, the system will drop out idling mode and goes back to standby mode.

7.3.7 启动模式 Start mode

- 1) 设定风轮转速设定值=风轮转速+1;
- 2) 如果风轮转速大于风轮转速设定值, 则重复过程 2, 直至提升风轮转速至并网转速;
- 3) 控制风轮转速, 使得风轮转速介于切入转速 $\pm 0.5rpm$ 范围内, 维持 5s 之后, 控制器向变流器发出并网信号, 进入“切入发电机模式”;
- 4) 如果风轮转速小于切入转速持续 120s, 则退出启动模式转入空转模式;



5) 如果风轮转速介于共振频率转速上下限之间持续 20s, 则退出启动模式转入空转模式。

- 1) Set the setting value of rotor speed = rotor speed+1;
- 2) If rotor speed is larger than the rotor speed setting value, repeat process 2, until rotor speed reaches the speed that wind turbine can connect to the power grid;
- 3) Control rotor speed within the cut-in speed $\pm 0.5\text{rpm}$ last for 5s, then send grid-connection signal, to go to cut-in generation mode;
- 4) If rotor speed is less than the cut-in speed last for 120s, the start mode will be exited and the idling mode will be entered;
- 5) If rotor speed between upper and lower limits of the resonance speed last for 20s, the start mode will be exited and the idling mode will be entered.

7.3.8 切入模式 Cut-in mode

- 1) 设定最大变桨速度为 $10^\circ/\text{s}$;
- 2) 控制风轮转速在切入转速 $\pm 0.5\text{rpm}$ 范围内;
- 3) 控制器向变流器发出励磁指令, 变流器收到励磁指令后对发电机进行励磁, 并且检查主断路器或接触器两端的电压和频率, 确认符合并网条件后进行并网, 将并网信号反馈给控制器;
- 4) 控制器收到变流器发出的并网确认信号后, 确认该信号并且进入发电模式。

- 1) Set the maximum pitch speed as $10^\circ/\text{s}$;
- 2) Control rotor speed within the cut-in speed $\pm 0.5\text{rpm}$;
- 3) Controller sends excitation instructions to the converter, and generator will be excited by the converter after receiving the messages. Then the coverter check the voltage and frequency of between the main breaker or contactor to confirm whether the grid-connection conditions are satisfied. And it will control the wind turbine to connect the power grid and feedback grid-connection signals to the controller if the grid-connection conditions are satisfied
- 4) The wind turbine will go to generation mode after the controller receive the grid-connection confirm signal sent from converter.

7.3.9 发电模式 Generation mode

- 1) 设定最大变桨速度为 $10^\circ/\text{s}$, 桨距角位置和风轮转速设定值通过计算求得;
- 2) 根据转矩表向变流器输出转矩要求和功率因数;
- 3) 当风机出于发电模式下, 若以下条件任意一个满足则风机脱网并转入空转模式:
 - 10min 平均风速小于 3m/s , 持续 120s
 - 有功功率为负值, 持续 120s

- 风轮转速小于 8.3rpm
- 4) 如果风轮转速介于共振频率转速上下限之间持续 20s, 则进入共振模式。
 - 1) Set the maximum pitch speed as $10^{\circ}/s$, and the setting value of pitch position and rotor speed acquired by calculation;
 - 2) Export torque requirement and power factor to converter according to the torque table;
 - 3) Wind turbine will be separated from power grid and go to idling mode when any of the following conditions is satisfied:
 - Average wind speed of 10 minute less than 3m/s for 120 seconds
 - Active power value negative for 120 seconds
 - Wind rotor speed less than 8.3 rpm
 - 6) If the wind rotor speed lies between upper and lower limits of the resonant speed for 20s, the wind turbine will go to resonant mode.

7.3.10 共振模式 Resonant mode

- 1) 对于 W2000-93-70 的风机, 由于塔架共振频率不能避免, 因此需要加入共振模式, 避免风机长时间工作在塔架共振频率范围内;
- 2) 在共振模式下, 设定风轮转速设定值为共振频率转速下限, 最大变桨速度为 $10^{\circ}/s$;
- 3) 在共振模式下, 如果桨距角大于共振频率桨距角初始值持续 20s, 则退出共振模式。
 - 1) For W2000-93-70, the tower resonant frequency can't be avoided, so the resonant mode shall be added to avoid the wind turbine works within the resonant frequency for a long time;
 - 2) Set the rotor speed as the lower limit of the resonant frequency, and set the maximum pitch speed as $10^{\circ}/s$ at the resonant mode;
 - 3) Resonant mode will be exited if the pitch angle exceeds the initial pitch angle of resonant frequency for 20s at the resonant mode.

7.3.11 变桨润滑模式 Pitch-lubrication mode

- 1) 变桨润滑程序在待机模式、空转模式和发电模式下执行;
- 2) 在空转模式或发电模式下, 如果桨距角变化值超过 24h 小于 15° , 则执行变桨润滑程序。
 - 1) Pitch-lubrication mode is executed at standby mode, idling mode and generation mode;
 - 2) If the change of the pitch angle less than 15° exceeds 24h at idling mode or generation mode, pitch-lubrication program will be executed.

7.4 系统控制 System control

W2000 风机系统控制主要包括制动、变桨、偏航、液压及安全系统的控制。

The control system of the W2000 wind turbine including: breaking control, pitch control, yaw



control, hydraulic control, and safety system control.

7.4.1 制动系统 Braking system

制动系统是风力发电机组安全保障的重要环节，通过风轮刹车片制动，由液压系统执行。制动功能按照失效保护的原则进行，即失电时处于制动保护状态。

制动系统将制动程序分为三个级别：正常停机（1）、快速停机（2）、紧急停机（3）。此外，对轻级别的故障风机将会给出报警信号，在停机程序中定义为 0。

风机在运行过程中，如果在同一个时间激活很多个状态代码，那么风机将执行这些状态代码中级别最高的制动程序。

当某一个制动程序执行时间超过设定值时，将启动高一级别的制动程序，这样就避免了制动失败时对风机造成的影响，提高了风机安全的可靠性。当正常停机程序或快速停机程序失败时，将启动紧急停机程序。

Braking system is an important link for wind turbine security, it works through rotor brake pad and implements with hydraulic system. The brake function is implemented according to failure protection principle, and it means the system will be at brake protection state when the electricity is cut off.

Brake program is divided into three levels: normal stop(1), fast stop(2) and emergency stop(3). Besides, the warning signals will be given out when lower level fault occurred, which is defined as 0 in the stop program.

If many state codes are activated at one time during wind turbine operation, the wind turbine will execute the highest level brake program between these state codes.

Higher brake program will be activated when the execution time exceeds the setting value, to avoid the impact of wind turbine caused by brake failure, and improve the reliability of the wind turbine safety. Emergency stop program will be activated if normal stop program and fast stop program failure.

7.4.1.1 正常停机 Normal stop

当出现温度越限、避雷器故障、振动故障、齿轮箱故障、发电机电刷故障、执行设备保护信号断开、风速计或风向标失效等故障时系统将执行正常停机。正常停机的流程图如图 19 示。

1) 桨叶以 4 %s 的速度变桨至 89°;

2) 当风轮转速低于 8.3rpm 时或瞬时输出功率小于 0kW(延时 2s)或 30s 平均输出功率小于 0kW(延时 0.5s)或 10m 平均输出功率小于 0kW(延时 0s)时，变流器脱网；

3) 当风轮转速低于 2rpm 时，正常停机结束。

The Normal stop is triggered if any of the following fault occurred: temperature exceeding, lightning arrester fault, oscillation fault, gearbox fault, generator brush fault, protection signals of

implementing device break, anemometer or vane fault, etc. The flow chart of normal stop is shown as fig19.

- 1) The blades rotated into 89° with the speed of 4 %s
- 2) If the rotor speed less than 8.3rpm, or the instantaneous power less than 0kw (delay 2s), or average power of 30 seconds less than 0kw(delay 0.5s), or average power of 10m less than 0kw(delay 0s), the converter will disconnect from the grid.
- 3) When the rotor speed less than 2rpm, normal stop is finished.

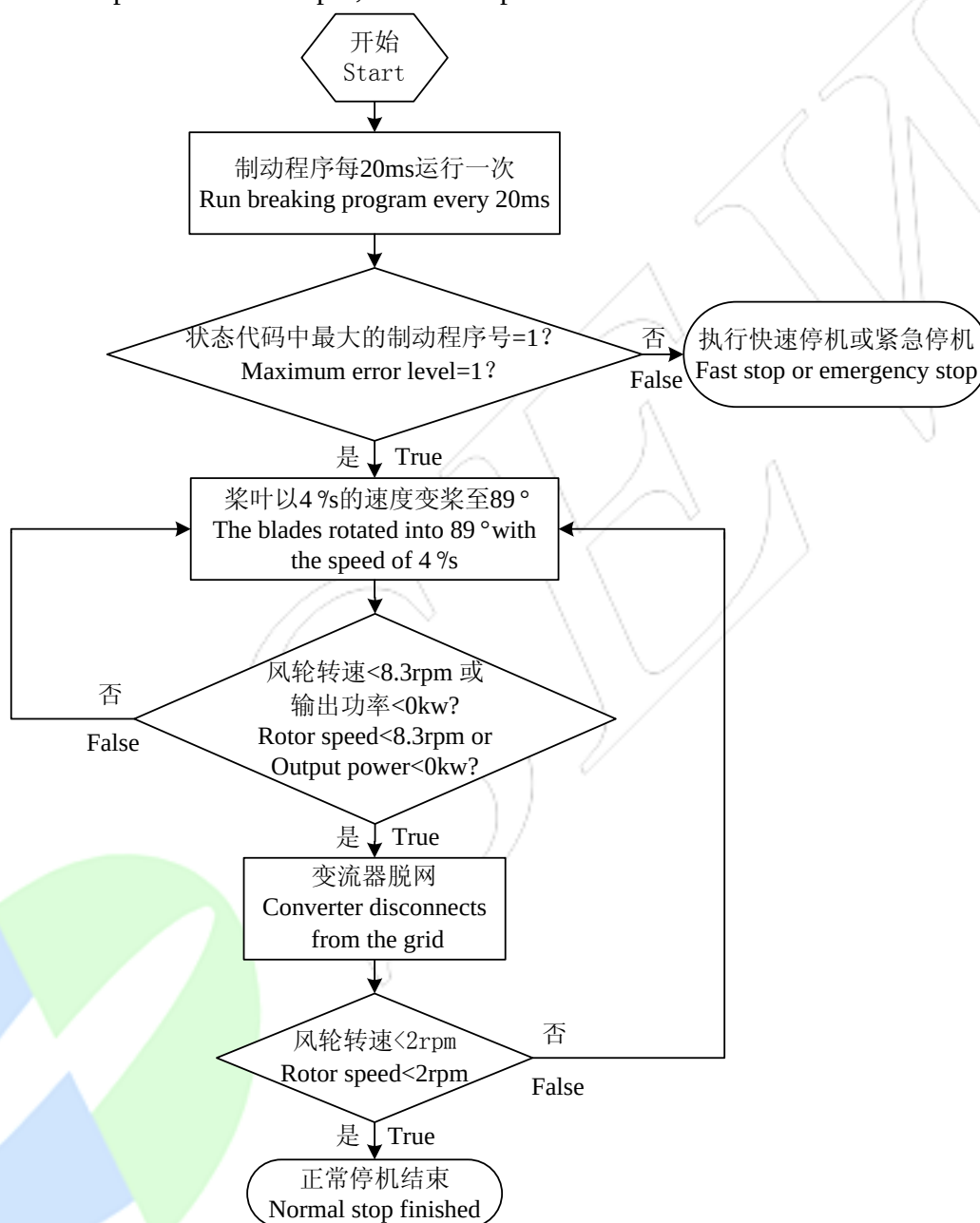


图 18 正常停机流程图

Fig18 Flow chart of normal stop



7.4.1.2 快速停机 Fast stop

当出现电网故障、通信故障、变流器并网断路器故障、熔断器故障、液压压力故障、风轮转速超限等故障时系统将执行快速停机。快速停机的流程图如图 20 示。

The Fast stop is triggered if a grid fault, a communication fault, a convertor cut-in fault, a fuse protector fault, a hydraulic pressure fault or a rotor speed over limitation fault occurs. The Fast stop runs as the flow chart 20.

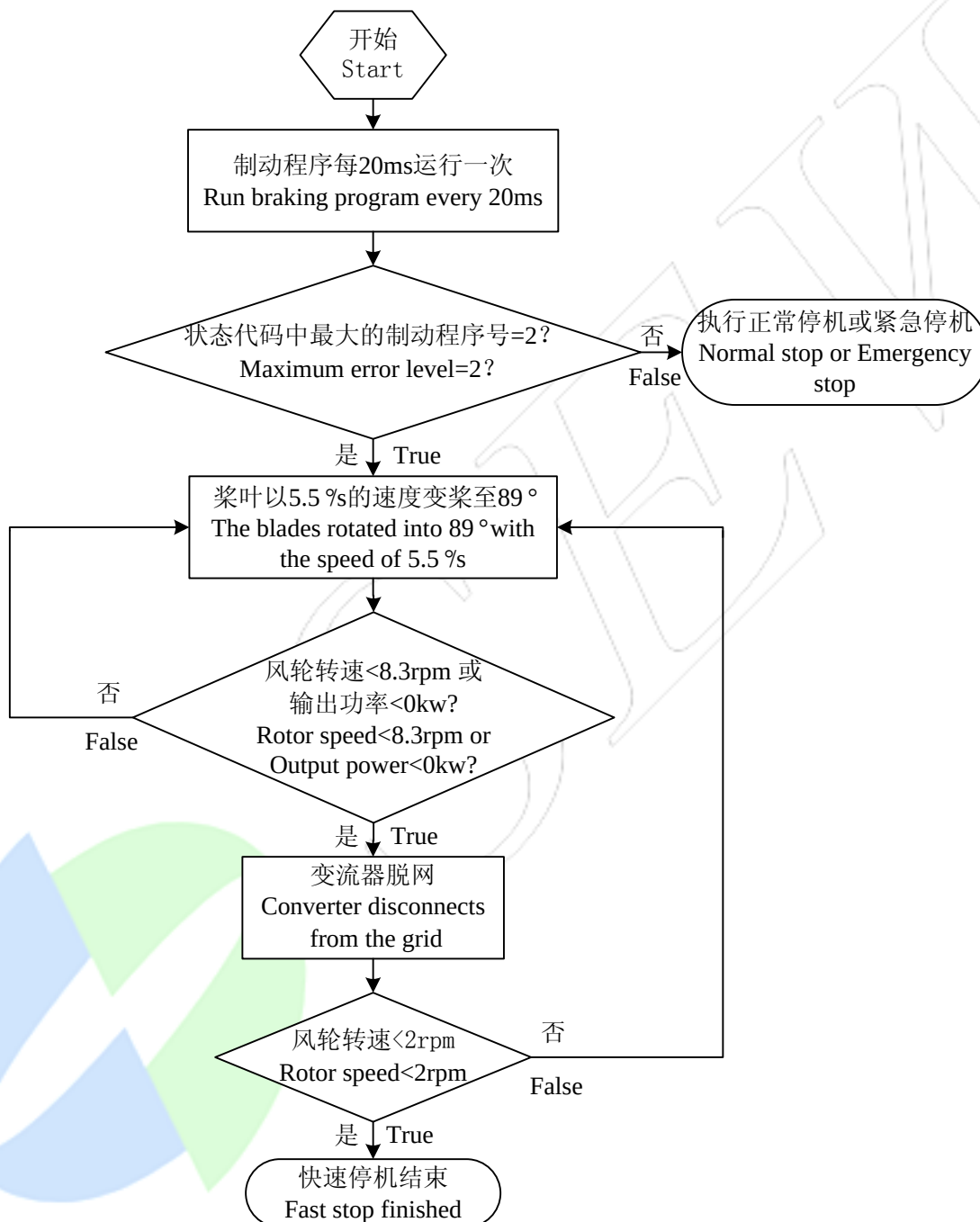


图 19 快速停机流程图

Fig 19 Flow chart of Fast stop

- 1) 桨叶以 5.5 %s 的速度变桨至 89°;
 - 2) 当风轮转速低于 8.3rpm 时或瞬时输出功率小于 0kW(延时 2s)或 30s 平均输出功率小于 0kW(延时 0.5s)或 10m 平均输出功率小于 0kW(延时 0s)时, 变流器脱网;
 - 3) 当风轮转速低于 2rpm 时, 快速停机结束。
- 1) The blades are rotated in the speed of 5.5 %s into the position of 89°.
 - 2) When the rotor speed is smaller than 8.3rpm, or instantaneous output power is less than 0kW (delay 2s), or mean output power of 30s is less than 0kW (delay 0.5s), or mean output power of 10m is less than 0kW (delay 0s), the convertor disconnects from the grid.
 - 3) When the rotor speed is less than 2rpm, Fast stop finishes.

7.4.1.3 紧急停机 Emergency stop

当出现变流器并网故障、变桨系统故障、风轮转速超限、安全链断开、制动程序 1 或 2 超时等故障时系统将执行紧急停机。紧急停机的流程图如图 21 示。

The Emergency stop is triggered if a convertor cut-in fault, a pitch system fault or a speed exceeding fault occurs, or safety chain is open, or parking program 1 or 2 runs out of time. The emergency stop runs as the flow chart 21.

- 1) 变流器脱网
 - 2) 桨叶以 7 %s 的速度变桨至 89°;
 - 3) 当风轮转速小于 5rpm 时, 液压系统执行风轮制动;
 - 4) 当风轮转速低于 2rpm 时, 快速停机结束。
- 1) The convertor disconnects from the grid;
 - 2) The blades are rotated in the speed of 7 %s into the position of 89°;
 - 3) When the rotor speed is smaller than 5rpm, rotor breaks by hydraulic system;
 - 4) When the rotor speed is smaller than 2rpm, Emergency stop finishes.

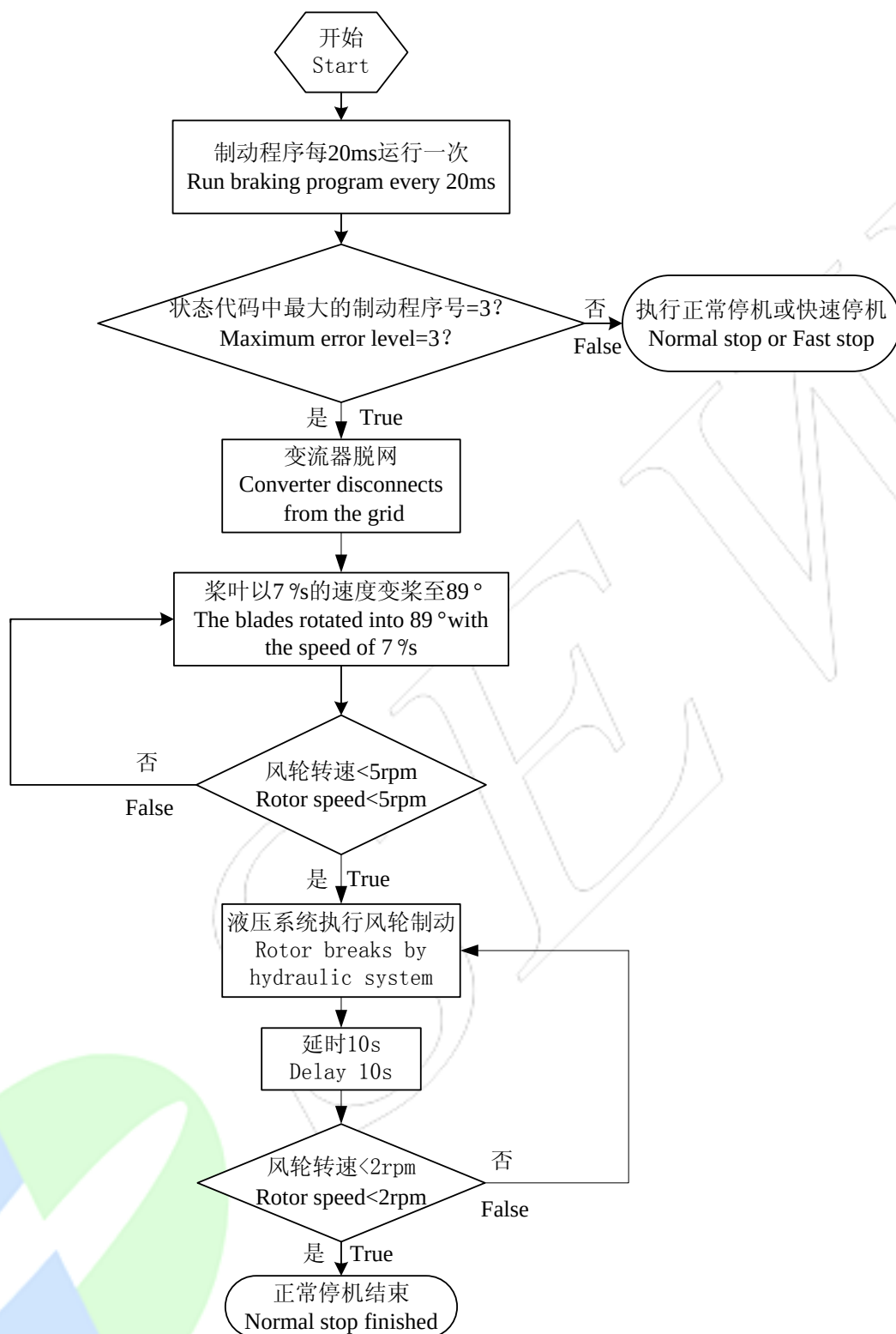


图 20 紧急停机流程图

Fig 20 Flow chart of Emergency stop

7.4.2 变桨系统 Pitch system

变桨控制系统的状态有：准备好，运行，停止，维护，紧停，报警。

在变桨控制过程中，将变桨控制器的控制区域划分为三块，如图 22 所示：其中控制区域 I 为低风速区；控制区域 II 为中风速区；控制区域 III 为高风速区，在此区域中，负载处于满载，达到额定功率状态。

The pitch control system has six statuses: ready, run, stop, service, emergency stop and warning.

In the pitch control procedure, the control area of the pitch controller is divided into three pieces, shown as Fig22. Zone I is the low wind speed zone; zone II is the middle wind speed zone; zone III is the high wind speed zone. In the high wind speed area, the wind turbine runs at full load, and the generator outputs the rated power.

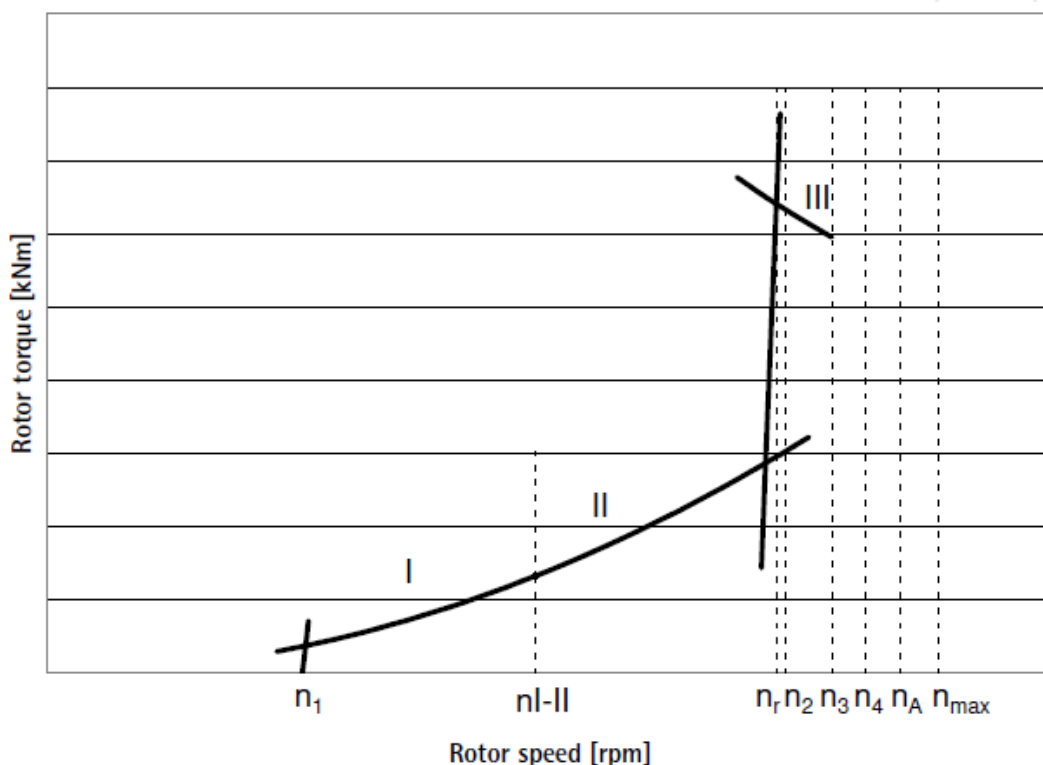


图 21 变桨控制器控制区域

Fig.21 The pitch controller operating zones

在控制区域 I(低风速区)变桨控制器未激活，变桨角度维持在一个最小的桨矩角上。在控制区域 III(高风速区)，风轮转速保持在额定转速，此时的桨矩角一般在桨矩控制器控制的桨矩角的 $\pm 10\%$ 范围内。在控制区域 II(中风速区)，对进行风轮转速进行 PID 控制。

进行变桨之前，预先设定变桨加速度和速度，并检测故障信号是否复位，变桨充电器电池是否报错，若一切正常，则变桨系统进入普通模式，调用变桨控制器。据当前的风速来查表插值计算变桨目标位置进行变桨。变桨控制流程图如图 23 所示。风机复位或者风机初始化，变桨系统进入普通模式，设定变桨目标位置。

In zone I (low wind speed zone), the pitch controller is not activated, and the pitch angles are kept at the minimum pitch angle. In zone III (high wind speed zone), the rotor speed is kept at the rated speed. The pitch angle in zone III is kept within $\pm 10\%$ of pitch angle set by the pitch controller. In

zone II (middle wind speed zone), the rotor speed is controlled by PID.

Before rotating blades, the pitch acceleration and the pitch speed is set, and whether the fault signals are reset and whether there are faults in the pitch chargers and batteries is detected. If everything is OK, the pitch system turns into the normal mode, and the pitch controller is activated. Blades are rotated into the position calculated by the table-look-up interpolation according to the wind speed. The pitch control flow chart is shown as Fig.23. After the wind turbine is reset or initialized, the pitch system turns into the normal mode and sets the target pitch angle.

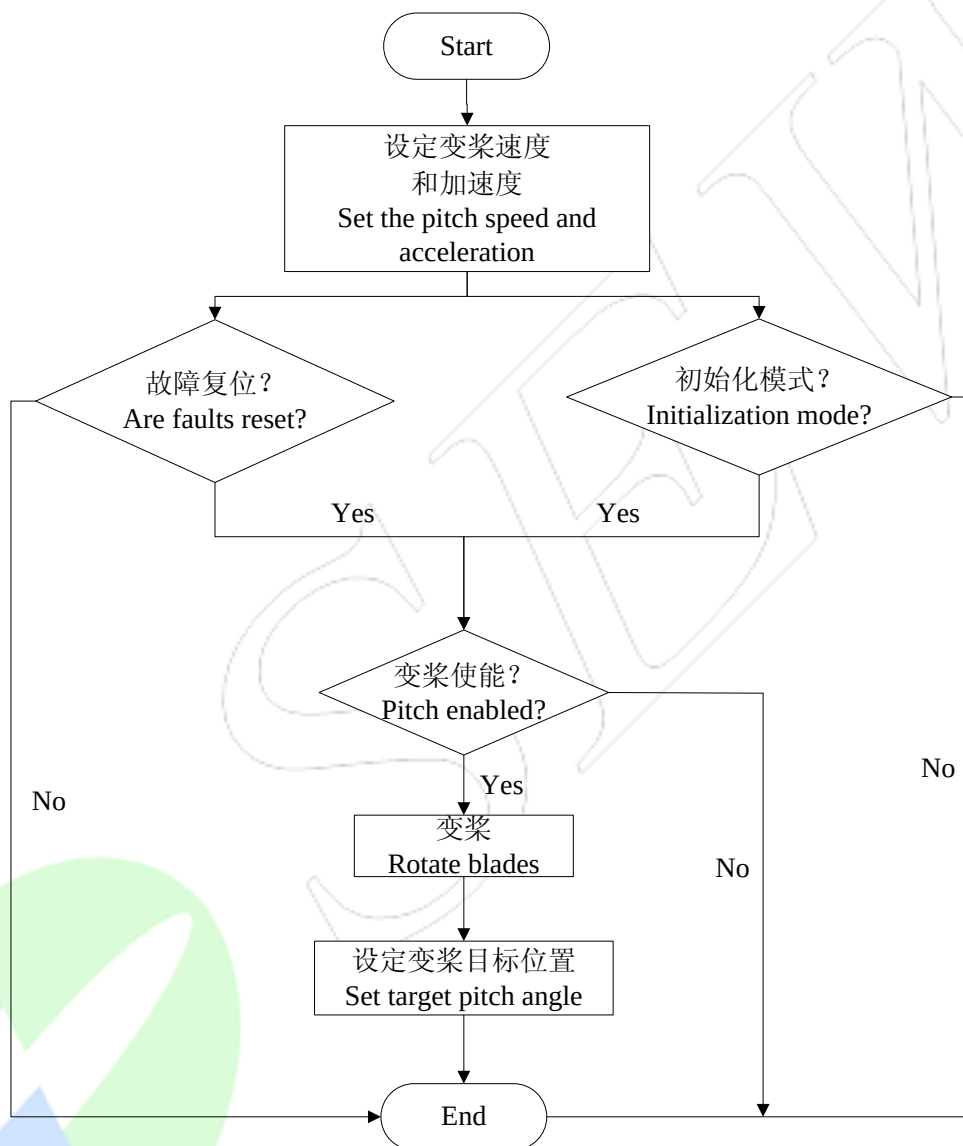


图 22 变桨控制流程图

Fig.22 Flow chart of pitch control

桨距控制的主要目的在于通过对转速偏差的 PID 调节来良好地控制风轮转速。如图 24 所示的变桨控制器原理，首先为防止变桨过程中风轮转速过速，通过 10s 平均风速风况下的最小桨距角；然后通过实际转速与转速设定值的偏差，分别计算 PID 各环节的输出；之后根据当前桨距角位置求得变桨非线性因数；通过变桨非线性因数与 PID 调节器的输出，决定变桨速度和桨距位置

设定点；最后与当前风况下的最小桨距角相比较，输出桨距位置设定点。

The main purpose of pitch control is to control rotor speed well by PID regulating speed difference. The principle of the pitch controller is shown as Fig.24. Firstly, mean wind speed of 10s is input to calculate the minimum pitch angle to prevent rotor speed from exceeding when rotating blades. Secondly, outputs of every PID node are calculated separately according to the difference between actual and set rotor speed. Next, nonlinear factor is obtained from the current pitch angle. Then pitch speed and target pitch angle is determined by nonlinear factor and output of PID regulator. At last, pitch angle is output by comparing target pitch angle and minimum pitch angle.

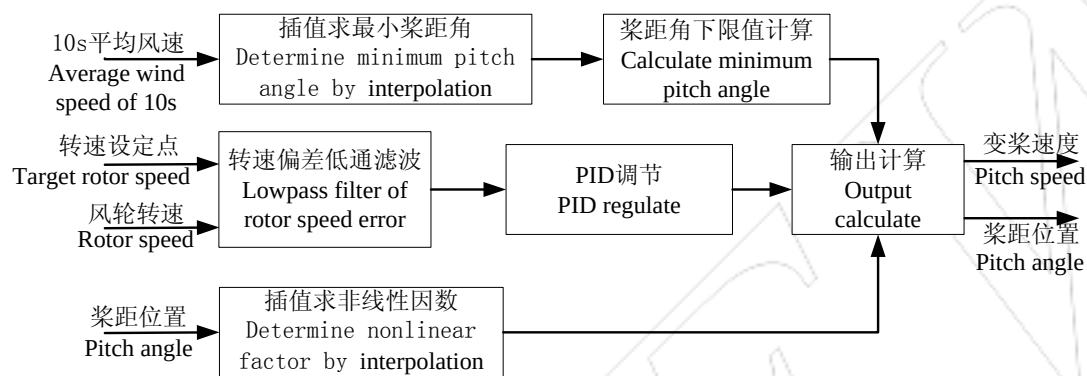


图 23 变桨控制原理图
Fig.23 Principle of pitch control

7. 4. 3 偏航系统 Pitch system

机舱与塔架之间安装有偏航系统，偏航调节的目的是使风力发电机组的叶轮始终能够正对风向，从而使风力发电机捕获最大的风能，同时也可以降低风力发电机组所承受的疲劳载荷，延长风机内各种构件的寿命。

Yaw system is installed between the nacelle and the tower. The purpose of yaw control is to ensure the blades always facing the prevailing wind direction, in order that the wind turbine can obtain maximum wind energy. Meanwhile, the fatigue load received by the wind turbine can be reduced, to lengthen the lives of the components in the wind turbine.

7. 4. 3. 1 自动偏航 Automatic yaw

7. 4. 3. 1. 1 自动偏航启动 Start of automatic yaw

在自动偏航下，控制系统将自动计算机舱位置与风向位置的偏差，并且进行修正。除了下面的状态和程序，自动偏航都可以执行：

- 1) 偏航全局故障
- 2) 偏航解缆故障
- 3) 风向标故障



- 4) 液压系统故障
- 5) 服务模式激活
- 6) 30s 平均风速小于偏航风速最小值(2.5m/s)
- 7) 安全系统故障

如果左偏航与右偏航的条件同时满足, 风机将不会执行自动偏航。

During automatic yaw, control system will calculate the difference between the nacelle position and the wind direction automatically, and yaw to the right position.

Automatic yaw will be executed excepted during the statuses and the procedures given below:

- 1) Global yaw fault
- 2) Cable unwinding fault
- 3) Wind vane fault
- 4) Hydraulic system fault
- 5) Service mode activated
- 6) Mean wind speed of 30s is less than minimum yaw wind speed (2.5m/s)
- 7) Safety system fault

If the condition of both yawing right and yawing left is met, the wind turbine will not yaw automatically.

7.4.3.1.2 自动偏航的方向 Direction of automatic yaw

1) 设定偏航控制参数

如果 60s 平均风速持续 25s 大于设定值则判断为高风速情况, 选择针对高风速情况的偏航控制参数。如果 60s 平均风速持续 25s 小于或等于设定值, 则判断为低风速情况, 采用针对低风速情况的控制参数。

2) 参考风向标的标准位置 180 度, 将 60s 风向平均值和所设定的偏差进行比较。如果风向标的平均值在一定时间内超过设定值, 则启动偏航电机进行偏航。

- 左偏航: 满足如下条件之一时, 风机进行左偏航。

- 当机组允许自动偏航时, 若 60s 平均风向持续 210s 低于左偏航风向设定值 $1(8^{\circ})$, 即风向与机舱风向标的夹角在区域 1 所示范围内时, 机组进行左偏航。

- 当机组允许自动偏航时, 若 60s 平均风向持续 20s 低于左偏航风向设定值 $2(15^{\circ})$, 即风向与机舱风向标的夹角在区域 3 所示范围内时, 机组进行左偏航。

- 右偏航: 满足如下条件之一时, 风机进行右偏航。

- 当机组允许自动偏航时, 若 60s 平均风向持续 210s 高于右偏航风向设定值 $1(8^{\circ})$, 即风向与机舱风向标的夹角在区域 2 所示范围内时, 机组进行右偏航。

- 当机组允许自动偏航时, 若 60s 平均风向持续 20s 高于右偏航风向设定值 $2(15^{\circ})$,

即风向与机舱风向标的夹角在区域 4 所示范围内时，机组进行右偏航。

3) 当风向与机舱风向标的夹角在区域 0 所示范围内时，系统发出停止偏航命令，机舱停止偏航。

1) Set yaw control parameter

The condition when the mean wind speed of 60s keeps bigger than the set value for 25s will be regarded as high wind speed condition, and in this condition, high wind speed yaw control parameter will be chosen. The condition when mean wind speed of 60s keeps smaller than the set value for 25s will be regarded as low wind speed condition, and in this condition, low wind speed yaw control parameter will be chosen.

2) The mean wind direction of 60s is compared with the set wind direction, according to the reverse wind vane standard direction. If the average value of the wind vane keeps bigger than the set value, the yaw electromotor will start working.

- Yaw left: The wind turbine will yaw left, if one of the conditions given below is met:

- When the wind turbine allows automatic yaw, if the mean wind direction angle of 60s keeps smaller than the set angle 1 (8°) for 210s, namely the angle between the wind direction and the wind vane direction is in zone 1 shown in the figure, the nacelle yaw left.

- When the wind turbine allows automatic yaw, if the mean wind direction angle of 60s keeps smaller than the set angle 2 (15°) for 20s, namely the angle between the wind direction and the wind vane direction is in zone 3 shown in the figure, the nacelle yaw left.

- Yaw right: The wind turbine will yaw right, if one of the conditions given below is met:

- When the wind turbine allows automatic yaw, if the mean wind direction angle of 60s keeps bigger than the set angle 1 (8°) for 210s, namely the angle between the wind direction and the wind vane direction is in zone 2 shown in the figure, the nacelle yaw right.

- When the wind turbine allows automatic yaw, if the mean wind direction angle of 60s keeps bigger than the set angle 2 (15°) for 20s, namely the angle between the wind direction and the wind vane direction is in zone 4 shown in the figure, the nacelle yaw right.

3) If the angle between the wind direction and the wind vane direction is in zone 0 shown in the figure, the control system will give stop yawing command, and the nacelle will stop yawing.

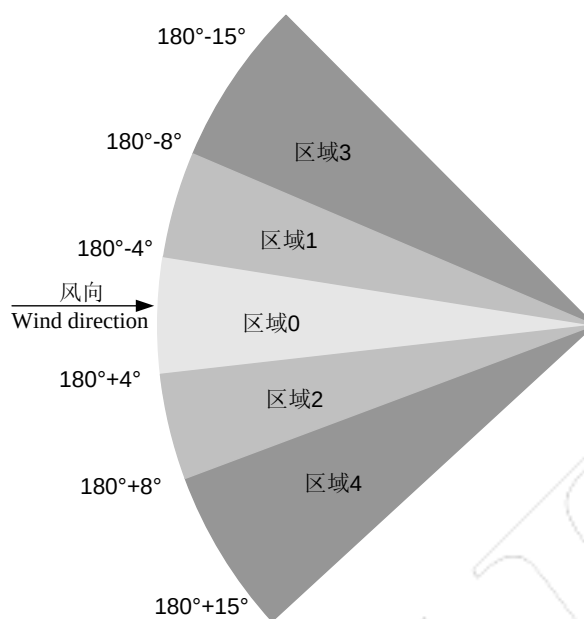


图 24 风机偏航方向示意图

Fig.24 Schematic diagram of yaw direction

7.4.3.2 手动偏航系统 Manual yaw

只要偏航系统、安全系统和液压系统未出现故障，在风机运行的各个状态都可以通过机舱控制柜门和 HMI 对风机进行手动偏航操作。

1) 机舱控制柜门上的手动偏航操作包括：

- 左偏航
- 右偏航

在松开上述按钮后，风机停止偏航。

2) HMI 上的手动偏航操作包括：

- 左偏航
- 右偏航
- 偏航停止
- 切换回自动偏航状态

If only yaw system fault, safety system fault and hydraulic system fault don't occur, manual yaw can be operated on the wind turbine though the door of the control cabinet in the nacelle and HMI during each status of the wind turbine running.

1) The manual yaw operations can be done though the door of the control cabinet in the nacelle includes:

- Yaw left
- Yaw right

2) The manual yaw operations can be done though HMI includes:\

- Yaw left

- Yaw right
- Stop yawing
- Turn back to automatic yaw status

7.4.3.3 偏航解缆 Cable unwinding

当偏航扭缆超过一定角度后，需要进行解缆，防止由于电缆扭转而导致断裂。

If the nacelle yaws more than a certain angle, Cable unwinding is needed to prevent cables from twisting and causing broken.

7.4.3.3.1 偏航解缆启动 Start of cable unwinding

当偏航位置的绝对值大于 710.0° 时，控制器延时 4s 输出扭缆信号，风机停机执行解缆操作。
解缆信号：

- 1) 当偏航位置大于 580.0° 时进行顺时针解缆；
- 2) 当偏航位置小于 -580.0° 时进行逆时针解缆。

If the absolute value of yaw angle is bigger than 710.0° , the controller will output cable unwinding signal, and then the wind turbine will stop to unwind cable. The cable unwinding signal will be triggered if:

- 1) When the yaw angle is bigger than 580.0° , the wind turbine will unwind cable clockwise;
- 2) When the yaw angle is smaller than -580.0° , the wind turbine will unwind cable anticlockwise;

7.4.3.3.2 偏航解缆结束 End of cable unwinding

当下述条件其中之一满足时，偏航解缆完成：

- 1) 风机未处于停止模式或服务模式下；
- 2) 系统出现全局故障或液压系统故障；
- 3) 顺时针解缆和逆时针解缆同时激活；
- 4) 偏航位置 $< 280.0^{\circ}$ 并且满足下述条件之一：
 - 偏航位置的绝对值 $< 40.0^{\circ}$ ；
 - 机舱与风向偏差的绝对值 $< 30.0^{\circ}$ 延时 10s。

Cable unwinding finishes if one of the conditions given below is met:

- 1) The wind turbine is not in stop mode or service mode;
- 2) Global or hydraulic system fault occurs;
- 3) Both clockwise untwisting and anticlockwise cable unwinding are triggered at the same time;
- 4) Yaw angle $< 280.0^{\circ}$; and one of the conditions given below is met:
 - The absolute value of yaw angle $< 40.0^{\circ}$;
 - The absolute value of the difference between the nacelle and the wind direction $< 30.0^{\circ}$; and



delay 10s

7.4.4 液压系统 Hydraulic system

7.4.4.1 液压风轮制动 Hydraulic rotor brake

液压风轮制动（亦称高速轴刹车）安装在驱动链的高速轴侧，是风机的二级刹车系统，通过液压控制来实现。液压风轮制动的主要功能是使风轮减速直至完全静止。在服务模式下，可以手动控制风轮制动使风轮静止；在紧急停机状态下，当风轮转速低于允许抱闸的设定值，液压风轮制动动作，使风轮快速静止。

当发生以下两种情况时，液压风轮制动抱闸。其它情况下，液压风轮制动都处于松闸状态。

1) 当系统执行紧急停机，并且主流程进入停机模式，且风轮转速小于允许风轮制动的风轮转速设定值的情况下，风轮制动抱闸。

2) 当服务开关激活，主流程进入服务模式时，且风轮转速小于允许风轮制动的风轮转速设定值的情况下，对风轮制动进行手动操作将制动闸关闭。

Hydraulic rotor brake (also called rotor brake caliper) integrated the high speed side of the drive train, is the secondary braking system. The brake works by hydraulic system. The primary function of Hydraulic rotor braking system is to bring the rotor to a complete stand-still. The Hydraulic rotor brake can be used to keep the rotor in a complete stand-still in the service mode. The Hydraulic rotor brake is also used during emergency stops to decelerate the wind turbine as quickly as possible after it has been decelerated to idling by the pitch system.

The Hydraulic rotor brake works if one of two conditions given below occurs. Otherwise, the hydraulic rotor brake doesn't work:

1) When the emergency stop program runs, the system has turned into stop mode, and rotor speed is smaller than the set speed which permits rotor braking, the rotor brake works.

2) When the service switch is on, the system has turned into service mode, and rotor speed is smaller than the set speed which permits rotor braking, manual rotor brake operation will cause the rotor brake working.

7.4.4.2 液压偏航制动 Hydraulic yaw brakes

液压偏航制动安装在偏航制动盘上，当风机运行需要自动偏航对风时或手动偏航时，液压偏航制动阀部分打开(留有备压)，液压偏航制动部分动作；当风机需要 90° 偏航、自动解缆或偏航润滑时，液压偏航制动阀全部打开，液压偏航制动全部动作。在其它情况下，液压偏航制动处于抱闸状态。

The hydraulic yaw brakes are integrated on the yaw brake disc. When the nacelle is yawing automatically or manually, the hydraulic yaw brake valves are partly open (still remain a certain pressure), the brakes are partly working. When the nacelle is yawing to 90°, unwinding cable or

lubricating yaw bearing, the brake valves are fully open and the brakes are fully working. In other circumstances, the hydraulic yaw brakes are fully closed.

7.4.4.3 液压泵马达控制 Control of hydraulic motor

液压泵控制的功能是保证液压系统的系统压力维持在设定的压力范围之内，当系统压力低于液压泵启动压力设定值(145bar)时，液压泵开启使系统压力上升；当系统压力高于液压泵停止压力设定值(160bar)时，液压泵停止使系统压力下降。

The hydraulic motor is controlled to ensure the pressure of the hydraulic system maintains in the given pressure range. When the system pressure is lower than the motor starting value (145bar), the motor is opened to increase the pressure. When the system pressure is higher than the motor stopping value (160bar), the motor is closed to decrease the pressure.

7.4.5 安全系统 Safety system

2MW 风机的安全系统通过安全链实现，安全链是独立于控制器的硬件保护措施，它总是优先于控制系统，即使控制系统发生异常，也不会影响安全链的正常动作。安全链采用反逻辑设计，将可能对风机造成致命伤害的重要机构部件的控制端串联成一个回路。当回路中任意一个触点发生故障，安全链断开，引起紧急停机，风机瞬间脱网，从而最大限度的保证风机的安全。

安全链中串接的安全信号通过安全输入端子和安全输出端子和 CPU 进行数据交互，安全系统的逻辑功能则通过安全逻辑端子实现。

安全链中串接的信号包括：

- 1) 塔底急停按钮：安装于塔底控制柜面板上；
- 2) 塔底主电源故障(400VAC)：通过相位监测器监测电网电压，当电网电压过低或过高时发出主电源故障信号；
- 3) 机舱紧停按钮：安装于机舱控制柜面板上；
- 4) 机舱远程紧停按钮：安装于机舱紧停按钮盒上；
- 5) 风轮超速 1 紧停：风轮转速超过 18.5rpm，超速模块发出信号；
- 6) 风轮超速 2 紧停：风轮转速超过 19rpm，超速模块发出信号；
- 7) 机舱振动紧停：机舱振动过大，机舱振动开关断开；
- 8) 变桨系统紧停：变桨系统安全链断开，通过断开安全继电器给控制器变桨紧停信号；
- 9) 偏航扭缆紧停：偏航扭缆限位开关动作；
- 10) 变流器紧停按钮：安装在变流器控制柜面板上；
- 11) 变流器并网开关故障：变流器并网开关失效。

安全链控制的流程图如图 26 所示。

The safety system of the 2MW wind turbine is implemented by safety chain. The safety system is the hardware protecting measure independent of control system, and will always have a higher priority



than the control system, so that even if faults occur in the control system, the safety chain works normally. The safety chain is designed by negative logic. The safety chain is combined with signals of the important components which may cause fatal damages in series. When a contact in the safety chain is opened, the safety chain will be opened, causing emergency stop and the wind turbine disconnecting from the grid, in order to guarantee the maximum safety of the wind turbine.

The safety signals combined in the safety chain interact with the data in CPU through input and output safety terminal. The logic function of the safety system is implemented by the safety logic terminal.

The signals combined in the safety chain are given below:

- 1) Emergency stop button at tower base: installed on the door of the control cabinet of tower base;
- 2) The main power fault of tower base (400VAC): the main power fault signal is triggered if the power of the grid monitored by the phase monitor is too low or too high;
- 3) Emergency stop button in the nacelle: installed on the door of the control cabinet of the nacelle;
- 4) Remote emergency stop button: installed on the button box of the nacelle;
- 5) Emergency stop of rotor speed exceeding 1: the signal is triggered if the rotor speed exceeds 18.5rpm;
- 6) Emergency stop of rotor speed exceeding 2: the signal is triggered if the rotor speed exceeds 19rpm;
- 7) Emergency stop of nacelle vibrating: the vibrating switch is open when the nacelle over vibrating;
- 8) Emergency stop of pitch system: the signal is triggered by open the safety relay if the safety chain of the pitch system is opened;
- 9) Emergency stop of cable unwinding: the cable unwinding limit switch is opened;
- 10) Emergency stop button of the convertor: installed on the panel of the control cabinet of the convertor;
- 11) Convertor connection fault: the convertor cut-in switch invalids.

The flow chart of the safety chain controlling is shown as Fig.26:

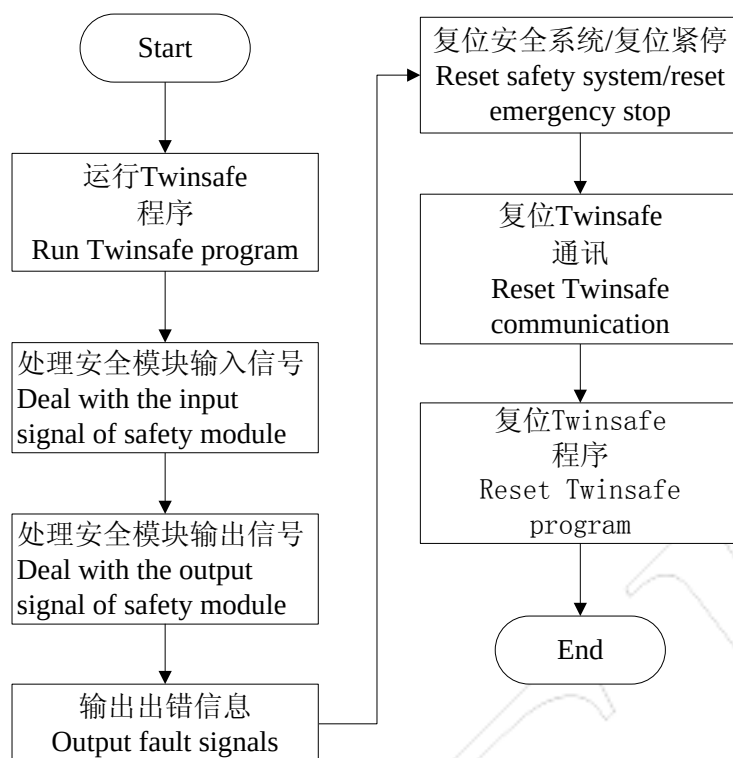


图 25 安全链控制流程图

Fig.25 Flow chart of the safety chain controlling

安全链断开后，只能通过塔底控制柜门上的复位按钮进行复位，机舱柜门上的复位按钮和远程复位都不能对安全链进行复位。

After the safety chain is opened, only the reset button on the door of the control cabinet of tower base is available to reset the safety chain, and the reset button on the door of the control cabinet of the nacelle and remote reset are unable to reset the safety chain.

7.5 电网检测 Grid monitoring

电网监测通过用于电力管理的倍福现场总线端子实现，主要对电网电压、电流、频率、有功功率进行监测，当上述量超出允许运行范围时，通过触发相应的状态代码执行不同的停机程序，以保证风机的正常运行。电网无功功率则是通过电网有功功率和功率因数计算获得的。

Grid monitoring is implemented by Beckhoff fieldbus terminal for electric power management monitoring voltage, current, frequency and active power of the grid. If these value runs out of operating range, corresponding state codes will be triggered causing corresponding stop procedure, to guarantee the normal operating of the wind turbine. Reactive power of the grid is calculated by active power and power factor of the grid.

7.5.1 电网电压检测 Grid voltage monitoring

- 1) 电网电压限值监测：见状态代码 100111~100113 和 100211~100213。
- 2) 电网电压不对称监测：见状态代码 100130。



1) Grid voltage limit monitoring: state code 100111~100113 and 100211~100213.

2) Grid voltage asymmetry monitoring: state code 100130.

7.5.2 电网电流检测 Grid current monitoring

1) 电网电流限值监测：见状态代码 100211~100213。

按下式计算电网电流极限值。

$$gGridIMax = \text{Min}[(gGridP + 1000), gInitGridIMax, 2100]$$

其中，gInitGridIMax 为电网电流初始设定最大值(A)，gGridP 为电网功率(kW)。

2) 电网电流不对称监测：见状态代码 100220。

1) Grid current limit monitoring: state code 100211~100213.

Grid current limit is calculated by the formula below:

$$gGridIMax = \text{Min}[(gGridP + 1000), gInitGridIMax, 2100]$$

gInitGridIMax represents the initial maximum grid current (A), and gGridP represents the grid power (kW).

2) Grid current asymmetry monitoring: state code 100220.

7.5.3 电网频率检测 Grid frequency monitoring

电网的频率为 A、B、C 三相频率的平均值。见状态代码 100311~100313 和 100321~100323。

Grid frequency is the mean frequency of phase A, phase B and phase C: state code 100311~100313 and 100321~100323.

7.5.4 电网有功功率检测 Grid active power monitoring

1) 电网有功功率监测：

- 电网实时有功功率监测：见状态代码 100410。

电网实时有功功率(gGridP)为 A、B、C 三相有功功率之和。

- 电网 30s 平均功率监测：见状态代码 100411。

- 电网 10m 平均功率监测：见状态代码 100412。

2) 电网功率差值监测：

比较电网实时有功功率和变流器测得的有功功率的差值的绝对值是否超限。见状态代码 100420。

1) Grid active power monitoring:

- Grid realtime active power monitoring: state code 100410.

Grid realtime active power (gGridP) represents the sum of the active power of phase A, phase B, phase C.

- Grid mean power of 30s monitoring: state code 100411.

- Grid mean power of 10m monitoring: state code 100412.

2) Grid power difference monitoring:

Whether the absolute value of the difference between the realtime active power of the grid and the convertor is over limit is monitored: state code 100420.

7.5.5 电网无功功率检测 Grid reactive power monitoring

电网 A、B、C 三相无功功率的计算公式如下所示:

$$GridQ1 = \sqrt{\left(\frac{gGridP1}{\max[\cos(gGrid \cos 1), 0.0001]} \right)^2 - (gGridP1)^2} \times |gGrid \cos 1|$$

$$GridQ2 = \sqrt{\left(\frac{gGridP2}{\max[\cos(gGrid \cos 2), 0.0001]} \right)^2 - (gGridP2)^2} \times |gGrid \cos 2|$$

$$GridQ3 = \sqrt{\left(\frac{gGridP3}{\max[\cos(gGrid \cos 3), 0.0001]} \right)^2 - (gGridP3)^2} \times |gGrid \cos 3|$$

其中 gGrid cos1, gGrid cos 2, gGrid cos3 分别为电网 A 相、B 相和 C 相的功率因数。电网无功功率为 A、B、C 三相无功功率的和。

The formulas to calculate the grid reactive power of phase A, phase B, phase C are given below:

$$GridQ1 = \sqrt{\left(\frac{gGridP1}{\max[\cos(gGrid \cos 1), 0.0001]} \right)^2 - (gGridP1)^2} \times |gGrid \cos 1|$$

$$GridQ2 = \sqrt{\left(\frac{gGridP2}{\max[\cos(gGrid \cos 2), 0.0001]} \right)^2 - (gGridP2)^2} \times |gGrid \cos 2|$$

$$GridQ3 = \sqrt{\left(\frac{gGridP3}{\max[\cos(gGrid \cos 3), 0.0001]} \right)^2 - (gGridP3)^2} \times |gGrid \cos 3|$$

gGrid cos1, gGrid cos 2 and gGrid cos3 represent the grid power factors of phase A, phase B, phase C respectively. The grid power factor is the sum of the power factors of phase A, phase B, phase C.

7.6 温度控制 Temperature control

温度控制主要是对发电机、齿轮箱、变压器、控制柜、机舱和塔底的各温度点进行监测, 当温度超出上限值时, 启动风扇或冷却水泵对其进行冷却; 当温度低于下限值时, 启动加热器对其



进行加热。当温度超出允许范围时，通过触发相应的状态代码执行不同的停机程序，以保证风机运行时各温度点在允许范围内。

Temperature control is implemented by monitoring the temperature of the generator, the gearbox, the voltage transformer, the control cabinets, the nacelle and tower base. When the temperature is higher than the upper limit, the fan or the cooling water pump is started to cool down. When the temperature is lower than the lower limit, the heater is started to heat. If the temperature is out of the permitted range, the corresponding state code is triggered, and the corresponding stop procedure is executed, to guarantee each temperature is within the permitted range during the wind turbine working.

7.6.1 齿轮箱温度控制 Control of gearbox temperature

1) 齿轮箱各轴承的温度监测:

- 齿轮箱低速轴温度监测: 见状态代码 080001。
- 齿轮箱高速轴温度监测: 见状态代码 080002。

2) 齿轮箱油温监测: 见状态代码 0800012。

3) 齿轮箱冷却水温监测: 见状态代码 0800020。

4) 齿轮箱冷却水泵控制: 当齿轮箱油温度高于齿轮箱冷却水泵启动温度设定值(50℃)时, 打开齿轮箱冷却水泵; 当齿轮箱油温度降低至齿轮箱冷却水泵停止温度设定值(45℃)时, 关闭齿轮箱冷却水泵。

5) 齿轮箱冷却水风扇控制: 当齿轮箱油温度高于齿轮箱冷却水风扇启动温度设定值(60℃)时, 打开齿轮箱冷却水风扇; 当齿轮箱油温度降低至齿轮箱冷却水风扇停止温度设定值(50℃)时, 关闭齿轮箱冷却水风扇。

6) 齿轮箱润滑油加热器控制: 当齿轮箱油温度低于齿轮箱润滑油加热器启动温度设定值(5℃)时, 打开齿轮箱润滑油加热器; 当齿轮箱油温度升高至齿轮箱润滑油加热器启动温度设定值(5℃)超过 3min 时, 关闭齿轮箱润滑油加热器。

1) Monitoring the temperature of each shaft of the gearbox:

- Monitoring the temperature of low speed shaft of the gearbox: state code 080001.
- Monitoring the temperature of high speed shaft of the gearbox: state code 080002.

2) Monitoring the temperature of the gearbox lubricant: state code 0800012.

3) Monitoring the temperature of the gearbox cooling water: state code 0800020.

4) The control of the gearbox cooling water pump: when the temperature of the gearbox lubricant is higher than the set temperature of starting the cooling water pump (50℃). The cooling water pump is turned on; when the temperature of the gearbox lubricant is lower than the set temperature of stopping the cooling water pump (45℃). The cooling water pump is turned off.

5) The control of the gearbox cooling fan: when the temperature of the gearbox lubricant is higher than the set temperature of starting the cooling fan (60℃). The cooling fan is turned on; when the

temperature of the gearbox lubricant is lower than the set temperature of stopping the cooling fan (50℃). The cooling fan is turned off.

6) The control of the gearbox heater: when the temperature of the gearbox lubricant is lower than the set temperature of starting the heater (5℃). The heater is turned on; when the temperature of the gearbox lubricant keeps higher than the set temperature of stopping the heater (5℃) for 3min. The heater is turned off.

7.6.2 发电机温度控制 Control of generator temperature

1) 发电机绕组温度监测:

发电机绕组温度为三相绕组温度的最大值。

- 发电机绕组温度报警: 见状态代码 070017, 070018 和 070019。
- 发电机绕组温度故障: 见状态代码 070011, 070013 和 070015。

2) 发电机轴承温度监测:

发电机轴承温度为前、后轴承温度的最大值。

- 发电机轴承温度报警: 见状态代码 070025 和 070026
- 发电机轴承温度故障: 见状态代码 070021 和 070022。

3) 发电机集电环温度监测: 见状态代码 070091。

4) 发电机冷却系统:

- 当发电机绕组温度大于发电机冷却风扇启动时绕组温度的初始设定值(80℃)超过 3min 时, 打开发电机内部冷却风扇; 若发电机绕组温度小于发电机冷却风扇停止时绕组温度的初始设定值(70℃)超过 3min 时, 关闭发电机内部冷却水风扇。

- 发电机进风口温度监测: 见状态代码 070036。
- 发电机出风口温度监测: 见状态代码 070037。

5) 发电机定子加热器:

在机舱温度(gTmpNac)小于发电机定子加热器启动设定值(10℃)且发电机绕组温度(gTmpGenWinding)满足下式时, 打开发电机定子加热器。在机舱温度大于发电机定子加热器启动设定值(10℃)或发电机绕组温度(gTmpGenWinding)不满足下式时, 延时 10min 关闭发电机定子加热器。

$$gTmpGenWinding - 1.0 < gTmpNac$$

1) Monitoring the temperature of the generator windings :

The temperature of the generator windings is the maximum winding temperature of each phase.

- Generator winding temperature warning: state code 070017,070018 and 070019.
- Generator winding temperature warning: state code 070011,070013 and 070015.

2) Monitoring the temperature of the generator shafts:



The temperature of the generator shafts is the maximum shaft temperature between front and rear shaft.

- Generator shafts temperature warning: state code 070025 and 070026.
 - Generator shafts temperature warning: state code 070021 and s070022.
- 3) Monitoring the temperature of the generator collecting ring: state code 070091.
- 4) Generator cooling system:
- When the generator winding temperature keeps higher than the set temperature of starting the generator cooling fan (80℃) for more than 3min, the generator inner cooling fan is turned on. When the generator winding temperature keeps lower than the set temperature of stopping the generator cooling fan (70℃) for more than 3min, the generator inner cooling fan is turned off.
 - Monitoring the temperature of the generator air inlet: state code 070036.
 - Monitoring the temperature of the generator air outlet: state code 070037.
- 5) Generator stator heater:

When the nacelle temperature (gTtmpNac) is lower than the set temperature of starting the generator stator heater (10℃) and the generator stator temperature (gTtmpGenWinding) satisfies the formula below, the generator stator heater is turned on. When the nacelle temperature is higher than the set temperature of starting the generator stator heater (10℃) or the generator stator temperature (gTtmpGenWinding) doesn't satisfy the formula below, the generator stator heater is turned off.

$$gTtmpGenWinding - 1.0 < gTtmpNac$$

7.6.3 变压器温度控制 Control of the voltage transformer temperature

- 1) 变压器超温开关监测：见状态代码 090001。
 - 2) 变压器温度监测：见状态代码 090002。
- 1) Monitoring the switch of the voltage transformer temperature overheating: state code 090001.
 - 2) Monitoring the voltage transformer temperature: state code 090002.

7.6.4 控制柜温度控制 Control of the control cabinets temperature

- 1) 控制柜加热：
 - 塔底控制柜加热：由温控开关 407S1 进行控制。
 - 机舱控制柜加热：由温控开关 204S8 进行控制。
- 2) 控制柜冷却：
 - 塔底控制柜冷却：

当塔底控制柜的温度高于塔底控制柜风扇启动温度设定值(40℃)超过 30s 时，打开塔底控制柜的风扇；当塔底控制柜的温度降低至塔底控制柜风扇停止温度设定值(35℃)以下超过 30s 时，关闭塔底控制柜的风扇。

- 机舱控制柜冷却:

当机舱控制柜的温度高于机舱控制柜风扇启动温度设定值 40°C 超过 30s 时, 打开机舱控制柜的风扇; 当机舱控制柜的温度降低至机舱控制柜风扇停止温度设定值(35°C)以下超过 30s 时, 关闭机舱控制柜的风扇。

3) 控制柜温度监测:

- 塔底控制柜温度监测: 见状态代码 110101。
- 机舱控制柜温度监测: 见状态代码 110102。
- 变流器控制柜温度监测: 见状态代码 030061。

1) Heating of control cabinets:

- Heating of the control cabinet of tower base: controlled by temperature switch 407S1.
- Heating of the control cabinet of the nacelle: controlled by temperature switch 204S8.

2) Cooling of control cabinets:

- Cooling of the control cabinet of tower base:\

When the temperature of the control cabinet of tower base keeps higher than the set temperature of starting the fan of the control cabinet of tower base (40°C) for more than 30s, the fan of the control cabinet of tower base is turned on. When the temperature of the control cabinet of tower base keeps lower than the set temperature of stopping the fan of the control cabinet of tower base (35°C) for more than 30s, the fan of the control cabinet of tower base is turned off.

- Cooling of the control cabinet of the nacelle:

When the temperature of the control cabinet of the nacelle keeps higher than the set temperature of starting the fan of the control cabinet of the nacelle (40°C) for more than 30s, the fan of the control cabinet of the nacelle is turned on. When the temperature of the control cabinet of the nacelle keeps lower than the set temperature of stopping the fan of the control cabinet of the nacelle (35°C) for more than 30s, the fan of the control cabinet of the nacelle is turned off.

3) Monitoring the temperature of the control cabinets:

- Monitoring the temperature of the control cabinet of tower base: state code 110101.
- Monitoring the temperature of the control cabinet of the nacelle: state code 110102.
- Monitoring the temperature of the control cabinet of the convertor: state code 030061.

7.6.5 机舱温度控制 Control of the nacelle temperature

1) 机舱冷却:

- 当机舱温度高于机舱冷却风扇启动温度设定值(40°C)超过 30s 时, 打开机舱冷却风扇; 当机舱温度低于机舱冷却风扇停止温度设定值(35°C)超过 30s 时, 机舱冷却风扇停止。
- 舱外温度监测: 见状态代码 270010。
- 机舱温度监测: 见状态代码 270011。



2) 机舱结冰：见状态代码 260005。

1) Cooling of the nacelle:

- When the nacelle temperature keeps higher than the set temperature of starting the nacelle cooling fan (40° C) for more than 30s, the nacelle cooling fan is turned on. When the nacelle temperature keeps lower than the set temperature of stopping the nacelle cooling fan (35° C) for more than 30s, the nacelle cooling fan is turned off.
- Monitor the temperature out of the nacelle: state code 270010.
- Monitor the nacelle temperature: state code 270011.

2) Nacelle icing: state code 260005.

7.6.6 塔底温度控制 Control of the tower base temperature

1) 塔底冷却：当塔底温度高于塔底冷却风扇启动温度设定值(40° C)超过 30s 时，打开塔底冷却风扇；当塔底温度低于塔底冷却风扇停止温度设定值(35° C)超过 30s 时，塔底冷却风扇停止。

2) 塔底温度监测：见状态代码 350011。

1) Cooling of tower base: When the tower base temperature keeps higher than the set temperature of starting the tower base cooling fan (40° C) for more than 30s, the tower base cooling fan is turned on. When the tower base temperature keeps lower than the set temperature of stopping the tower base cooling fan (35° C) for more than 30s, the tower base cooling fan is turned off.

2) Monitoring the tower base temperature: state code 350011.

7.7 润滑控制 Control of lubrication

7.7.1 主轴润滑控制 Control of rotor bearing lubrication

1) 在控制器开机运行后，主轴承润滑计时器清零并开始计数。

2) 当计时器计时达到设定值(12 小时)之后，系统将启动主轴承润滑。

3) 主轴承润滑泵运行达到设定值(3 分钟)后关闭，计时器复位。运行时间

1) After the control is turned on, the timer of rotor bearing lubrication is reset and starts counting.

2) After the timer reaches the set time (12 hours), rotor bearing lubrication system starts working.

3) The rotor bearing lubrication system stops working after the working time of it reaches the set time (3 min). The timer then is reset.

7.7.2 发电机润滑控制 Control of generator lubrication

目前机型配套的发电机本身带有润滑控制，不需要控制器进行控制。

The generator assorted with the current type of wind turbine recently has the control of lubrication itself, so the controller does not need to control generator lubrication.

7.7.3 齿轮箱润滑控制 Control of gearbox lubrication

齿轮箱润滑泵分成高速模式和低速模式两个工作模式。

1) 高速模式：当下述条件其中一个满足时，齿轮箱润滑泵工作在高速模式。

- 齿轮箱油温大于 40 °C；
- 风轮转速大于 10.5rpm。

2) 低速模式：当下述条件其中一个满足时，齿轮箱润滑泵工作在低速模式。

- 齿轮箱油温大于 40°C 且风机运行不处于服务模式 and 停机模式；
- 风轮转速大于 1.2rpm 且小于 10.5rpm。

The gearbox lubrication pump has two working mode; high speed mode and low speed mode

1) High speed mode: When one of the conditions below is satisfied, the gearbox lubrication pump works in the high speed mode.

- The temperature of the gearbox lubricant is higher than 40° C;
- The rotor speed is higher than 10.5rpm.

2) Low speed mode: When one of the conditions below is satisfied, the gearbox lubrication pump works in the low speed mode.

- The temperature of the gearbox lubricant is lower than 40° C, and wind turbine is not in service mode or stop mode;
- The rotor speed is higher than 1.2rpm and lower than 10.5rpm.

7.7.4 偏航润滑控制 Control of yaw bearing lubrication

当偏航系统运行一段时间后，需要进行偏航润滑。在下列 3 种条件其中之一满足时，系统将打开润滑泵对风机进行偏航润滑：

- 1) 系统运行计时器超过设定值；
- 2) 偏航系统运行计时器超过设定值；
- 3) 在偏航解缆激活的条件下，距离上次解缆的时间大于两次偏航解缆间隔时间的设定值。启动偏航润滑泵，偏航 360° 之后，关闭润滑泵。偏航润滑计时器复位。

When the yaw system has worked for a certain time, yaw bearing lubrication is needed. If one of three conditions below is satisfied, the lubrication pump is turned on and starts yaw bearing lubrication:

- 1) The working timer counts over the set time;
- 2) The yaw system working timer counts over the set time;
- 3) When cable unwinding is active, the interval time from the last cable unwinding is bigger than the set time between two cable unwinding.

After the yaw bearing lubrication pump is turned on, the nacelle yaws 360° . And then the yaw bearing lubrication pump will be turned off, and the yaw bearing lubrication timer will be reset.



7.7.5 变桨润滑控制 Control of pitch bearing lubrication

控制器开机运行后，变桨润滑计时器开始计时。当计时器计时达到一定时间后，启动变桨润滑。当变桨润滑完成后，对变桨计时器进行复位。

After the controller is turned on, the pitch bearing lubrication timer starts counting. After the timer come to a certain time, the pitch bearing lubrication starts working. After the pitch bearing lubrication finishes, the timer will be reset.

7.8 手动控制 Manual control

手动控制包括手动启动、手动停机、手动复位以及服务模式下对执行机构的手动控制四个部分。通过操作塔底控制柜、机舱控制柜或 HMI 操作界面上的相关动作部分，实现手动控制的功能。

Manual control includes manual start, manual stop, manual reset and manual control of executive equipment during service mode. Manual control is implemented by operating corresponding action in the control cabinet of tower base and the nacelle, or on the HMI operating interface.

7.8.1 手动启动 Manual start

7.8.1.1 功能描述 Function description

手动启动的功能是通过塔底启动按钮、机舱启动按钮或 HMI 上的手动启动操作，发出启动指令。风机处于静止模式，而且满足进入启动模式的条件，当接收到启动指令，使风机从静止模式进入到启动模式。

Manual start is to give start signal through the start button of tower base and the nacelle or through the manual start operation on HMI. If start signal is received and the condition to turn into start mode is satisfied during standstill mode, the turbine will turn into start mode from stop mode.

7.8.1.2 操作方法 Operating method

风机处于静止模式时，当塔底启动按钮、机舱启动按钮、HMI 上的手动启动操作中任意一种情况触发，则发出风机启动信号。

The start signal is given out through the start button of tower base and the nacelle or through the manual start operation on HMI during standstill mode.

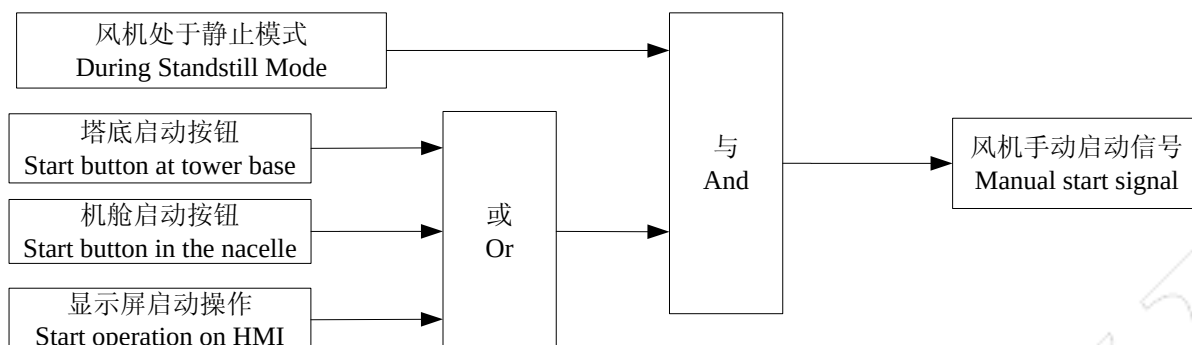


图 26 手动启动功能示意图

Fig.26 schematic diagram of manual start

7.8.2 手动停机 Manual stop

7.8.2.1 功能描述 Function description

手动停机的功能是通过塔底停机按钮、机舱停机按钮或 HMI 上的手动停机操作，发出停机指令。手动停机使风机正常停机。

Manual stop is to give stop signal through the stop button of tower base and the nacelle or through the manual stop operation on HMI. If stop signal is received, normal stop will be activated.

7.8.2.2 操作方法 Operating method

当塔底停机按钮、机舱停机按钮或 HMI 的手动停机操作中任意一种情况触发，则发出风机停机信号。

The stop signal is given out if the stop button of tower base or the nacelle or through the manual stop operation on HMI is triggered.

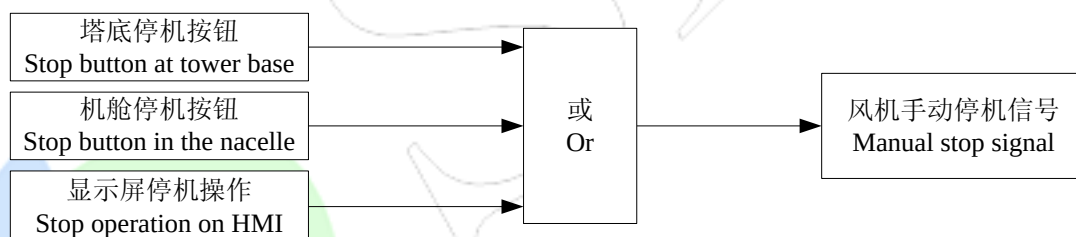


图 27 手动停机功能示意图

Fig.27 schematic diagram of manual stop

7.8.3 手动复位 Manual reset

7.8.3.1 功能描述 Function description

当塔底复位按钮、机舱复位按钮或 HMI 上的手动复位操作中任意一种情况触发，则发出风机软复位信号。HMI 手动复位只能复位一般级别的故障，而塔底复位按钮和机舱复位按钮既可以复位一般级别的故障，又可以复位严重级别的故障。

Manual stop is to give reset signal through the reset button of tower base and the nacelle or through

the manual reset operation on HMI. The manual reset operation on HMI can only reset general level error. The reset buttons of tower base and the nacelle can reset both general and serious level error.

7.8.3.2 操作方法 Operating method

1) 通过 HMI 复位操作来复位一般级别的故障。

当风机处于静止模式，此时在 HMI 上进行复位操作，风机软复位信号为 TRUE 可复位一般级别的故障。

2) 通过机舱复位按钮或塔底复位按钮来复位一般级别的故障。

当风机处于静止模式、服务模式或停机模式三种模式之一，此时触发机舱复位按钮或塔底复位按钮，风机软复位信号为 TRUE 可复位一般级别的故障。

3) 通过机舱复位按钮或塔底复位按钮来复位严重级别的故障。

当风机处于静止模式、服务模式两种模式之一，此时触发机舱复位按钮或塔底复位按钮，风机软复位严重故障信号为 TRUE 可复位严重级别的故障。

1) Reset general level errors by reset operation in HMI.

After doing the reset operation in HMI during standstill mode, the soft reset signal turns TURE, and general level errors are reset.

2) Reset general level errors by reset buttons of tower base or the nacelle.

After pressing the reset buttons of tower base or the nacelle during standstill mode, service mode or stop mode, the soft reset signal turns TURE, and general level errors are reset.

3) Reset serious level errors by reset buttons of tower base or the nacelle.

After pressing the reset buttons of tower base or the nacelle during standstill mode, service mode, the soft reset serious level error signal turns TURE, and serious level errors are reset.



上海电气 上海电气风电设备有限公司 SHANGHAI ELECTRIC WINDPOWER EQUIPMENT CO.,LTD.		W2000 风电机组	MD-030.00-A	
操作运行手册 Operating Manual			共 130 页	第 78 页
机舱复位按钮 Reset button of the nacelle 塔底复位按钮 Reset button of tower base 静止模式 During standstill mode 维护模式 During service mode 停机模式 During stop mode 显示屏复位操作 Reset operations in HMI 静止模式 During standstill mode 或 Or 或 Or 与 And 或 Or 与 And 或 Or 风机软复位严重故障 Soft reset 机舱复位按钮 Reset button of the nacelle 塔底复位按钮 Reset button of tower base 静止模式 During standstill mode 维护模式 During service mode 或 Or 或 Or 与 And 风机软复位严重故障 Soft reset serious error				
图 28 手动复位功能示意图 Fig.28 schematic diagram of manual reset				
7.8.4 服务模式下的手动控制 Manual control during service mode 在服务模式下，为了方便调试，在程序中设置了对执行机构的手动控制方式，通过 HMI 对各个机构进行操作，对执行机构进行测试，便于系统调试。 The manual control of executive equipment is set in the program for the convenience of debugging during service mode. Executive equipment can be operated and test through HMI to debug the system.				
7.8.4.1 功能描述 Function description 在服务模式下，可以进行手动变桨、手动刹车和手动控制各执行设备的启停的功能。手动偏航在风机运行的各个状态都可以进行操作。 执行设备主要有： 1) 泵：				



- 液压油泵
- 齿轮箱润滑泵(高速和低速)
- 齿轮箱冷却水泵
- 主轴承润滑泵
- 偏航轴承润滑泵
- 偏航齿轮润滑泵

2) 风扇:

- 塔底冷却风扇
- 塔底柜内冷却风扇
- 机舱冷却风扇
- 机舱柜内冷却风扇
- 齿轮箱水冷风扇
- 发电机空冷风扇
- 发电机滑环风扇

3) 加热器:

- 液压泵加热器
- 齿轮箱润滑油加热器
- 齿轮箱润滑油泵加热器
- 齿轮箱水冷却风扇加热器
- 齿轮箱冷却水泵加热器
- 发电机定子加热器

Manual pitch, manual brake, manual control of starting or stopping equipment is available during service mode. Manual yaw is available during all modes.

Executive equipment includes:

1) Pumps:

- Hydraulic pumps
- Gearbox lubrication pump (high speed and low speed)
- Gearbox cooling water pump
- Rotor bearing lubrication pump
- Yaw bearing lubrication pump
- Yaw bearing gear lubrication pump

2) Fans:

- Tower base cooling fan
- Cooling fan in the control cabinet of tower base

- Nacelle cooling fan
- Cooling fan in the control cabinet of the nacelle
- Gearbox cooling fan
- Generator cooling fan
- Generator slip ring cooling fan

3) Heaters:

- Hydraulic pump heaters
- Gearbox lubricant heater
- Gearbox lubrication pump heater
- Gearbox cooling fan heater
- Gearbox cooling water pump heater
- Generator stator heater

7.8.4.2 操作方法 Operating method

- 1) 当机舱维护开关或塔底维护开关任意一个动作，则发出维护开关激活信号。当接收到维护开关激活信号，则主流程延时 3s 进入服务模式；
- 2) 当接收到左偏航或右偏航按钮信号，则手动偏航动作，左偏航或右偏航同时按下，偏航不动作；
- 3) 当接收到手动风轮制动按钮信号或 HMI 上的手动刹车信号，则执行手动刹车操作；
- 4) 在服务模式下，当手动变桨控制激活且安全系统未动作，则可以根据 HMI 上手动输入的变量，控制变桨位置、变桨速度、以及风轮转速限制；限制风轮转速
- 5) 在服务模式下，当接收到 HMI 上的各执行设备的启停控制信号，则可以手动控制各执行设备的启停。

1) If either of the service switch of the nacelle or tower base is on, the service switch signal will be triggered. 3s after the triggered service switch signal has been received, service mode will be turned into.

2) When signal of yaw left or yaw right are received, manual yaw will be operated. If both signal of yaw left and yaw right are received, manual yaw will not be operated.

3) When manual brake signal from the manual brake button or HMI are received, manual brake will be operated.

4) If manual pitch is activated and the safety system signal is not triggered during service mode, the pitch angle, pitch speed, rotor speed limit can be controlled by input valuables through HMI, to limit rotor speed.

5) If starting and stopping signals of executive equipment is received through HMI during service mode, executive equipment will be manual started or stopped.



7.9 变频器冷却系统 Converter cooling system

7.9.1 功能描述 Function description

变频器冷却系统为变频器 IGBT 冷却, 冷却水泵出口压力低于 0.6bar 时, 冷却系统压力开关闭合, 发出报警信号。当变频器换热器进水口温度达到 55℃时, 变频器发出报警信号, 当温度达到 65℃时, 变频器发出停机信号, 风机停机。

Converter cooling system cools IGBT in the converter. When the export pressure of the cooling water pump is lower than 0.6bar, cooling system pressure switch will be closed and a warning signal will be given out. When the temperature of the converter import heat exchanger reaches 55℃, a warning signal will be given out. When the temperature of the converter import heat exchanger reaches 65℃, a brake signal will be given out, and stop mode will be activated.

7.9.2 安装、维护操作方法 Operations of installing and maintenance

- 水冷却系统开机调试前检查确认设备安装已完成, 管路连接正确, 电气接线及供电正确
- 系统预充液:

充液前先将连接软管接到水泵装置上的充液球阀, 启动充液泵对系统进行预充液, 充液过程中应在系统最高点(变频器顶端排气阀)手动排气。充液压力 3.4bar, 充液过程中系统进行自动排气, 检查系统是否有泄露; 关闭充液球阀, 启动水泵电机, 自动排气阀进行系统排气, 观察压力表, 压力表指针不稳定, 压力下降, 停泵, 再次打开充液球阀充液, 充液后关闭球阀启动水泵, 观察压力表, 反复这样操作直至压力表读数稳定, 表明排气基本完毕, 充液完成, 并且在冷却系统循环一段时间后停泵观察介质预充压力, 当预充介质压力低于 3bar 时应及时补液。充液泵扬程必须大于 35m, 充液泵出水口软管内径为 $\Phi 15$ (带相应的卡箍)。

- 泵启动前需要注意下列问题:

- 1) 确认电机转向: 分别点动水泵电机和风扇电机, 确认电机转向与标识一致
- 2) 不允许在系统未充液的情况下启动泵。
- 3) 启动水泵后注意观察水泵的运转情况, 如有噪音、压力波动等现象应停泵排气, 如此反复多次确保水泵工作时回路充满工作介质。
- 4) 系统应加注规定的冷却介质
- 5) 系统需要泄压、排液时, 可逆时针旋转安全阀后面的红色按钮进行泄压排液, 或者打开充液球阀排液, 待压力下降为 0 时, 寻找管路的最低点继续进行排液, 尽量将管路中的介质排干净, 把介质排放到合适的容器里。

- Equipment being installed completely, tubes and wires being connected correctly and power being supplied correctly must be checked before cooling water system being turned on.

- System prefilled liquid:

First connect the connecting hose to the filling ball valve of the pump, and then start the filling pump to prefill the system. During filling procedure, manual exhausting should be operated on the top of the

system (the exhaust valve on the top of the convertor). The filling pressure should be set at 3.4bar. During filling, automatic exhausting will be processing, and whether leak happens should be checked. Then close the filling ball valve, start the electric motor of the pump, and system exhausting will be processing by automatic exhausting valve. Watch the pressure gauge. If the pointer of the pressure gauge is not stable or the pressure decreases, stop the pump and open the filling ball valve again. After filling close the filling ball valve and watch the pressure gauge. Repeat these operation till the reading of the pressure gauge is stable, which intends exhausting is complete. After filling completes, cooling system should be cycled for a certain time, and then stop the pump to watch the prefilling pressure. If the prefilling pressure is lower than 3bar, add liquid immediately. The head of the filling pump must exceed 35m. and the inner diameter of the filling pump export hose should be $\Phi 15$ (with corresponding hoop).

- Pay attention to the question below before starting the pump:

- 1) Confirm the rotation direction of the electric motor: start the pump motor and the fan motor, to ensure the rotation direction is consistent with the identifier.

- 2) Starting the pump is not permitted before filling.

- 3) After starting the pump, watch the pump running. If noise or unstable pressure occurs, stop the pump and exhaust. Repeat doing so until ensure the loop filled with working liquid during the pump working.

- 4) Required cooling liquid should be filled in.

- 5) When relieving pressure and discharging liquid is needed, the red button behind the safety valve can be turned anticlockwise, or open the filling ball valve to discharge liquid. After the pressure has decreased to 0, discharging liquid can be continued through the lowest point of the pipelines. Discharge liquid in the pipelines as much as possible, and liquid should be discharged to a appropriate container.

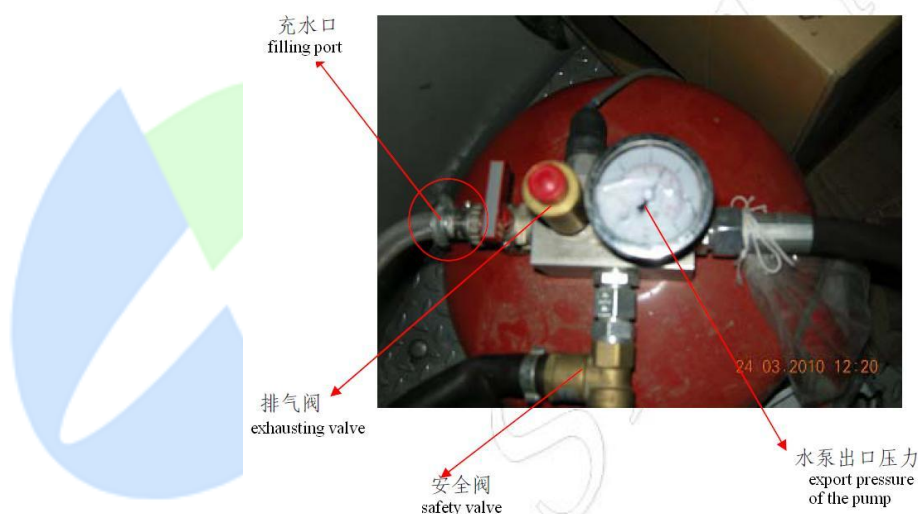


图 29 冷却水泵
Fig29 Cooling pump



附录 A 状态代码列表 Appendix I State code list

风机在运行过程中出现各种各样的故障，不同故障对风机的影响有轻重之分，级别低的故障只需给控制器发送报警信号，级别高的故障就需要给控制器发送停机信号。为此，定义了 29 类状态代码类别用于记录风机的各故障情况。停机程序分为三类：1 表示正常停机，2 表示快速停机，3 表示紧急停机。此外，对于级别低的故障以停机程序 0 表示报警。

当风机运行过程中，在同一时间发生多个故障，那么应根据故障统计结果启动级别最高的停机程序。另外，当某一级停机程序执行时间超时，将启动高一级别的停机程序，这样就避免了停机失败时对风机造成的影响，提高了风机的安全可靠性能。

All kinds of errors will happen during the wind turbine running. Different errors have different impact on the wind turbine. Only warning signals will be sent to the controller if low level errors occur, while stop signals will be sent to the controller if high level errors occur. So, 29 kinds of state code are defined to record all the errors. Stop programs are divided into 3 programs: stop program 1 represents normal stop; stop program 2 represents fast stop; stop program 3 represents emergency stop. In addition, stop program 0 which represents warning signals is sent when low level errors occur.

During the running of the wind turbine, if more than one error occur at one time, the highest level stop program among these errors will start running. In addition, if the stop program of some level runs out of time, higher level stop program will start running, to avoid the infection on the wind turbine when stopping fails, and to improve the safety and reliability.

表 A.1 Table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
1 机舱振动故障 Nacelle libration errors (010000)					
1.1	010001	机舱振动有效值超限 Nacelle libration effective value over limit	1	振动有效值>限值(0.15g) Libration virtual value >limit (0.15g)	手动复位 Manual reset
1.2	010002	机舱振动瞬时值超限 Nacelle libration instantaneous value over limit	1	振动瞬时值>限值(0.35g)超过 20s Libration instantaneous value > limit (0.35g) lasts for over 20s	手动复位 Manual reset
1.3	010011	机舱振动报警 Nacelle libration warning	0	振动瞬时值>限值(0.28g) Libration instantaneous value > limit (0.28g)	-

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操作运行手册 Operating Manual				共 130 页	第 84 页
表 A.1 (续) Continued table A.1					
序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
2 变流器故障 Convortor errors (030000)					
2.1	030001	变流器测得的发电机 转速超限 Generator speed measured from conveter over limit	2	变流器测得的发电机转速超限信 号(di)=0 Signal of generator speed measured from conveter over limit (di) = 0	自动复位 Auto reset
2.2	030002	变流器 ups 报警 Convortor ups warning	1	变流器 ups 报警信号(di)=0 Converter ups warning signal (di)=0	自动复位 Auto reset
2.3	030003	变流器并网断路器跳 闸 Conveter grid cut-in breaker open	2	变流器并网跳闸信号(di)=0 signal of conveter grid cut-in breaker open (di) =0	自动复位 Auto reset
2.4	030004	变流器 CSC 状态 Convortor CSC state	3	变流器 CSC 状态信号(di)=0 Signal of conveter CSC state (di) =0	自动复位 Auto reset
2.5	030005	变流器系统故障 Conveter system error	2	变流器系统故障信号(di)=0 Signal of converter system error (di) =0	自动复位 Auto reset
2.6	030006	变流器并网故障 Convortor cut-in error	3	变流器并网故障信号(di)=0 Signal of converter cut-in error (di)=0	自动复位 Auto reset
2.7	030011	变流器准备好并网信 号超时 Signal of convortor ready for cut-in over time	1	在 300s 内未收到变流器 准备好并网信号(di)=0 Haven't received the Signal of convortor ready for cut-in for 300s	手动复位 Manual reset
2.8	030061	变流器控制柜温度超 限 Temperature of conveter control cabinet over limit	1	变流器控制柜温度(ai)>限值 (55℃)超过 5s Temperature of conveter control cabinet (ai) > limit (55℃) lasts for over 5s	手动复位 Manual reset
2.9	030081	变流器控制柜风扇保 护 Fan protection of convortor control cabinet	1	变流器控制柜风扇保护信号 (di)=0 超过 5s Fan protections signal of convortor control cabinet (di)=0 lasts for over 5s	手动复位 Manual reset



操作运行手册 Operating Manual

共 130 页

第 85 页

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
3 风轮制动故障 Rotor brake errors (040000)					
3.1	040010	风轮制动垫磨损 Rotor brake pad wear off	1	风轮制动垫磨损信号(di)=0 超过 15s Signal of rotor brake pad wear off (di)=0 lasts for over 15s	手动复位 Manual reset
3.2	040030	风轮制动垫报警 Rotor brake pad warning	0	风轮制动垫报警信号(di)=0 Signal of rotor brake pad warning (di)=0	-
4 液压故障 Hydraulic errors (050000)					
4.1	050001	液压油泵加热器保护 Protection of hydraulic pump heater	1	液压油泵加热器保护信号(di)=0 Signal of protection of hydraulic pump heater (di)=0	手动复位 Manual reset
4.2	050002	液压泵马达保护 Protection of hydraulic motor	1	液压泵马达保护 ok 信号(di)=0 Ok signal of protection of hydraulic motor (di)=0	手动复位 Manual reset
4.3	050010	液压泵反馈时间超限 Hydraulic pump feedback time over limit	1	功率生产模式下, 偏航制动阀未打开的情况下, 若液压泵马达使能信号(do)=1 超过 80s During power production mode, when yaw brake valve is not open, enable signal of hydraulic pump (do)=1 lasts for over 80s	手动复位 Manual reset
4.4	050020	液压油位低 Hydraulic oil level low	1	液压油位 ok 信号(di)=0 超过 6s Ok signal of hydraulic oil level (di)=0 lasts for over 6s	手动复位 Manual reset
4.5	050025	液压油温超限 Hydraulic oil temperature over limit	1	液压油温 ok 信号(di)=0 超过 5s Ok signal of hydraulic oil temperature (di)=0 lasts for over 5s	手动复位 Manual reset
4.6	050030	液压风轮制动压力故障 Hydraulic rotor brake pressure error	2	液压风轮制动阀未打开的情况下制动压力开关信号(di)=0 超过 6s When hydraulic rotor brake valve is not open, switch signal of brake pressure (di)=0 lasts for over 6s	手动复位 Manual reset

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
4.7	050035	液压系统压力低 Hydraulic system pressure low	2	在开始、运行、发电模式，偏航制动阀关闭且不在偏航状态，液压主压力(ai)<限值(135par)超过15s During start, run, generating mode, when yaw brake valve is closed and yaw is not activated, hydraulic main pressure (ai)<limit (135par) lasts for over 15s	手动复位 Manual reset
4.8	050036	液压系统压力高 Hydraulic system pressure high	2	液压主压力(ai)>限值(170par)超过1s Hydraulic main pressure (ai)>limit (170par) lasts for over 1s	手动复位 Manual reset
4.9	050038	液压过滤器入口压力故障 Import pressure of the hydraulic filter error	1	液压过滤器入口压力开关信号(di)=0 Switch signal of the import pressure of the hydraulic filter (di)=0	手动复位 Manual reset
4.10	050039	液压过滤器出口压力故障 Export pressure of the hydraulic filter error	1	液压过滤器出口压力开关信号(di)=0 Switch signal of the export pressure of the hydraulic filter (di)=0	手动复位 Manual reset

5 偏航故障 Yaw errors (060000)

5.1	060001	偏航位置故障 Yaw position error	1	偏航位置>=限值(7500)超过 2s Yaw position >= limit(7500) lasts for over 2s	手动复位 Manual reset
5.2	060002	顺时针偏航故障 Yaw clockwise error	1	偏航右停止位信号(di)=1 Signal of yaw right stop (di)=1	手动复位 Manual reset
5.3	060003	逆时针偏航故障 Yaw anticlockwise error	1	偏航左停止位信号(di)=1 Signal of yaw left stop (di)=1	手动复位 Manual reset
5.4	060010	偏航速度低 Yaw speed low	1	顺时针偏航:偏航速度<-0.40/s 逆时针偏航:偏航速度<-0.40/s Yaw clockwise: yaw speed<-0.40/s Yaw anticlockwise: yaw speed<-0.40/s	手动复位 Manual reset
5.5	060100	偏航保护 Yaw protection	1	偏航保护信号(di)=0 Signal of yaw protection (di)=0	手动复位 Manual reset
5.6	060101	偏航马达保护 1 Yaw motor protection 1	1	偏航马达 1 保护 ok 信号(di)=1 Ok signal of yaw motor protection (di)=1	手动复位 Manual reset



操作运行手册 Operating Manual

共 130 页

第 87 页

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
5.7	060102	偏航马达保护 2 Yaw motor protection 2	1	偏航马达 2 保护 ok 信号(di)=1 Ok signal of yaw motor protection 2 (di)=1	手动复位 Manual reset
5.8	060103	偏航马达保护 3 Yaw motor protection 3	1	偏航马达 3 保护 ok 信号(di)=1 Ok signal of yaw motor protection 3 (di)=1	手动复位 Manual reset
5.9	060104	偏航马达保护 4 Yaw motor protection 4	1	偏航马达 4 保护 ok 信号(di)=1 Ok signal of yaw motor protection 4 (di)=1	手动复位 Manual reset
5.10	060201	偏航制动保护 Yaw brake protection	1	偏航制动保护 ok 信号(di)=0 Ok signal of yaw brake protection (di)=0	手动复位 Manual reset
5.11	060300	偏航软启动器故障 Yaw softstart error	1	偏航软启动器 ready 信号(di)=0 Ready signal of yaw softstart (di)=0	手动复位 Manual reset
5.12	060301	偏航软启动器斜率错误 Yaw softstart slope fault	1	偏航软启动器斜率信号(di)=0 Signal of yaw softstart slope (di)=0	手动复位 Manual reset
5.13	060605	偏航轴承润滑油油位低 Yaw bearing lubricant level low	0	偏航轴承润滑油油位满信号 (di)=0 超过 10s Full signal of yaw bearing lubricant (di)=0 lasts for over 10s	-
5.14	060606	偏航轴承润滑故障 Yaw bearing lubrication error	1	偏航轴承润滑泵停止信号(di)=0 Stop signal of yaw bearing lubrication pump (di)=0	手动复位 Manual reset
5.15	060801	偏航齿轮润滑油油位低 Yaw bearing gear lubricant level low	0	偏航齿轮润滑油油位满信号 (di)=0 超过 10s Full signal of yaw bearing gear lubricant (di)=0 lasts for over 10s	-
5.16	060802	偏航齿轮润滑故障 Yaw bearing gear lubrication error	1	偏航齿轮润滑泵停止信号(di)=0 Stop signal of yaw bearing gear lubrication pump (di)=0	手动复位 Manual reset
6 发电机故障 Generator errors (070000)					
6.1	070011	发电机绕组 u1 温度超限 Generator winding u1 temperature over limit	1	发电机绕组 u1 温度(ai)>限值 (165℃)超过 5s Generator winding u1 temperature (ai)>limit(165℃) lasts for over 5s	手动复位 Manual reset
6.2	070013	发电机绕组 v1 温度超限 Generator winding v1 temperature over limit	1	发电机绕组 v1 温度(ai)>限值 (165℃)超过 5s Generator winding v1 temperature (ai)>limit(165℃) lasts for over 5s	手动复位 Manual reset

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
6.3	070015	发电机绕组 w1 温度超限 Generator winding w1 temperature over limit	1	发电机绕组 w1 温度(ai)>限值(165℃)超过 5s Generator winding v1 temperature (ai)>limit(165℃) lasts for over 5s	手动复位 Manual reset
6.4	070017	发电机绕组 u1 温度报警 Generator winding u1 temperature warning	0	发电机绕组 u1 温度(ai)>限值(155℃) Generator winding u1 temperature (ai)>limit(155℃)	-
6.5	070018	发电机绕组 v1 温度报警 Generator winding v1 temperature warning	0	发电机绕组 v1 温度(ai)>限值(155℃) Generator winding v1 temperature (ai)>limit(155℃)	-
6.6	070019	发电机绕组 w1 温度报警 Generator winding w1 temperature warning	0	发电机绕组 w1 温度(ai)>限值(155℃) Generator winding w1 temperature (ai)>limit(155℃)	-
6.7	070021	发电机前轴承温度超限 Generator front bearing temperature over limit	1	发电机前轴承温度(ai)>限值(95℃)超过 5s Generator front bearing temperature (ai)> limit(95℃) lasts for over 5s	手动复位 Manual reset
6.8	070022	发电机后轴承温度超限 Generator rear bearing temperature over limit	1	发电机后轴承温度(ai)>限值(95℃)超过 5s Generator rear bearing temperature (ai)> limit(95℃) lasts for over 5s	手动复位 Manual reset
6.9	070025	发电机前轴承温度报警 Generator front bearing temperature warning	0	发电机前轴承温度(ai)>限值(85℃) Generator front bearing temperature (ai)> limit(85℃)	-
6.10	070026	发电机后轴承温度报警 Generator rear bearing temperature warning	0	发电机后轴承温度(ai)>限值(85℃) Generator rear bearing temperature (ai)> limit(85℃)	-
6.11	070036	发电机进风口温度超限 Generator air inlet temperature over limit	1	发电机进风口温度(ai)>限值(80℃)超过 5s Generator air inlet temperature (ai)> limit(80℃) lasts for over 5s	手动复位 Manual reset
6.12	070037	发电机出风口温度超限 Generator air outlet temperature over limit	1	发电机出风口温度(ai)>限值(80℃)超过 5s Generator air outlet temperature (ai)> limit(80℃) lasts for over 5s	手动复位 Manual reset



操作运行手册 Operating Manual

共 130 页

第 89 页

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
6.13	070056	发电机定子加热器保护 1 Generator stator heater protection 1	1	发电机定子加热器 1 保护信号 (di)=0 Signal of generator stator heater 1 protection (di)=0	手动复位 Manual reset
6.14	070057	发电机定子加热器保护 2 Generator stator heater protection 2	1	发电机定子加热器 2 保护信号 (di)=0 Signal of generator stator heater 2 protection (di)=0	手动复位 Manual reset
6.15	070059	发电机内部冷却风扇保护 Generator inner cooling fan protection	1	发电机内部冷却风扇保护信号(di)=0 Signal of generator inner cooling fan protection (di)=0	手动复位 Manual reset
6.16	070070	发电机电刷故障 Generator brush error	1	发电机电刷 ok 信号(di)=0 Ok signal of generator brush (di)=0	手动复位 Manual reset
6.17	070091	发电机滑环温度超限 Generator slip temperature over limit	1	发电机滑环温度(ai)>限值(80℃) 超过 5s Generator slip temperature (ai) > limit(80℃) lasts for over 5s	手动复位 Manual reset
7 齿轮箱故障 Gearbox errors (080000)					
7.1	080001	齿轮箱高速轴发电机侧温度超限 Gearbox high speed shaft generator side temperature over limit	1	齿轮箱高速轴发电机侧温度(ai)>限值 (100℃)超过 5s Gearbox high speed shaft generator side temperature (ai) > limit (100℃) lasts for over 5s	手动复位 Manual reset
7.2	080002	齿轮箱高速轴风轮侧温度超限 Gearbox high speed shaft rotor side temperature over limit	1	齿轮箱高速轴风轮侧温度(ai)>限值 (100℃)超过 5s Gearbox high speed shaft rotor side temperature (ai) > limit (100℃) lasts for over 5s	手动复位 Manual reset
7.3	080012	齿轮箱油温超限 Gearbox oil temperature over limit	1	齿轮箱油温(ai)>限值(80℃)超过 5s Gearbox oil temperature (ai)> limit (80℃) lasts for over 5s	手动复位 Manual reset
7.4	080014	齿轮箱入口油温超限 Gearbox import oil temperature over limit	1	齿轮箱入口油温(ai)>限值(80℃) 超过 5s Gearbox import oil temperature (ai)> limit (80℃) lasts for over 5s	手动复位 Manual reset

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
7.5	080020	齿轮箱冷却水温超限 Gearbox cooling water temperature over limit	1	齿轮箱冷却水温(ai)>限值(20℃) 超过 5s Gearbox cooling water temperature (ai)> limit (20℃) lasts for over 5s	手动复位 Manual reset
7.6	080023	齿轮箱冷却水压力异常 Gearbox cooling water pressure fault	1	齿轮箱冷却水泵打开时, 压力 ok 信号 (di)=0 超过 20s When gearbox cooling water pump is opened, ok signal of pressure (di)=0 lasts for over 20s	手动复位 Manual reset
7.7	080030	齿轮箱油位低 Gearbox oil level low	1	齿轮箱油位 ok 信号(di)=0 超过 10s Ok signal of gearbox oil level (di)=0 lasts for over 10s	手动复位 Manual reset
7.8	080040	齿轮箱润滑油压力异常 Gearbox oil pressure fault	1	齿轮箱润滑油泵打开的情况下, 润滑油 压力开关 ok 信号(di)=0 超过 12s When gearbox lubrication pump is opened, ok signal of pressure (di)=0 lasts for over 12s	手动复位 Manual reset
7.9	080050	齿轮箱过滤器油压力异常 Gearbox filter oil pressure fault	1	齿轮箱润滑油泵打开的情况下, 过滤器油 压力开关 ok 信号(di)=0 超过 12s When gearbox lubrication pump is opened, ok signal of filter oil pressure (di)=0 lasts for over 12s	手动复位 Manual reset
7.10	080071	齿轮箱润滑油加热器保护 Gearbox lubricant heater protection	1	齿轮箱润滑油加热器保护信号 (di)=0 Signal of gearbox lubricant heater protection (di)=0	手动复位 Manual reset
7.11	080072	齿轮箱润滑油泵马达保护 Gearbox lubrication pump motor protection	1	齿轮箱润滑油泵马达保护信号 (di)=0 Signal of gearbox lubrication pump motor protection (di)=0	手动复位 Manual reset
7.12	080073	齿轮箱润滑油泵加热器保护 Gearbox lubrication pump heater protection	1	齿轮箱润滑油泵加热器保护信号 (di)=0 Signal of gearbox lubrication pump heater protection (di)=0	手动复位 Manual reset
7.13	080075	齿轮箱冷却水风扇保护 Gearbox cooling water fan protection	1	齿轮箱冷却水风扇保护信号 (di)=0 Signal of gearbox cooling water fan protection (di)=0	手动复位 Manual reset



操作运行手册 Operating Manual

共 130 页

第 91 页

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
7.14	080080	齿轮箱冷却水泵保护 Gearbox cooling water pump protection	1	齿轮箱冷却水泵保护信号(di)=0 Signal of gearbox cooling water pump protection (di)=0	手动复位 Manual reset
7.15	080081	齿轮箱冷却水泵加热器保护 Gearbox cooling water pump heater protection	1	齿轮箱冷却水泵加热器保护信号(di)=0 Signal of gearbox cooling water pump heater protection (di)=0	手动复位 Manual reset
8 变压器故障 Converter errors (090000)					
8.1	090001	变压器超温开关动作 Action of voltage transformer over temperature switch	1	变压器超温开关信号(di)=0 超过 30s Signal of voltage transformer over temperature switch (di)=0 lasts over 30s	手动复位 Manual reset
8.2	090002	变压器超温 Voltage transformer over temperature	1	变压器温度(ai)>限值(90℃) Voltage transformer temperature (ai) > limit (90℃)	手动复位 Manual reset
8.3	090011	变压器轻瓦斯报警 Voltage transformer light gas warning	0	变压器轻瓦斯报警信号(di)=1 Signal of voltage transformer light gas warning (di)=1	-
8.4	090012	变压器重瓦斯故障 Voltage transformer heavy gas warning	1	变压器重瓦斯故障信号(di)=1 Signal of voltage transformer heavy gas warning (di)=1	手动复位 Manual reset
9 电网故障 Grid errors (100000)					
9.1	100111	电网 A 相电压高于上 限值 Grid voltage of phase A over upper limit	2	电网 A 相电压>限值(440V) 超过 100ms Grid voltage of phase A > limit(440V) last for over 100ms	手动复位 Manual reset
9.2	100112	电网 B 相电压高于上 限值 Grid voltage of phase B over upper limit	2	电网 B 相电压>限值(440V) 超过 100ms Grid voltage of phase B > limit(440V) lasts for over 100ms	手动复位 Manual reset
9.3	100113	电网 C 相电压高于上 限值 Grid voltage of phase C over upper limit	2	电网 C 相电压>限值(440V) 超过 100ms Grid voltage of phase C > limit(440V) lasts for over 100ms	手动复位 Manual reset
9.4	100121	电网 A 相电压低于下 限值 Grid voltage of phase A under lower limit	2	电网 A 相电压<限值(360V) 超过 100ms Grid voltage of phase A < limit(360V) lasts for over 100ms	手动复位 Manual reset

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
9.5	100122	电网 B 相电压低于下 限值 Grid voltage of phase B under lower limit	2	电网 B 相电压<限值(360V) 超过 100ms Grid voltage of phase B < limit(360V) lasts for over 100ms	手动复位 Manual reset
9.6	100123	电网 C 相电压低于下 限值 Grid voltage of phase C under lower limit	2	电网 C 相电压<限值(360V) 超过 100ms Grid voltage of phase C < limit(360V) lasts for over 100ms	手动复位 Manual reset
9.7	100130	电网电压不对称 Grid voltage asymmetry	2	三相电网电压两两差值 最大值>限值(50v)超过 2s The maximum difference of grid voltage of three phases between each other > limit (50V) lasts for over 2s	手动复位 Manual reset
9.8	100211	电网 A 相电流高于上 限值 Grid current of phase A over upper limit	2	电网 A 相电流>电网电流极限值 Grid current of phase A > grid current limit	手动复位 Manual reset
9.9	100212	电网 B 相电流高于上 限值 Grid current of phase B over upper limit	2	电网 B 相电流>电网电流极限值 Grid current of phase B > grid current limit	手动复位 Manual reset
9.10	100213	电网 C 相电流高于上 限值 Grid current of phase C over upper limit	2	电网 C 相电流>电网电流极限值 Grid current of phase C > grid current limit	手动复位 Manual reset
9.11	100220	电网电流不对称 Grid voltage asymmetry	2	三相电网电流两两差值最大值 >限值(100A)超过 2s The maximum difference of grid current of three phases between each other > limit (100A) lasts for over 2s	手动复位 Manual reset
9.12	100311	电网 A 相频率高于上 限值 Grid frequency of phase A over upper limit	2	电网 A 相电压>限值(51.5Hz) 超过 100ms Grid frequency of phase A > limit(51.5Hz) lasts for over 100ms	手动复位 Manual reset
9.13	100312	电网 B 相频率高于上 限值 Grid frequency of phase B over upper limit	2	电网 B 相电压>限值(51.5Hz) 超过 100ms Grid frequency of phase B > limit(51.5Hz) lasts for over 100ms	手动复位 Manual reset



操作运行手册 Operating Manual

共 130 页

第 93 页

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
9.14	100313	电网 C 相频率高于上限值 Grid frequency of phase C over upper limit	2	电网 C 相电压>限值(51.5Hz) 超过 100ms Grid frequency of phase C > limit(51.5Hz) lasts for over 100ms	手动复位 Manual reset
9.15	100321	电网 A 相频率低于下限值 Grid frequency of phase A under lower limit	2	电网 A 相电压<限值(48.5Hz) 超过 100ms Grid frequency of phase A < limit(48.5Hz) lasts for over 100ms	手动复位 Manual reset
9.16	100322	电网 B 相频率低于下限值 Grid frequency of phase B under lower limit	2	电网 B 相电压<限值(48.5Hz) 超过 100ms Grid frequency of phase B < limit(48.5Hz) lasts for over 100ms	手动复位 Manual reset
9.17	100323	电网 C 相频率低于下限值 Grid frequency of phase C under lower limit	2	电网 C 相电压<限值(48.5Hz) 超过 100ms Grid frequency of phase C < limit(48.5Hz) lasts for over 100ms	手动复位 Manual reset
9.18	100410	电网实时有功功率超限 Grid realtime active power over limit	2	电网实时有功功率>限值 (1600kW)超过 1s Grid realtime active power > limit(1600kW) lasts for over 1s	手动复位 Manual reset
9.19	100411	电网 30s 平均有功功率超限 Grid mean active power of 30s over limit	2	电网 30s 平均有功功率>限值(2200kW)超过 5s Grid mean active power of 30s > limit(2200kW) lasts for over 5s	手动复位 Manual reset
9.20	100412	电网 10m 平均有功功率超限 Grid mean active power of 10m over limit	2	电网 10m 平均有功功率>限值(2100kW)超过 10s Grid mean active power of 10m > limit(2100kW) lasts for over 10s	手动复位 Manual reset
9.21	100420	电网功率与变换器功率偏差超限 Difference between grid power and convertor power over limit	2	电网的实时有功功率与变换器测得的实时有功功率差值的绝对值>限值(50kW)超过 10s The absolute value of realtime power difference between the grid and the convertor > limit(50kW) lasts for over 10s	手动复位 Manual reset
9.22	100510	电网实时无功功率超限 Grid realtime reactive power over limit	2	电网实时无功功率>限值 (120kVar)超过 120s Grid realtime reactive power > limit(120kVar) lasts for over 120s	手动复位 Manual reset

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操作运行手册 Operating Manual				共 130 页	第 94 页
表 A.1 (续) Continued table A.1					
序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
10 柜内冷却故障 Control cabinet cooling errors (110000)					
10.1	110101	塔底控制柜温度超限 Temperature of control cabinet of tower base over limit	1	塔底柜温(ai)>限值(55℃)超过 5s Temperature of control cabinet of tower base (ai) > limit (55℃) lasts for over 5s	手动复位 Manual reset
10.2	110102	机舱控制柜温度超限 Temperature of control cabinet of the nacelle over limit	1	机舱柜温(ai)>限值(55℃)超过 5s Temperature of control cabinet of the nacelle (ai) > limit (55℃) lasts for over 5s	手动复位 Manual reset
11 柜内加热故障 Control cabinet heating errors (120000)					
11.1	120001	塔底控制柜温度控制开关保护 Temperature control switch protection of control cabinet of tower base	1	塔底控制柜温度控制开关保护信号(di)=0 超过 5s Temperature control switch protection signal of control cabinet of tower base (di)=0 lasts for over 5s	手动复位 Manual reset
11.2	120002	机舱控制柜温度控制开关保护 Temperature control switch protection of control cabinet of the nacelle	1	机舱控制柜温度控制开关保护信号(di)=0 超过 5s Temperature control switch protection signal of control cabinet of the nacelle (di)=0 lasts for over 5s	手动复位 Manual reset
12 UPS 故障 UPS errors (130000)					
12. 1	130101	塔底控制柜 UPS 电池故障 UPS battery error of control cabinet of tower base	1	塔底控制柜 UPS 电池信号(di)=0 UPS battery signal of control cabinet of tower base (di)=0	手动复位 Manual reset
12. 2	130102	机舱控制柜 UPS 电池故障 UPS battery error of control cabinet of the nacelle	1	机舱控制柜 UPS 电池信号(di)=0 UPS battery signal of control cabinet of the nacelle (di)=0	手动复位 Manual reset
12. 3	130201	塔底控制柜 UPS 供电异常 UPS power supply error of control cabinet of tower base	1	塔底控制柜 UPS 供电信号(di)=0 UPS power supply signal of control cabinet of tower base (di)=0	手动复位 Manual reset



操作运行手册 Operating Manual

共 130 页

第 95 页

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
12. 4	130202	机舱控制柜 UPS 供电异常 UPS power supply error of control cabinet of the nacelle	1	机舱控制柜 UPS 供电信号(di)=0 UPS power supply signal of control cabinet of the nacelle (di)=0	手动复位 Manual reset
13 电源故障 Power supply errors (160000)					
13.1	160001	塔底 400VAC 供电保护 Tower base 400VAC power supply protection	2	塔底 400VAC 供电保护信号(di)=0 Signal of tower base 400VAC power supply protection (di)=0	手动复位 Manual reset
13.2	160025	机舱控制柜 24VDC 供电异常 Nacelle 24VAC power supply fault	2	机舱控制柜 24VDC 供电保护信号(di)=0 Signal of nacelle 24VAC power supply protection (di)=0	手动复位 Manual reset
13.3	160026	机舱控制柜外部传感器 24VDC 供电异常 External sensor of control cabinet of the nacelle 24VDC power supply fault	2	机舱控制柜外部传感器 24VDC 信号(di)=0 External sensor of control cabinet of the nacelle 24VDC signal (di)=0	手动复位 Manual reset
13.4	160061	变桨 400VAC 供电保护 Pitch 400VAC power supply protection	2	变桨 400VAC 供电保护信号(di)=0 Signal of pitch 400VAC power supply protection (di)=0	手动复位 Manual reset
13.5	160062	变桨 230VAC 供电保护 Pitch 230VAC power supply protection	2	变桨 230VAC 供电保护信号(di)=0 Signal of pitch 230VAC power supply protection (di)=0	手动复位 Manual reset
14 风速风向故障 Wind speed and direction errors (170000)					
14.1	170030	风向与机舱偏差超限 Difference between wind direction and the nacelle over limit	1	在运行和发电模式下， 风向与机舱偏差值>70° 超过 22s Difference between wind direction and the nacelle>70° last for over 22s during run and generating mode	手动复位 Manual reset

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操作运行手册 Operating Manual				共 130 页	第 96 页
表 A.1 (续) Continued table A.1					
序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
14.2	170041	风速计 1 故障 Anemometer 1 error	1	(风速 2－风速 1)>1.5m/s；风轮转速>9rpm 时前后两个 500ms 周期内读到的风速计 1 的读数差值>0.05m/s 或风速计 1<0.5m/s 超过 120s (wind speed2－wind speed1)>1,5m/s when rotor speed > 9rpm, anemometer 1 reading difference between former and latter 500m/s > 0.05m/s or anemometer 1 reading < 0.5m/s lasts for over 120s	手动复位 Manual reset
14.3	170042	风速计 2 故障 Anemometer 2 error	1	(风速 1－风速 2)>1.5m/s；风轮转速>9rpm 时前后两个 500ms 周期内读到的风速计 2 的读数差值>0.05m/s 或风速计 2<0.5m/s 超过 120s (wind speed1－wind speed2)>1,5m/s when rotor speed > 9rpm, anemometer 2 reading difference between former and latter 500m/s > 0.05m/s or anemometer 2 reading < 0.5m/s lasts for over 120s	手动复位 Manual reset
14.4	170051	风向标 1 故障 Vane 1 error	1	10s 平均风速>3m/s 时(25s 平均风向 2-25s 平均风向 1)>20° 超过 20s；前后两个 500ms 周期内读到的风向标 1 的读数差值>0.2° 超过 2.5min When mean wind speed of 10s >3m/s, (aveage wind direction2 of 25s – mean wind direction1 of 25s) > 20 °; vane 1 reading difference between former and latter 500ms > 0.2 °lasts for over 2.5min	手动复位 Manual reset



表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
14.5	170052	风向标 2 故障 Vane 2 error	1	10s 平均风速>3m/s 时(25s 平均风向 1-25s 平均风向 2)>20° 超过 20s; 前后两个周期 500ms 内读到的风向标 2 的读数差值>0.2° 超过 2.5min When mean wind speed of 10s >3m/s, (aveage wind direction1 of 25s – mean wind direction2 of 25s) > 20 °; vane 2 reading difference between former and latter 500ms > 0.2 °lasts for over 2.5min	手动复位 Manual reset
14.6	170058	风向标故障 Vanes error	2	风向标 1 和风向标 2 同时出现故障。 Vane 1 and vane 2 error occurs at the same time	手动复位 Manual reset
14.7	170101	桨叶位置判断暴风 Storm estimated by pitch angle	1	发电模式下, 10min 平均变桨位置>20° 或 10s 平均变桨位置>25° 或 1s 平均变桨位置>33° 超过 10s During generating mode, mean pitch angle of 10min>20 °or mean pitch angle of 10s >25 °or mean pitch angle of 1s >33 °lasts for over 10s	手动复位 Manual reset
14.8	170102	暴风 storm	1	风速>=瞬间最大风速(33m/s)或 10s 平均风速>=限值(25 m/s)或 10min 平均风速>=限值(20m/s)超过 10s 或者从变桨桨叶位置判断暴风 Wind speed >= instantaneous maximum wind speed(33m/s) or mean wind speed of 10s >= limit(25m/s) or mean wind speed of 10min >= limit(20m /s) lasts for over 10s, or determined by pitch angle	手动复位 Manual reset

 上海电气 上海电气风电设备有限公司 SHANGHAI ELECTRIC SHANGHAI ELECTRIC WINDPOWER EQUIPMENT CO.,LTD.		W2000 风电机组		MD-030.00-A	
操作运行手册 Operating Manual				共 130 页	第 98 页
表 A.1 (续) Continued table A.1					
序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
14.9	170200	风力不足 Wind power shortage	0	发电模式下风轮转速<10rpm 超过 15min 或风轮转速<6rpm 超过 5min 或风轮转速<4rpm rotor speed<10rpm lasts for over 15min or rotor speed<6rpm lasts for over 5min or rotor speed<4rpm	-
15 熔断器故障 Fuse errors (180000)					
15.1	180001	塔底控制柜熔断器断 开 Fuse in the tower base control cabinet open	2	塔底控制柜熔断器信号(di)=0 Signal of Fuse in the tower base control cabinet (di)=0	手动复位 Manual reset
15.2	180002	机舱控制柜熔断器断 开 Fuse in the nacelle control cabinet open	2	机舱控制柜熔断器信号(di)=0 Signal of Fuse in the nacelle control cabinet (di)=0	手动复位 Manual reset
16 避雷保护故障 Lightning protection errors (190000)					
16.1	190001	塔底避雷保护动作 Lightning protection action of tower base	1	塔底避雷信号(di)=0 Signal of lightning protection of tower base (di)=0	手动复位 Manual reset
16.2	190002	机舱避雷保护动作 Lightning protection action of the nacelle	1	机舱避雷信号(di)=0 Signal of lightning protection of the nacelle (di)=0	手动复位 Manual reset
16.3	190011	发电机定子避雷保护 动作 Lightning protection action of generator stator	1	发电机定子避雷信号(di)=0 Signal of lightning protection of generator stator (di)=0	手动复位 Manual reset
16.4	190012	发电机转子避雷保护 动作 Lightning protection action of generator rotor	1	发电机转子避雷信号(di)=0 Signal of lightning protection of generator rotor (di)=0	手动复位 Manual reset
16.5	190031	箱变避雷保护动作 Lightning protection action of voltage transformer	1	箱变避雷信号(di)=0 Signal of lightning protection of voltage transformer (di)=0	手动复位 Manual reset
17 初始化故障 Initialization errors (200000)					
17.1	200001	读取 InitDef.txt 文件出 错 Read InitDef.txt error	1	读取 InitDef.txt 文件出错 Read InitDef.txt error	-



操作运行手册 Operating Manual

共 130 页

第 99 页

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
17.2	200002	读取 InitRun.txt 文件出错 Read InitRun.txt error	1	读取 InitRun.txt 文件出错 Read InitRun.txt error	
17.3	200021	读取 NLPitchTable.txt 文件出错 Read NLPitchTable.txt error	1	读取 NLPitchTable.txt 文件出错 Read NLPitchTable.txt error	-
17.4	200022	读取 OptiPitchTable.txt 文件出错 Read OptiPitchTable.txt error	1	读取 OptiPitchTable.txt 文件出错 Read OptiPitchTable.txt error	-
17.5	200023	读取 TorqueTable.txt 文件出错 Read TorqueTable.txt error	1	读取 TorqueTable.txt 文件出错 Read TorqueTable.txt error	-
17.6	200051	读取 StatusList.txt 文件出错 Read StatusList.txt error	1	读取 StatusList.txt 文件出错 Read StatusList.txt error	-
17.7	200052	读取 StatusCount.txt 文件出错 Read StatusCount.txt error	1	读取 StatusCount.txt 文件出错 Read StatusCount.txt error	-
18 停机故障 Parking errors (220000)					
18.1	220001	停机超时 Parking over time	3	发出停机命令后风轮转速>10rpm 超过 35s 或风轮转速>4rpm 超过 90s After stop signal is triggered, rotor speed > 10rpm last for over 35s, or rotor speed > 4rpm lasts for over 90s	手动复位 Manual reset
19 系统故障 System errors (250000)					
19.1	250101	CPU 占用率 70% CPU possession rate 70%	1	CPU 占用率>70%超过 2min CPU possession rate > 70% lasts for over 2min	手动复位 Manual reset
19.2	250102	CPU 占用率 86% CPU possession rate 86%	1	CPU 占用率>86%超过 30s CPU possession rate > 86% lasts for over 30s	手动复位 Manual reset
20 结冰故障 Icing errors (260000)					

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操作运行手册 Operating Manual				共 130 页	第 100 页
表 A.1 (续) Continued table A.1					
序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
20.1	260005	机舱结冰传感器 Nacelle icing sensor	1	机舱结冰传感器有信号(暂无) Nacelle icing sensor signal occurs (temporarily none)	手动复位 Manual reset
21 机舱冷却故障 Nacelle cooling errors (270000)					
21.1	270005	机舱冷却风扇保护 Nacelle cooling fan protection	1	机舱冷却风扇保护信号(di)=0 超过 10s Signal of nacelle cooling fan protection (di)=0 lasts for over 10s	手动复位 Manual reset
21.2	270010	机舱外温度低 Temperature outside the nacelle low	1	机舱外温度(ai)<限值(30℃)超过 5min Temperature outside the nacelle (ai) < limit (30℃) lasts for over 5min	自动复位 Auto reset
21.3	270011	机舱温度超限 Nacelle temperature over limit	1	机舱温度(ai)>限值(85℃)超过 30s Nacelle temperature (ai) > limit (85℃) lasts for over 30s	手动复位 Manual reset
21 风轮位置故障 Rotor position errors (280000)					
22.1	280005	风轮位置超限 Rotor position over limit	1	风轮初始位置大于 360 °或小于 0 ° Rotor initial position bigger than 360 °or smaller than 0 °	手动复位 Manual reset
21 主轴轴承故障 Main bearing errors (290000)					
23.1	290050	主轴轴承润滑油油位低 Main bearing lubricant level low	0	主轴轴承润滑油油位满信号 (di)=0 超过 10s Full signal of main bearing lubricant (di)=0 lasts for over 10s	-
23.2	290060	主轴轴承润滑故障 Main bearing lubrication error	1	主轴轴承润滑泵停止信号(di)=0 Stop signal of main bearing lubrication pump (di)=0	手动复位 Manual reset
24 桨距故障 Pitch errors (300000)					
桨距故障由变桨系统通过 Profibus 通信传输, 控制器解析状态字, 根据通信协议中定义的数字量故障信号(0 或 1)和模拟量值是否超限激活相应的状态代码。					
24.1	300101	桨叶 1 的电池主开关断开 Main switch of blade 1 battery open	3	桨叶 1 的电池主开关信号=0 超过 500ms Main switch signal of blade 1 battery = lasts for over 500ms	手动复位 Manual reset
24.2	300102	桨叶 2 的电池主开关断开 Main switch of blade 2 battery open	3	桨叶 2 的电池主开关信号=0 超过 500ms Main switch signal of blade 2 battery = lasts for over 500ms	手动复位 Manual reset



操作运行手册 Operating Manual

共 130 页

第 101 页

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
24.3	300103	桨叶 3 的电池主开关断开 Main switch of blade 3 battery open	3	桨叶 3 的电池主开关信号=0 超过 500ms Main switch signal of blade 3 battery = lasts for over 500ms	手动复位 Manual reset
24.4	300111	轴柜 1 电池过电压 Axle cabinet 1 battery over voltage	3	轴柜 1 电池过电压信号=1 超过 500ms Signal of axle cabinet 1 battery over voltage = 1 lasts for over 500ms	手动复位 Manual reset
24.5	300112	轴柜 2 电池过电压 Axle cabinet 2 battery over voltage	3	轴柜 2 电池过电压信号=1 超过 500ms Signal of axle cabinet 2 battery over voltage = 1 lasts for over 500ms	手动复位 Manual reset
24.6	300113	轴柜 3 电池过电压 Axle cabinet 3 battery over voltage	3	轴柜 3 电池过电压信号=1 超过 500ms Signal of axle cabinet 3 battery over voltage = 1 lasts for over 500ms	手动复位 Manual reset
24.7	300114	轴柜 1 电池欠电压 Axle cabinet 1 battery under voltage	3	轴柜 1 电池欠电压信号=1 超过 500ms Signal of axle cabinet 1 battery under voltage = 1 lasts for over 500ms	手动复位 Manual reset
24.8	300115	轴柜 2 电池欠电压 Axle cabinet 2 battery under voltage	3	轴柜 2 电池欠电压信号=1 超过 500ms Signal of axle cabinet 2 battery under voltage = 1 lasts for over 500ms	手动复位 Manual reset
24.9	300116	轴柜 3 电池欠电压 Axle cabinet 3 battery under voltage	3	轴柜 3 电池欠电压信号=1 超过 500ms Signal of axle cabinet 3 battery under voltage = 1 lasts for over 500ms	手动复位 Manual reset
24.10	300121	桨叶 1 电池充电器的断路器断开 Blade 1 battery charger breaker open	3	桨叶 1 电池充电器的断路器信号=0 超过 500ms Signal of blade 1 battery charger breaker = 0 lasts for over 500ms	手动复位 Manual reset
24.11	300122	桨叶 2 电池充电器的断路器断开 Blade 2 battery charger breaker open	3	桨叶 2 电池充电器的断路器信号=0 超过 500ms Signal of blade 2 battery charger breaker = 0 lasts for over 500ms	手动复位 Manual reset

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
24.12	300123	桨叶 3 电池充电器的 断路器断开 Blade 3 battery charger breaker open	3	桨叶 3 电池充电器的断路器信号 =0 超过 500ms Signal of blade 3 battery charger breaker = 0 lasts for over 500ms	手动复位 Manual reset
24.13	300124	变换器 1 启动电阻 超温 Convertor 1 starting resistor over temperature	3	变换器 1 启动电阻超温信号=1 Signal of convertor 1 starting resistor over temperature = 1	手动复位 Manual reset
24.14	300125	变换器 2 启动电阻 超温 Convertor 2 starting resistor over temperature	3	变换器 2 启动电阻超温信号=1 Signal of convertor 2 starting resistor over temperature = 1	手动复位 Manual reset
24.15	300126	变换器 3 启动电阻 超温 Convertor 3 starting resistor over temperature	3	变换器 3 启动电阻超温信号=1 Signal of convertor 3 starting resistor over temperature = 1	手动复位 Manual reset
24.16	300131	桨叶 1 控制连接断开 Blade 1 control connection open	3	桨叶 1 控制连接信号=0 超过 500ms Signal of blade 1 control connection = 0 lasts for over 500ms	手动复位 Manual reset
24.17	300132	桨叶 2 控制连接断开 Blade 2 control connection open	3	桨叶 2 控制连接信号=0 超过 500ms Signal of blade 2 control connection = 0 lasts for over 500ms	手动复位 Manual reset
24.18	300133	桨叶 3 控制连接断开 Blade 3 control connection open	3	桨叶 3 控制连接信号=0 超过 500ms Signal of blade 3 control connection = 0 lasts for over 500ms	手动复位 Manual reset
24.19	300141	电池盒 1 温度超限 Battery 1 temperature over limit	3	电池盒 1 温度>上限值(65℃)或电 池盒 1 温度<下限值(-10℃)超过 20s Battery 1 temperature > upper limit(65℃) or battery 1 temperature < lower limit(-10℃) lasts for over 20s	手动复位 Manual reset



操作运行手册 Operating Manual

共 130 页

第 103 页

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
24.20	300142	电池盒 2 温度超限 Battery 2 temperature over limit	3	电池盒 2 温度>上限值(65℃)或电 池盒 2 温度<下限值(-10℃) 超过 20s Battery 2 temperature > upper limit(65℃) or battery 2 temperature < lower limit(-10℃) lasts for over 20s	手动复位 Manual reset
24.21	300143	电池盒 3 温度超限 Battery 3 temperature over limit	3	电池盒 3 温度>上限值(65℃)或电 池盒 3 温度<下限值(-10℃)超过 20s Battery 3 temperature > upper limit(65℃) or battery 3 temperature < lower limit(-10℃) lasts for over 20s	手动复位 Manual reset
24.22	300150	变桨电池运行信号 Pitch battery operating signal	3	变桨电池运行信号(di)=0 Pitch battery operating signal (di)=0	手动复位 Manual reset
24.23	300201	通信超时 Communication over time	3	160ms 内未收到变桨控制器的数 据 Haven't received pitch controller data for 160ms	手动复位 Manual reset
24.24	300311	桨叶 1 变换器故障 Blade 1 convertor error	1	桨叶 1 变换器故障信号=1 超过 1s Blade 1 convertor error signal =1 lasts for over 1s	手动复位 Manual reset
24.25	300312	桨叶 2 变换器故障 Blade 2 convertor error	1	桨叶 2 变换器故障信号=1 超过 1s Blade 2 convertor error signal =1 lasts for over 1s	手动复位 Manual reset
24.26	300313	桨叶 3 变换器故障 Blade 3 convertor error	1	桨叶 3 变换器故障信号=1 超过 1s Blade 3 convertor error signal =1 lasts for over 1s	手动复位 Manual reset
24.27	300321	桨叶 1 变换器直流侧 断路器断开 Blade 1 convertor DC side breaker open	1	桨叶 1 变换器直流侧断路器信号 =1 Blade 1 convertor DC side breaker signal = 1	手动复位 Manual reset
24.28	300322	桨叶 2 变换器直流侧 断路器断开 Blade 2 convertor DC side breaker open	1	桨叶 2 变换器直流侧断路器信号 =1 Blade 2 convertor DC side breaker signal = 1	手动复位 Manual reset

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操作运行手册 Operating Manual				共 130 页	第 104 页
表 A.1 (续) Continued table A.1					
序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
24.29	300323	桨叶 3 变换器直流侧 断路器断开 Blade 3 convertor DC side breaker open	1	桨叶 3 变换器直流侧断路器信号 =1 Blade 3 convertor DC side breaker signal = 1	手动复位 Manual reset
24.30	300331	桨叶 1 开关电源的 断路器断开 Blade 1 switching power supply breaker open	1	桨叶 1 开关电源的 断路器信号=1 Blade 1 switching power supply breaker signal =1	手动复位 Manual reset
24.31	300332	桨叶 2 开关电源的 断路器断开 Blade 2 switching power supply breaker open	1	桨叶 2 开关电源的 断路器信号=1 Blade 2 switching power supply breaker signal =1	手动复位 Manual reset
24.32	300333	桨叶 3 开关电源的 断路器断开 Blade 3 switching power supply breaker open	1	桨叶 3 开关电源的 断路器信号=1 Blade 3 switching power supply breaker signal =1	手动复位 Manual reset
24.33	300334	桨叶 1 电机断路器 断开 Blade 1 electric motor breaker open	1	桨叶 1 电机断路器=0 Blade electric motor breaker =0	手动复位 Manual reset
24.34	300335	桨叶 2 电机断路器 断开 Blade 2 electric motor breaker open	1	桨叶 2 电机断路器=0 Blade 2 electric motor breaker =0	手动复位 Manual reset
24.35	300336	桨叶 3 电机断路器 断开 Blade 3 electric motor breaker open	1	桨叶 3 电机断路器=0 Blade 3 electric motor breaker =0	手动复位 Manual reset
24.36	300341	桨叶 1 的变换器电源故 障 Blade 1 convetor power supply error	1	桨叶 1 的变换器电源信号=0 Blade 1 convetor power supply signal =0	手动复位 Manual reset
24.37	300342	桨叶 2 的变换器电源故 障 Blade 2 convetor power supply error	1	桨叶 2 的变换器电源信号=0 Blade 2 convetor power supply signal =0	手动复位 Manual reset



操作运行手册 Operating Manual

共 130 页

第 105 页

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
24.38	300343	桨叶 3 的变换器电源故障 Blade 3 convertor power supply error	1	桨叶 3 的变换器电源信号=0 Blade 3 convertor power supply signal =0	手动复位 Manual reset
24.39	300347	桨叶 1 变换器直流侧熔断器断开 Blade 1 convertor DC side fuse open	1	桨叶 1 变换器直流侧熔断器信号=1 Blade 1 convertor DC side fuse signal = 1	手动复位 Manual reset
24.40	300348	桨叶 2 变换器直流侧熔断器断开 Blade 2 convertor DC side fuse open	1	桨叶 2 变换器直流侧熔断器信号=1 Blade 2 convertor DC side fuse signal = 1	手动复位 Manual reset
24.41	300349	桨叶 3 变换器直流侧熔断器断开 Blade 3 convertor DC side fuse open	1	桨叶 3 变换器直流侧熔断器信号=1 Blade 3 convertor DC side fuse signal = 1	手动复位 Manual reset
24.42	300401	桨叶 1 的加热器或风扇的断路器断开 Break of blade 1 heater or fan open	1	桨叶 1 的加热器或风扇的断路器信号=1 Signal of break of blade 1 heater or fan =1	手动复位 Manual reset
24.43	300402	桨叶 2 的加热器或风扇的断路器断开 Break of blade 2 heater or fan open	1	桨叶 2 的加热器或风扇的断路器信号=1 Signal of break of blade 2 heater or fan =1	手动复位 Manual reset
24.44	300403	桨叶 3 的加热器或风扇的断路器断开 Break of blade 3 heater or fan open	1	桨叶 3 的加热器或风扇的断路器信号=1 Signal of break of blade 3 heater or fan =1	手动复位 Manual reset
24.45	300511	桨叶 1 的电机温度超限 Blade 1 electric motor temperature over limit	1	桨叶 1 的电机温度>上限值(140℃)或桨叶 1 的电机温度<下限值(-60℃)超过 5s Blade 1 electric motor temperature > upper limit (140℃) or < lower limit (-60℃) lasts for over 5s	手动复位 Manual reset
24.46	300512	桨叶 2 的电机温度超限 Blade 2 electric motor temperature over limit	1	桨叶 2 的电机温度>上限值(140℃)或桨叶 2 的电机温度<下限值(-60℃)超过 5s Blade 2 electric motor temperature > upper limit (140℃) or < lower limit (-60℃) lasts for over 5s	手动复位 Manual reset

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
24.47	300513	桨叶 3 的电机温度超限 Blade 3 electric motor temperature over limit	1	桨叶 3 的电机温度>上限值(140℃) 或桨叶 3 的电机温度<下限值 (-60℃)超过 5s Blade 3 electric motor temperature > upper limit (140℃) or < lower limit (-60℃) lasts for over 5s	手动复位 Manual reset
24.48	300611	桨叶 1 的限位开关触发 Blade 1 limit switch triggered	3	桨叶 1 的限位开关信号=1 Blade 1 limit switch signal =1	手动复位 Manual reset
24.49	300612	桨叶 2 的限位开关触发 Blade 2 limit switch triggered	3	桨叶 2 的限位开关信号=1 Blade 2 limit switch signal =1	手动复位 Manual reset
24.50	300613	桨叶 3 的限位开关触发 Blade 3 limit switch triggered	3	桨叶 3 的限位开关信号=1 Blade 3 limit switch signal =1	手动复位 Manual reset
24.51	300621	桨叶 1 的 91°限位开关 触发 Blade 1 91°limit switch triggered	3	桨叶 1 的 91°限位开关信号=1 Blade 1 91°limit switch signal =1	手动复位 Manual reset
24.52	300622	桨叶 2 的 91°限位开关 触发 Blade 2 91°limit switch triggered	3	桨叶 2 的 91°限位开关信号=1 Blade 2 91°limit switch signal =1	手动复位 Manual reset
24.53	300623	桨叶 3 的 91°限位开关 触发 Blade 3 91°limit switch triggered	3	桨叶 3 的 91°限位开关信号=1 Blade 3 91°limit switch signal =1	手动复位 Manual reset
24.54	300631	桨叶 1 的 96° 限位开关触发 Blade 1 96°limit switch triggered	3	桨叶 1 的 96°限位开关信号=1 Blade 1 96°limit switch triggered	手动复位 Manual reset
24.55	300632	桨叶 2 的 96°限位开关 触发 Blade 2 96°limit switch triggered	3	桨叶 2 的 96°限位开关信号=1 Blade 2 96°limit switch triggered	手动复位 Manual reset
24.56	300633	桨叶 3 的 96°限位开关 触发 Blade 3 96°limit switch triggered	3	桨叶 3 的 96°限位开关信号=1 Blade 3 96°limit switch signal =1	手动复位 Manual reset



操作运行手册 Operating Manual

共 130 页

第 107 页

表 A.1 (续) Continued table A.1

序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
24.57	300641	桨叶 1 的编码器偏差超限 Blade 1 coder difference over limit	1	桨叶 1 的 2 个编码器编码器 读数差值超过限值(0.2 °) Difference between 2 coder readings of blade 1 over limit (0.2 °)	手动复位 Manual reset
24.58	300642	桨叶 2 的编码器偏差超限 Blade 2 coder difference over limit	1	桨叶 2 的 2 个编码器编码器 读数差值超过限值(0.2 °) Difference between 2 coder readings of blade 2 over limit (0.2 °)	手动复位 Manual reset
24.59	300643	桨叶 3 的编码器偏差超限 Blade 3 coder difference over limit	1	桨叶 3 的 2 个编码器编码器 读数差值超过限值(0.2 °) Difference between 2 coder readings of blade 3 over limit (0.2 °)	手动复位 Manual reset
24.60	300670	桨距角两两偏差超限 Pitch angle difference between each other over limit	3	3 个桨叶的主编码器两两差值的 最大值超过限值(0.2 °) Maximum difference of 3 pitch angle coder between each other exceeds limit (0.2 °)	手动复位 Manual reset
24.61	300681	桨叶 1 桨距角偏差超限 Pitch 1 angle difference over limit	3	启动模式下, 桨叶 1 的桨距角与 设定桨距角的偏差超过限值(0.2 °) During start mode, difference between pitch 1 angle and set pitch angle exceeds limit (0.2 °)	手动复位 Manual reset
24.62	300682	桨叶 2 桨距角偏差超限 Pitch 2 angle difference over limit	3	启动模式下, 桨叶 2 的桨距角与 设定桨距角的偏差超过限值(0.2 °) During start mode, difference between pitch 2 angle and set pitch angle exceeds limit (0.2 °)	手动复位 Manual reset
24.63	300683	桨叶 3 桨距角偏差超限 Pitch 3 angle difference over limit	3	启动模式下, 桨叶 3 的桨距角与 设定桨距角的偏差超过限(0.2 °) During start mode, difference between pitch 3 angle and set pitch angle exceeds limit (0.2 °)	手动复位 Manual reset
24.64	300705	变桨手动维护 Pitch manual maintenance	1	启动, 运行, 发电或停机模式下, 变桨手动维护信号=1 During start, run, generation or stop mode, pitch manual maintenance signal=1	手动复位 Manual reset

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操作运行手册 Operating Manual				共 130 页	第 108 页
表 A.1 (续) Continued table A.1					
序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
24.65	300709	变桨紧停 Pitch emergency stop	1	启动, 运行, 发电或停机模式下, 变桨紧停信号=1 During start, run, generation or stop mode, pitch emergency stop signal=1	手动复位 Manual reset
24.66	300711	桨叶 1 的轴柜温度超限 Axle cabinet 1 temperature over limit	1	桨叶 1 的轴柜温度>上限值(90℃) 或桨叶 1 的电机温度<下限值(0℃) 超过 25s Axle cabinet 1 temperature > upper limit (90℃) or < lower limit (0℃) lasts for over 25s	手动复位 Manual reset
24.67	300712	桨叶 2 的轴柜温度超限 Axle cabinet 3 temperature over limit	1	桨叶 2 的轴柜温度>上限值(90℃) 或桨叶 2 的电机温度<下限值(0℃) 超过 25s Axle cabinet 2 temperature > upper limit (90℃) or < lower limit (0℃) lasts for over 25s	手动复位 Manual reset
24.68	300713	桨叶 3 的轴柜温度超限 Axle cabinet 3 temperature over limit	1	桨叶 3 的轴柜温度>上限值(90℃) 或桨叶 3 的电机温度<下限值(0℃) 超过 25s Axle cabinet 3 temperature > upper limit (90℃) or < lower limit (0℃) lasts for over 25s	手动复位 Manual reset
24.69	300720	轮毂温度超限 Rotor hub temperauter over limit	1	轮毂温度>上限值(90℃)或轮毂温 度<下限值(0℃)超过 25s Rotor hub temperature > upper limit (90℃) or < lower limit (0℃) lasts for over 25s	手动复位 Manual reset
25 塔底冷却故障 Tower base cooling errors (350000)					
25.1	350005	塔底冷却风扇保护 Tower base cooling fan protection	1	塔底冷却风扇保护信号(di)=0 超 过 10s Tower base cooling fan protection signal (di)=0 lasts for over 10s	手动复位 Manual reset
25.2	350011	塔底温度超限 Tower base temperature over limit	1	塔底温度(ai)>限值(85℃)超过 30s Tower base temperature (ai) > limit (85℃) lasts for over 30s	手动复位 Manual reset
26 航灯故障 Aircraft light errors (650000)					
26.1	650001	航灯保护 Aircraft light protection	1	航标灯保护 ok 信号(di)=0 超过 5s Ok signal of aircraft light protection (di)=0 lasts for over 5s	手动复位 Manual reset

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操作运行手册 Operating Manual				共 130 页	第 109 页
表 A.1 (续) Continued table A.1					
序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
27 风轮转速故障 Rotor speed errors (700000)					
27.1	700002	风轮转速比较故障 Rotor speed comparison erroe	1	风轮转速>4.5rpm 时, 风轮转速 1、风轮转速 2、发电机转速/变速比三个值的差分>限值(1rpm)超过 3s When rotor speed>4.5tpm, difference between rotor speed 1, rotor speed 2, and generator speed/shift rate > limit(1rpm) lasts for over 3s	手动复位 Manual reset
27.2	700003	风轮转速超过临界转速 Rotor speed over critical speed	2	风轮转速>限值(18.5rpm) Rotor speed > limit (18.5rpm)	手动复位 Manual reset
27.3	700004	风轮转速超过紧停转速 Rotor speed over emergency stop speed	3	风轮转速>限值(19rpm) Rotor speed > limit (19rpm)	手动复位 Manual reset
28 主流程故障 Main flow errors (760000)					
28.1	760011	自检 1 故障 Self-test 1 error	1	20s 内桨距角未达到 70°±2°范围内 Pitch angle doesn't reach in the range of 70°±2° in 20s	手动复位 Manual reset
28.2	760012	自检 2 故障 Self-test 1 error	1	系统主压力大于 170bar 或小于 135bar 或风轮转速大于 1rpm System main pressure > 170bar or <135bar or rotor speed >1rpm	手动复位 Manual reset
28.3	760013	自检 3 故障 Self-test 1 error	1	风轮转速 1 小于 0.1rpm 且风轮转速 2 小于 0.1rpm Rotor speed 1 < 0.1rpm and rotor speed 2 < 0.1 rpm	手动复位 Manual reset
29 安全系统故障 Safety system errors (800000)					
29.1	800001	塔底紧停开关触发 Tower base emergency switch triggered	3	塔底紧停按钮触发(Safety di)=0 Tower base emergency button triggered (Safety di)=0	手动复位 Manual reset
29.2	800002	机舱紧停开关触发 Nacelle emergency switch triggered	3	机舱紧停按钮触发(Safety di)=0 Nacelle emergency button triggered (Safety di)=0	手动复位 Manual reset

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操作运行手册 Operating Manual				共 130 页	第 110 页
表 A.1 (续) Continued table A.1					
序号 No.	状态代码 State code	状态代码名称 Designation	停机程序 Stop program	激活条件 Activate condition	复位方式 Reset method
29.3	800003	机舱远程紧停开关触发 Nacelle remote emergency switch triggered	3	机舱远程紧停按钮触发(Safety di)=0 Nacelle remote emergency button triggered (Safety di)=0	手动复位 Manual reset
29.4	800011	发电机超速 Generator over speed	3	安全链发电机超速信号触发 (Safety di)=0 Safety chain generator over speed signal (Safety di)=0	手动复位 Manual reset
29.5	800015	机舱振动开关动作 Nacelle libration switch action	3	安全链机舱振动信号触发(Safety di)=0 Safety chain nacelle libration signal (Safety di)=0	手动复位 Manual reset
29.6	800017	偏航扭缆限位开关动作 Unwinding switch action	3	安全链偏航扭缆信号触发(Safety di)=0 Safety chain unwinding signal (Safety di)=0	手动复位 Manual reset
29.7	800020	变桨系统紧停动作 Pitch system emergency stop action	3	安全链变桨紧停信号触发(Safety di)=0 Safety chain pitch emergency stop signal (Safety di)=0	手动复位 Manual reset
29.8	800030	PLC 紧停动作 PLC emergency action	3	安全链 PLC 紧停信号触发(Safety di)=1 Safety chain PLC emergency stop signal (Safety di)=0	手动复位 Manual reset
29.9	800051	安全输入模块故障 Safety input module error	3	由安全模块监测 Detected by safety module	手动复位 Manual reset
29.10	800052	安全输出模块故障 Safety output module error	3	由安全模块监测 Detected by safety module	手动复位 Manual reset
29.11	800053	安全模块通信故障 Safety module communication error	3	由安全模块监测 Detected by safety module	手动复位 Manual reset
29.12	800070	安全系统紧停 Safety system emergency stop	3	安全输出模块(Safety do)=0 Safety output module (Safety do)=0	手动复位 Manual reset



附录 B 数据记录 Appendix B Data records

数据记录对历史数据进行统计后存储在 CF 卡上, 历史数据包括操作数据表、功率曲线表和风玫瑰图表。此外, 状态代码列表存储最近发生的 100 条状态代码, 状态代码统计表对所有状态代码的激活和复位次数进行统计。

Data records are stored in CF card, after history data statistic. History data includes operating data table, power curve table and wind rose diagram. In addition, state code lists stores 100 state codes recently occurred, and analyse the active and reset times of all the state codes statistically.

附录 B.1 操作数据表 Appendix B.1 Operation data table

1) 操作数据表按时间存成下述 3 个 txt 文件, 存储在 OpertionData 文件夹下:

- TotalYears_Operation_Data.txt—总报表, 风机从开始运行以来的运行数据。
- Years_Operation_Data_20xx.txt—年报表, 风机某一年(20xx)的运行数据, 每月一条记录, 存整个一年的数据(12 个月)。
- Months_Operation_Data_20xx_xx.txt—月报表, 风机某年某月(20xx_xx)的运行数据, 每天一条记录, 存整个一个月的数据(28 天, 29 天, 30 天或 31 天)。

2) 操作数据表存储的内容包括:

- 日期 Date
- 电量 Energy(kWh)
- 可利用率 Availability
- 工作时间 WorkTime(min)
- 故障时间 FaultTime(min)
- 停机时间 StopTime(min)
- 发电时间 PD.Time(min)
- 偏航时间 YawTime(min)
- 顺时针偏航时间 CWYawTime(min)
- 逆时针偏航时间 CCWYawTime(min)
- 维护时间 ServiceTime(min)
- 停机次数 StopNr
- 并网次数 CutinNr
- 偏航次数 YawNr
- 主断路器闭合次数 CircuitOnNr
- 维护次数 ServiceNr

3) 操作数据表每 5min 存储一次。

1) Operation data table is divided into 3 txt files below, and stored in the OpertionData folder:

- TotalYears_Operation_Data.txt—Total report form, includes operation data from the first running of the wind turbine.
- Years_Operation_Data_20xx.txt—Annual report form, includes operation data of the wind turbine in one year (20xx). Records of the whole year (12months) are recorded, in form of a record each month.
- Months_Operation_Data_20xx_xx.txt—Monthly report form, includes operation data of the wind turbine in one month (20xx_xx). Records of the whole moth(28 days, 29 days, 30 days or 31 days) are recorded, in form of a record each day.

2) Data recorded in operation data table includes:

- Date
- Energy(kWh)
- Availability
- WorkTime(min)
- FaultTime(min)
- StopTime(min)
- PD.Time(min)
- YawTime(min)
- CWYawTime(min)
- CCWYawTime(min)
- ServiceTime(min)
- StopNr
- CutinNr
- YawNr
- CircuitOnNr
- ServiceNr

3) Operation data table is stored every 5min.

附录 B.2 功率曲线表 Appendix B.2 Power curve table

1) 功率曲线数据存成下述 3 个 txt 文件, 存储在 PowerCurve 文件夹下:

- PowerCurve.txt—总报表, 风机从开始运行以来的功率曲线数据。
- PowerCurve_20xx.txt—年报表, 风机某一年(20xx)的功率曲线数据。
- PowerCurve_20xx_xx.txt—月报表, 风机某年某月(20xx_xx)的功率曲线数据。
- 根据 0~25m/s 的风速分成 50 个区间, 每个 txt 文件分 50 行保存相关数据。数据每 1 小时



更新一次。每到下一月，生成新的月报表，每到下一年，生成新的年报表。

2) 功率曲线表包括以下数据：

- 等级 Class
- 区间 Range
- 发生次数 Values
- 时间 Times
- 平均风速 MeanWindSpeed(m/s)
- 平均功率 MeanPower(kW)

3) 功率曲线表每 5min 存储一次。

1) Power curve table is divided into 3 txt files below, and stored in the PowerCurve folder:

- PowerCurve.txt—Total report form, includes power curve data from the first running of the wind turbine.
- PowerCurve_20xx.txt—Annual report form, includes power curve data of the wind turbine in one year (20xx).
- PowerCurve_20xx_xx.txt—Monthly report form, includes power curve data of the wind turbine in one month (20xx_xx).
- The wind speed is divided into 50 intervals, and every txt file saves data by 50 lines accordingly.
- Data are updated every 1 hour. A new monthly report form will be created, as soon as a new month comes; a new annual report form will be created, as soon as a new year comes.

2) Power curve table includes data below:

- Class
- Range
- Values
- Times
- MeanWindSpeed(m/s)
- MeanPower(kW)

3) Power curve table is stored every 5min.

附录 B.3 风玫瑰图表 Appendix B.3 Wind rose table

1) 风玫瑰图数据存成 3 个 txt 文件，存储在 WindRose 文件夹下：

- WindRose.txt—总报表，风机从开始运行以来的风玫瑰图数据；
- WindRose_20xx.txt—年报表，风机某一年(20xx)的风玫瑰图数据；
- WindRose_20xx_xx.txt—月报表，风机某年某月(20xx_xx)的风玫瑰图数据。

 上海电气 上海电气风电设备有限公司 SHANGHAI ELECTRIC SHANGHAI ELECTRIC WINDPOWER EQUIPMENT CO.,LTD.		W2000 风电机组	MD-030.00-A	
操作运行手册 Operating Manual			共 130 页	第 114 页
• 根据 0~360° 的风向分成 12 个区间，每个 txt 文件分 20 行保存相关数据。数据每 1 小时更新一次。每到下一月，生成新的月报表，每到下一年，生成新的年报表。 2) 风玫瑰图表包括以下数据： • 等级 Class • 区间 Range • 发生次数 Values • 时间 Times • 平均风速 MeanWindSpeed(m/s) • 平均功率 MeanPower(kW) • 平均机舱位置 MeanNacellePossion • 风速的三次方 WindSpeed_3 3) 风玫瑰图表每 5min 存储一次。 1) Wind rose table is divided into 3 txt files below, and stored in the WindRosefolder: • WindRose.txt—Total report form, includes wind rose data from the first running of the wind turbine. • WindRose _20xx.txt—Annual report form, includes wind rose data of the wind turbine in one year (20xx). • WindRose _20xx_xx.txt—Monthly report form, includes wind rose data of the wind turbine in one month (20xx_xx). • The wind direction is divided into 12 intervals, and every txt file saves data by 20 lines accordingly. Data are updated every 1 hour. A new monthly report form will be created, as soon as a new month comes; a new annual report form will be created, as soon as a new year comes. 2) Wind rose table includes data below: • Class • Range • Values • Times • MeanWindSpeed(m/s) • MeanPower(kW) • MeanNacellePossion • WindSpeed_3 3) Wind rose table is stored every 5min.				
附录 B. 4 状态代码列表 AppendixB.4 List of state code				



1) 状态代码列表记录最近发生的 100 条状态代码信息，存储在 Init 文件夹下的 StatusList.txt 文件中，文件抬头内容包括公司名，风机类型以及风机号。

2) 状态代码列表包括以下数据：

- 序号 Class
- 状态代码编号 StatusCode
- 最近一次激活的时间 ActiveTime
- 最近一次复位的时间 ResetTime

3) 状态代码列表每 1min 存储一次。

1) State code list records 100 state codes occurred lately, and saves them in the file named StatusList.txt at Init folder. The header information of the file includes company name, wind turbine types and wind turbine number.

2) State code list includes the following data:

- Serial number: Class
- State code number: StatusCode
- The last activated time: ActiveTime
- The last reset time: ResetTime

3) State code list saves every 1 minute.

附录 B. 5 状态代码统计表 Appendix B.5 State code statistics

1) 状态代码统计表记录所有状态代码的激活和复位信息，存储在 Init 文件夹下 StatusCount.txt 文件中，文件抬头内容包括公司名，风机类型以及风机号。

2) 状态代码统计表包括以下数据：

- 序号 Class
- 状态代码编号 StatusCode
- 最近一次激活的时间 LastActiveTime
- 当天激活次数 ActiveCount
- 最近一次复位的时间 LastResetTime
- 当天复位次数 ResetCount
- 激活总次数 ActiveTotal
- 复位总次数 ResetTotal

状态代码统计表每 1min 存储一次。

1) State code list records all activated and reset information of the state code, and saves them in the file named StatusCount.txt at Init folder. The header information of the file includes company name, wind turbine types and wind turbine number.

2) State code list includes the following data:

- Serial number: Class
- State code number: StatusCode
- The last activated time: LastActiveTime
- Activated number of the day: ActiveCount
- The last reset time: LastResetTime
- Reset time of the day: ResetCount
- Total activated number: ActiveTotal
- Total reset number: ResetTotal

State code list saves every 1 minute.





附录 C I/O 列表 AppendixC I/O List

说明:

DI: 数字量输入

DO: 为数字量输出

AI: 模拟量输入

AO: 模拟量输出。

端口: 指该 I/O 量在电气原理图的位置。

I/O 列表分为机舱和塔底两部分。

Instruction:

DI: Digital signals input

DO: Digital signals output

AI: Analog signals input

AO: Analog signals output

Port: Position of the I/O signals located at the electric schematics

The I/O list is comprised of I/O list of the nacelle and tower base.

附录 C.1 塔底 I/O 列表 AppendixC.1 I/O list of tower base

注: 塔底 I/O 信号由于各项目箱变配置及要求不同而可能不同。

Annotation: the I/O signals located at tower base may be different because of the configuration and requirement of the transformer is different.

附录 C.1.1 塔底 DI 列表 AppendixC.1.1 DI list of tower base

附录 C.1.1.1 普通 DI 信号列表 AppendixC.1.1.1 Normal DI signals list

表 C.1 Table C.1

端口 Port	说明 Instruction
213DI1-1	塔底控制柜 UPS 电池 ok UPS battery in control cabinet located at tower base ok
213DI1-4	塔底启动按钮 Start button at tower base
213DI1-5	塔底控制柜 UPS 供电模式 UPS power-supply mode of control cabinet at tower base

表 C.1 (续) Continued table C.1

端口 Port	说明 Instruction
213DI1-8	塔底停止按钮 Stop button at tower base
213DI2-1	塔底复位按钮 Reset button at tower base
213DI2-4	塔底紧停开关 Emergency switch at tower base
213DI2-5	塔底远程紧停 Remotely emergency stop at tower base
213DI2-8	备用 Reserve
213DI3-1	塔底服务模式开关 Service mode switch at tower base
213DI3-4	备用 Reserve
213DI3-5	塔底箱变保护 Protection of transformer at tower base
213DI3-8	塔底控制柜加热器保护 ok Protection of heater in control cabinet at tower base ok
213DI4-1	塔底熔断器反馈 Fuse feedback at tower base
213DI4-4	塔底风扇保护 ok Protection of fan at tower base ok
213DI4-5	塔底避雷保护 Lightning protection at tower base
213DI4-8	备用 Reserve
213DI5-1	变压器超温开关动作信号 Transformer over-temperature
213DI5-4	塔底主电源 ok Main power at tower base ok
213DI5-5	箱变轻瓦斯报警信号 Light gas warning of the transformer cabinet
213DI5-8	箱变重瓦斯故障信号 Severe gas warning of the transformer cabinet



附录 C. 1. 1. 2 安全 DI 信号列表 AppendixC.1.1.2 List of safety DI signals

表 C.2 Table C.2

端口 Port	说明 Instruction
214DI1-1/2	安全系统塔底紧停按钮 Emergency stop button of safety system at tower base
214DI1-3/4	安全系统 PLC 紧停 PLC emergency stop of safety system
214DI1-5/6	安全系统塔底复位按钮 Reset button of safety system at tower base
214DI1-7/8	备用 Reserve

附录 C. 1. 2 塔底 DO 信号列表 AppendixC.1.2 List of DO signals at tower base

表 C.3 Table C.3

端口 Port	说明 Instruction
214DO1-1	塔底控制柜冷却风扇 Cooling fan in control cabinet at tower base
214DO1-4	塔底故障灯 Fault lamps at tower base
214DO1-5	塔底准备就绪灯 Ready lamp at tower base
214DO1-8	塔底操作灯 Operation lamp at tower base
214DO2-1	塔底服务模式灯 Service mode lamp at tower base
214DO2-4	塔底紧停开关激活灯 Emergency stop switch activated lamp at tower base
214DO2-5	塔底安全链断开灯 Safety chain open lamp at tower base
214DO2-8	塔底风轮锁定灯 Rotor-locked lamp at tower base
214DO3-1	塔底并网灯 Grid-connected lamp at tower base
214DO3-4	塔底风扇 Fan at tower base
214DO3-5	安全系统 PLC 紧停 PLC emergency stop of safety system
214DO3-8	塔底脉冲测试 Pulse-test at tower base
214DO4-1	塔底 400VAC 主电源保护 Protection of 400VAC main source at tower base

表 C.3 (续) Continued table C.3

端口 Port	说明 Instruction
214DO4-4	备用 Reserve
214DO4-5	备用 Reserve
214DO4-8	备用 Reserve

附录 C. 1. 3 塔底 AI 信号列表 Appendix C.1.2 List of AI signals at tower base

表 C.4 Table C.4

端口 Port	说明 Instruction
214AI1-1/2	塔底控制柜温度 Temperature of control cabinet at tower base
214AI1-3/4	塔底环境温度 Environment temperature at tower base
214AI1-5/6	箱变温度 Transformer temperature
214AI1-7/8	备用 Reserve
214AI2-L1	L1 相电压 Phase voltage of L1
214AI2-L2	L2 相电压 Phase voltage of L2
214AI2-L3	L3 相电压 Phase voltage of L3
214AI2-N	中性点 N Neutral point N
214AI2-I1	L1 相电流 Phase current of L1
214AI2-I2	L2 相电流 Phase current of L2
214AI2-I3	L3 相电流 Phase current of L3
214AI2-In	电流 In Current In



附录 C. 2 机舱 I/O 列表 AppendixC.2 I/O list of nacelle

附录 C. 2. 1 机舱 DI 信号列表 AppendixC.2.1 DI signals list of nacelle

附录 C. 2. 1. 1 普通 DI 信号列表 AppendixC.2.1.1 Normal DI signals list

表 C.5 Table C.5

端口 Port	说明 Instruction
453DI1-1	齿轮箱油位 ok Oil level of gearbox ok
453DI1-4	齿轮箱润滑泵马达保护 ok Protection of gearbox lubrication pump motor ok
453DI1-5	齿轮箱冷却水泵保护 ok Protection of gearbox cooling pump ok
453DI1-8	备用 Reserve
453DI2-1	备用 Reserve
453DI2-4	备用 Reserve
453DI2-5	齿轮箱润滑油过滤器压力 ok Gearbox lubrication oil filter pressure ok
453DI2-8	齿轮箱冷却水压力 ok Pressure of gearbox cooling water ok
453DI3-1	齿轮箱加热器保护 ok Protection of gearbox heater ok
453DI3-4	齿轮箱冷却水泵加热器保护 ok Protection of gearbox cooling water pump heater ok
453DI3-5	齿轮箱润滑油加热器保护 ok Protection of gearbox lubrication heater ok
453DI3-8	齿轮箱润滑压力 ok Pressure of gearbox lubrication ok
453DI4-1	机舱停止按钮 Stop button in nacelle
453DI4-4	机舱紧停开关 Emergency stop in nacelle
453DI4-5	机舱复位按钮 Reset button in nacelle
453DI4-8	机舱服务开关 Service switch in nacelle
453DI5-1	机舱远程风轮制动按钮 Remote rotor brake button in nacelle
453DI5-4	备用 Reserve

表 C.5 (续) Continued table C.5

端口 Port	说明 Instruction
453DI5-5	机舱远程紧停开关 Remote emergency stop in nacelle
453DI5-8	机舱启动按钮 Start button in nacelle
453DI6-1	机舱左偏航按钮 Yaw left button in nacelle
453DI6-4	机舱前变桨按钮 Pitch forward button in nacelle
453DI6-5	机舱右偏航按钮 Yaw right button in nacelle
453DI6-8	机舱后变桨按钮 Pitch backward button in nacelle
453DI7-1	变桨旁路极限开关 Pitch bypass utmost switch
453DI7-4	手动变桨开关 Manually pitch switch
453DI7-5	变桨 400VAC 供电保护 ok Protection 400VAC power supply for pitch system ok
453DI7-8	变桨 230VAC 供电保护 ok Protection 230VAC power supply for pitch system ok
453DI8-1	备用 Reserve
453DI8-4	备用 Reserve
453DI8-5	变桨电池运行信号 Pitch battery running signals
453DI8-8	备用 Reserve
453DI9-1	齿轮箱冷却水风扇保护 ok Protection of gearbox cooling water fan ok
453DI9-4	发电机定子避雷器保护 Lightning protection of generator stator
453DI9-5	变桨 91 度限位开关 91 degree limit switch for pitch
453DI9-8	备用 Reserve
454DI1-1	风轮锁定 Rotor locked
454DI1-4	机舱控制柜加热器风扇保护 ok Protection of heater fan in control cabinet in nacelle ok



表 C.5 (续) Continued table C.5

端口 Port	说明 Instruction
454DI1-5	风轮解锁 Rotor unlocked
454DI1-8	液压过滤器入口压力 ok Pressure of hydraulic filter entrance ok
454DI2-1	液压油温 ok Oil temperature of the hydraulic system ok
454DI2-4	液压油泵加热器保护 ok Protection of hydraulic pump heater ok
454DI2-5	液压油位 ok Oil level of hydraulic system
454DI2-8	液压泵马达保护 ok Protection of hydraulic motor ok
454DI2.1-1	风轮制动垫报警 Rotor brake pad warning
454DI2.1-4	液压风轮制动压力 ok Pressure of rotor brake hydraulic ok
454DI2.1-5	风轮制动垫停止 Rotor brake pad stop
454DI2.1-8	备用 Reserve
454DI3-1	偏航保护 ok Yaw protection ok
454DI3-4	偏航软启动器 ok Yaw soft start ok
454DI3-5	偏航制动保护 ok Protection of yaw brake ok
454DI3-8	偏航软启动器斜率 ok Slope of yaw soft-start ok
454DI4-1	发电机定子加热器 1 保护 ok Protection of generator stator heater1 ok
454DI4-4	液压过滤器出口压力 ok Pressure of hydraulic filter exit ok
454DI4-5	发电机定子加热器 2 保护 ok Protection of generator stator heater2 ok
454DI4-8	发电机内部冷却风扇保护 ok Protection of cooling fan inside generator ok
454DI4.1-1	左偏航停止位置 Stop position of yaw left
454DI4.1-4	发电机电刷 ok Generator brush ok

表 C.5 (续) Continued table C.5

端口 Port	说明 Instruction
454DI4.1-5	右偏航停止位置 Stop position of yaw right
454DI4.1-8	发电机转子避雷保护 Lightning protection of generator rotor
454DI5-1	偏航轴承润滑油位满 Oil level of yaw bearing lubrication full
454DI5-4	航灯保护 ok Protection of aircraft lights
454DI5-5	偏航轴承润滑泵停止 Yaw bearing lubrication pump stop
454DI5-8	偏航齿轮润滑油位满 Oil level of yaw bearing gear lubrication full
454DI6-1	机舱控制柜内 24VDC 供电 ok 24VDC power supply for control cabinet in nacelle ok
454DI6-4	偏航齿轮润滑泵停止 Yaw bearing gear lubrication pump stop
454DI6-5	机舱外部传感器 24VDC 供电 ok 24VDC power supply for sensors outside the nacelle ok
454DI6-8	机舱控制柜避雷保护 Lightning protection of control cabinet in nacelle
454DI6.1-1	机舱控制柜 UPS 电池 ok UPS battery in nacelle ok
454DI6.1-4	机舱风扇保护 Protection of fans in nacelle
454DI6.1-5	机舱控制柜 UPS 供电模式 Control cabinet in nacelle at UPS power supply mode
454DI6.1-8	桨叶 1 锁定 Pitch 1 locked
454DI7-1	偏航马达 1 保护 ok Protection of yawing motor 1 ok
454DI7-4	偏航马达 2 保护 ok Protection of yawing motor 2 ok
454DI7-5	偏航马达 3 保护 ok Protection of yawing motor 3 ok
454DI7-8	偏航马达 4 保护 ok Protection of yawing motor 4 ok
454DI8-1	主轴承润滑油位满 Oil level of rotor bearing lubrication full
454DI8-4	桨叶 2 锁定 Blade 2 locked



表 C.5 (续) Continued table C.5

端口 Port	说明 Instruction
454DI8-5	主轴承润滑泵停止 Rotor bearing lubrication pump stop
454DI8-8	桨叶 3 锁定 Blade 3 locked

附录 C. 2. 1. 2 安全 DI 信号列表 AppendixC.2.1.2 Safety DI signals list

表 C.6 Table C.6

端口 Port	说明 Instruction
456DI1-1/2	安全系统机舱紧停按钮 Emergency stop button in nacelle of safety system
456DI1-3/4	安全系统机舱远程紧停开关 Remote emergency stop switch of safety system
456DI1-5/6	安全系统风轮超速 1 紧停 Rotor speed exceeds emergency stop1 of safety system
456DI1-7/8	安全系统机舱振动紧停 Nacelle vibration emergency stop of safety system
456DI2-1/2	安全系统偏航扭缆紧停 Yawing twist cable emergency stop of safety system
456DI2-3/4	安全系统风轮超速 2 紧停 Rotor speed exceeds emergency stop2 of safety system
456DI2-5/6	安全系统变桨紧停 Pitch emergency stop of safety system
456DI2-7/8	安全系统机舱复位按钮 Reset button in nacelle of safety system

附录 C. 2. 2 机舱 DO 信号列表 AppendixC.2.2 DO signals list of nacelle

附录 C. 2. 2. 1 普通 DO 信号列表 AppendixC.2.2.1 Normal DO signals list

表 C.7 Table C.7

端口 Port	说明 Instruction
455DO1-1	机舱故障灯 Fault lamp in nacelle
455DO1-4	机舱服务模式灯 Service mode lamp in nacelle
455DO1-5	机舱运行模式灯 Operation mode lamp in nacelle
455DO1-8	机舱安全链断开灯 Safety chain open lamp in nacelle

表 C.7 (续) Continued table C.7

端口 Port	说明 Instruction
455DO2-1	机舱紧停激活灯 Emergency stop activated lamp in nacelle
455DO2-4	机舱控制柜风扇 Fan in control cabinet in nacelle
455DO2-5	机舱风轮锁定灯 Rotor locked lamp in nacelle
455DO2-8	机舱脉冲测试灯 Pulse test lamp in nacelle
455DO2.1-1	齿轮箱高速润滑油泵 Gearbox high-speed lubrication pump
455DO2.1-4	机舱风扇 Nacelle fan
455DO2.1-5	齿轮箱低速润滑油泵 Gearbox low-speed lubrication pump
455DO2.1-8	齿轮箱润滑泵启动 Gearbox lubrication pump start
455DO3-1	齿轮箱润滑油泵加热器 Gearbox lubrication pump heater
455DO3-4	备用 Reserve
455DO3-5	齿轮箱润滑油加热器 Gearbox lubrication oil heater
455DO3-8	齿轮箱冷却水泵 Gearbox cooling water pump
455DO4-1	发电机定子加热器锁定 Heater of generator stator locked
455DO4-4	发电机滑环风扇 Generator slip ring fan
455DO4-5	齿轮箱冷却水泵加热器 Gearbox cooling water pump heater
455DO4-8	备用 Reserve
455DO4.1-1	风轮位置零位设置 Zero setting of rotor position
455DO4.1-4	偏航维护零位设置 Zero setting for yawing service
455DO4.1-5	备用 Reserve
455DO4.1-8	备用 Reserve



表 C.7 (续) Continued table C.7

端口 Port	说明 Instruction
455DO5-1	发电机定子加热器 1 Generator stator heater1
455DO5-4	备用 Reserve
455DO5-5	发电机定子加热器 2 Generator stator heater2
455DO5-8	发电机内部冷却风扇 Cooling fan inside the generator
455DO6-1	主轴承润滑泵 Rotor bearing lubrication pump
455DO6-4	液压油泵加热器 Hydraulic oil pump heater
455DO6-5	偏航齿轮润滑泵 Yaw gear lubrication pump
455DO6-8	液压泵马达 Hydraulic motor
456DO5-1	左偏航 Yaw left
456DO5-4	液压偏航制动阀完全打开 Hydraulic yaw brake valves completely open
456DO5-5	右偏航 Yaw right
456DO5-8	液压偏航制动阀部分打开 Hydraulic yaw brake valves partly open
456DO5.1-1	偏航轴承润滑泵 Yaw bearing lubrication pump
456DO5.1-4	偏航制动马达 Yaw brake motor
456DO5.1-5	液压风轮制动阀打开 Hydraulic rotor brake valves open
456DO5.1-8	液压风轮制动阀关闭 Hydraulic rotor brake valves close
456DO6-1	备用 Reserve
456DO6-4	备用 Reserve
456DO6-5	齿轮箱冷却水风扇 Gearbox cooling fan
456DO6-8	备用 Reserve

附录 C. 2. 2. 2 安全 DO 信号列表 Appendix C.2.2.2 Safety DO signals list

表 C.8 Table C.8

端口 Port	说明 Instruction
456DO3-1	安全系统变桨紧停 Pitch emergency stop of safety system
456DO3-4	备用 Reserve
456DO3-5	安全系统偏航紧停 Yawing emergency stop of safety system
456DO3-8	备用 Reserve

附录 C. 2. 2. 3 机舱 AI 信号列表 Appendix C.2.2.3 AI signals list in nacelle

表 C.9 Table C.9

端口 Port	说明 Instruction
457AI3-1/2	机舱 X 轴振动 Nacelle vibration of X axes
457AI3-3/4	风轮转速 2 Rotor speed 2
457AI3-5/6	机舱 Y 轴振动 Nacelle vibration of Y axes
457AI3-7/8	风轮转速 1 Rotor speed 1
457AI3.1-1/2	风向 1 Wind direction 1
457AI3.1-3/4	备用 Reserve
457AI3.1-5/6	风速 2 Wind speed 2
457AI3.1-7/8	备用 Reserve
457AI3-1/2	风向 2 Wind direction 2
457AI4-3/4	风速 1 Wind speed 1
457AI4-5/6	备用 Reserve
457AI4-7/8	备用 Reserve
457AI5-1/2	液压系统主压力 Main pressure of hydraulic system



表 C.9 (续) Continued table C.9

端口 Port	说明 Instruction
457AI5-3/4	备用 Reserve
457AI5-5/6	备用 Reserve
457AI5-7/8	备用 Reserve
458AI1-1/2	机舱控制柜温度 Temperature of control cabinet in nacelle
458AI1-3/4	齿轮箱高速轴发电机侧温度 Temperature of gearbox high speed side
458AI1-5/6	备用 Reserve
458AI1-7/8	齿轮箱低速轴风轮侧温度 Temperature of gearbox low speed side
458AI2-1/2	齿轮箱入口油温 Temperature of the oil at the gearbox entrance
458AI2-3/4	备用 Reserve
458AI2-5/6	备用 Reserve
458AI2-7/8	齿轮箱油温 Oil temperature of gearbox
458AI2.1-1/2	齿轮箱冷却水温 Cooling water temperature of gearbox
458AI2.1-3/4	备用 Reserve
458AI2.1-5/6	机舱外温度 Temperature outside the nacelle
458AI2.1-7/8	机舱温度 Temperature in the nacelle
458AI3-1/2/3	发电机定子绕组 U1 相温度 U1 phase of the generator stator temperature
458AI3-5/6/7	发电机定子绕组 V1 相温度 V1 phase of generator stator temperature
458AI4-1/2/3	发电机定子绕组 W1 相温度 W1 phase of generator stator temperature
458AI4-5/6/7	发电机前轴承温度 Front generator bearing temperature
458AI4.1-1/2/3	发电机后轴承温度 Behind generator bearing temperature

表 C. 9 (续) Continued table C.9

端口 Port	说明 Instruction
458AI4.1-5/6/7	发电机进风口温度 Temperature of generator wind entrance
458AI5-1/2/3	发电机出风口温度 Temperature of generator wind exit
458AI5-5/6/7	发电机滑环温度 Generator clip ring temperature